Adaptive Finite Element Methods for Elliptic PDEs

A Short Course

Section Notes (9 Sections)

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Contents

1 Basic Theory of PDE

We motivate the variational framework through a model minimization problem, derive its Euler–Lagrange equations, and introduce the function spaces (weak derivatives, Sobolev spaces) in which the theory is well-posed.

1.1 The minimization problem

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with $\partial\Omega$ smooth enough to integrate by parts (say C^1 or polygonal). Given a continuous datum $f \in C(\bar{\Omega})$, consider the *energy* (Dirichlet) functional

$$J[v] = \tfrac{1}{2}\int_\Omega |\nabla v|^2 \,\mathrm{d}x - \int_\Omega f\,v\,\mathrm{d}x.$$

We work, for now, in the classical admissible set of functions that are smooth up to the boundary and satisfy the homogeneous Dirichlet condition,

$$\mathscr{A} = \left\{ v \in C^1(\bar{\Omega}) : v = 0 \text{ on } \partial\Omega \right\}, \tag{1}$$

and consider the minimization problem

$$\text{find } u \in \mathscr{A} \;\text{ such that }\; J[u] = \min_{v\in\mathscr{A}} J[v]. \tag{2}$$

Note $J[v]$ only involves first derivatives, so it is well defined for every $v \in \mathscr{A} \subset C^1(\bar{\Omega})$.

1.2 Variational formulation

Suppose $u \in \mathscr{A}$ minimizes J. Fix a *test function* $v \in C_0^2(\bar{\Omega})$ (smooth, vanishing on $\partial\Omega$), so that $u + tv \in \mathscr{A}$ for all $t \in \mathbb{R}$. The scalar function $g(t) = J[u + tv]$ has a minimum at $t = 0$, and

$$g(t) = J[u] + t\Big(\int_\Omega \nabla u \cdot \nabla v - \int_\Omega f v \Big) + \tfrac{t^2}{2}\int_\Omega |\nabla v|^2 .$$

Thus $g'(0) = 0$ gives the *variational (weak) formulation*: find $u \in \mathscr{A}$ such that

$$\mathscr{B}[u, v] := \int_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x = \int_\Omega f v \,\mathrm{d}x =: \langle f, v\rangle \qquad \forall\, v \in C_0^2(\bar{\Omega}). \tag{3}$$

Conversely, since J is convex (the quadratic term $\frac{t^2}{2}\int_\Omega |\nabla v|^2 \geq 0$), any $u \in \mathscr{A}$ satisfying (3) is a minimizer. So minimization and the variational identity are equivalent on $\mathscr{A}$.

1.3 Euler–Lagrange equations

If a minimizer $u \in C^2(\Omega)$, we may integrate (3) by parts. Since $\operatorname{div}(v\nabla u) = \nabla u \cdot \nabla v + v\,\Delta u$, the Divergence Theorem gives

$$\int_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x = \int_\Omega \operatorname{div}(v\nabla u)\,\mathrm{d}x - \int_\Omega v\,\Delta u\,\mathrm{d}x = \int_{\partial\Omega} v\,\frac{\partial u}{\partial n}\,\mathrm{d}s - \int_\Omega (\Delta u)\,v\,\mathrm{d}x,$$

where n is the outward unit normal on $\partial\Omega$. The boundary term drops because $v = 0$ on $\partial\Omega$, so (3) becomes $-\int_\Omega (\Delta u)\,v\,\mathrm{d}x = \int_\Omega f\,v\,\mathrm{d}x$ for all $v \in C_0^2(\bar{\Omega})$, i.e.

$$\int_\Omega (-\Delta u - f)\,v\,\mathrm{d}x = 0 \qquad \forall\, v \in C_0^2(\bar{\Omega}).$$

As $-\Delta u - f$ is continuous and this holds for all such v, the fundamental lemma of the calculus of variations forces the integrand to vanish pointwise. The *strong (Euler–Lagrange) form* is therefore the Poisson problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ \quad\; u = 0 & \text{on } \partial\Omega. \end{cases} \tag{4}$$

More generally, for $J[v] = \frac{1}{2}\int_\Omega A\nabla v\cdot\nabla v - \int_\Omega fv$ with a symmetric positive-definite piecewise smooth coefficient $A(x)$, the same computation yields $-\operatorname{div}(A\nabla u) = f$.

1.4 Why classical spaces are not enough

We have shown: *if* a classical minimizer $u \in \mathscr{A}$ exists, it solves (4). But does a minimizer exist in $\mathscr{A}$? The direct method of the calculus of variations suggests how to build one: J is bounded below (complete the square and use the Poincaré inequality), so take a *minimizing sequence* $v_n \in \mathscr{A}$ with $J[v_n] \to \inf_{\mathscr{A}} J$. The natural quantity controlled along the sequence is the *energy*

$$\|v\|_{\mathscr{B}[,]} := \Big(\int_\Omega |\nabla v|^2 \,\mathrm{d}x\Big)^{1/2},$$

and one checks (v_n) is Cauchy in this norm. The trouble is that $\big(\mathscr{A},\ \|\cdot\|_{\mathscr{B}[,]}\big)$ is *not complete*: the limit of a sequence of smooth functions with bounded energy need not be C^2, or even C^1. A Cauchy minimizing sequence may converge to a function with a corner or mild singularity — exactly the behavior we expect near reentrant corners (Section 4) — which lies outside $\mathscr{A}$.

This is the same obstruction seen from the operator side. The clean existence tool for a symmetric problem is the following classical result.

Theorem 1.1 (Riesz Representation)**.** *Let $\mathbb{V}$ be a Hilbert space with inner product $\langle\cdot,\cdot\rangle_{\mathbb{V}}$ and induced norm $\|\cdot\|_{\mathbb{V}}$, and let $F\colon \mathbb{V} \to \mathbb{R}$ be a bounded linear functional. Then there exists a* unique $u_F \in \mathbb{V}$ *such that*

$$F(v) = \langle u_F, v\rangle_{\mathbb{V}} \qquad \forall\, v \in \mathbb{V},$$

and $\|u_F\|_{\mathbb{V}} = \|F\|_{\mathbb{V}^*} := \sup_{v\neq 0} |F(v)|/\,\|v\|_{\mathbb{V}}$.

In words: every bounded linear functional on $\mathbb{V}$ is *represented* by taking the inner product with a unique fixed element $u_F \in \mathbb{V}$. Because $\mathscr{B}[\cdot,\cdot]$ is *symmetric*, it is itself an inner product on any subspace where it is coercive, so the variational problem (3) asks precisely for the Riesz representative of $F(v) = \langle f, v\rangle_{L^2}$ with respect to this inner product. To apply Theorem 1.1 we need a function space that

1. is *complete* for the energy norm (so minimizing/Galerkin sequences have limits in the space), and
2. requires only *one* derivative, in an integral sense, since that is all $\mathscr{B}[u,v]$ and $J[v]$ ever use.

The remedy is to enlarge $\mathscr{A}$ to the *completion* of $C_0^\infty(\bar{\Omega})$ under the energy norm. This completion is the Sobolev space $H_0^1(\Omega)$ introduced below; it is a Hilbert space, and on it the Riesz theorem applies with coercivity supplied by the Poincaré inequality $\|v\|_{L^2} \le C_\Omega \|\nabla v\|_{L^2}$ and continuity by Cauchy–Schwarz. The remainder of this Section builds the machinery (weak derivatives, Sobolev spaces, embeddings) needed to make this rigorous. Section 1.11 extends the existence theory to nonsymmetric problems, where the Riesz argument no longer applies and one must invoke the Lax–Milgram theorem.

1.5 Weak derivatives

The variational formulation (3) only requires *one* derivative of u and v, and only in an integral sense. We first make precise what "a derivative in an integral sense" means.

Before introducing the concept of weak derivative, we observe that if u and v are C^1 functions on a Lipschitz domain Ω, we have, applying the Divergence Theorem to uve_i, the following *integration by parts* formula:

$$\int_\Omega u\frac{\partial v}{\partial x_i} = \int_{\partial\Omega} u\,v\,n_i - \int_\Omega \frac{\partial u}{\partial x_i}v. \tag{5}$$

Definition 1.2 (weak derivative)**.** Let $u \in L^1_{\text{loc}}(\Omega)$ and $\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{N}_0^d$ a multi-index with $|\alpha| = \alpha_1 + \cdots + \alpha_d$. We say $g \in L^1_{\text{loc}}(\Omega)$ is the *weak derivative* $D^\alpha u$ if

$$\int_\Omega u\,D^\alpha\varphi\,\mathrm{d}x = (-1)^{|\alpha|}\int_\Omega g\,\varphi\,\mathrm{d}x \qquad \forall\,\varphi \in C_c^\infty(\Omega).$$

The identity comes from a repeated integration by parts with boundary terms absent because φ is compactly supported in Ω (the function all its derivatives vanish in a neighborhood of the boundary). Weak derivatives are unique up to sets of measure zero; when the classical derivative exists and is locally integrable it coincides with the weak one.

Example 1.3 (Classical functions)**.** If $u \in C^k(\Omega)$, the weak derivative $D^\alpha u$ equals the classical derivative for $|\alpha| \le k$.

Example 1.4 (Piecewise smooth functions)**.** Let $\Omega = \Omega_1 \cup \gamma \cup \Omega_2$ where $\gamma = \partial\Omega_1 \cap \partial\Omega_2$ is a smooth interface, and suppose $u_i \in C^1(\overline{\Omega}_i)$ with matching traces $u_1|_\gamma = u_2|_\gamma$. Then the weak gradient is the piecewise classical gradient:

$$\nabla u(x) = \begin{cases} \nabla u_1(x), & x \in \Omega_1, \\ \nabla u_2(x), & x \in \Omega_2. \end{cases} \tag{6}$$

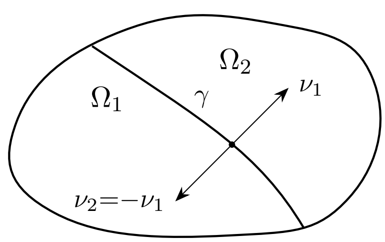


Figure 1: Two subdomains meeting at interface γ with opposite outward normals. Provided traces match, the weak gradient is the piecewise classical gradient.

Proof of (6). Integrate by parts on each subdomain:

$$\begin{aligned} -\int_\Omega u\nabla\varphi\,\mathrm{d}x &= -\int_{\Omega_1} u_1\nabla\varphi\,\mathrm{d}x - \int_{\Omega_2} u_2\nabla\varphi\,\mathrm{d}x \\ &= \int_{\Omega_1} \nabla u_1\,\varphi\,\mathrm{d}x + \int_{\Omega_2} \nabla u_2\,\varphi\,\mathrm{d}x - \int_\gamma u_1\varphi\,\nu_1\,\mathrm{d}s - \int_\gamma u_2\varphi\,\nu_2\,\mathrm{d}s. \end{aligned}$$

Since $u_1 = u_2$ on γ and $\nu_2 = -\nu_1$, the interface terms cancel.

A key application: *hat functions* ϕ_z (continuous, piecewise linear, equal to 1 at node z like the one depicted in Figure 4 satisfy (6), so their weak gradients are the piecewise-constant gradients $\nabla\phi_z|_T$.

Example 1.5 (Radial functions)**.** Let $\Omega = B(0,1) \subset \mathbb{R}^d$ and $f(x) = \rho(|x|)$ where $\rho\colon (0,1] \to \mathbb{R}$ is smooth with $\int_0^1 |\rho'(r)|\, r^{d-1}\, \mathrm{d}r < \infty$. Then $\nabla f(x) = \rho'(|x|)\, x/\,|x|$ for $x \neq 0$. (Exercise)

Example 1.6 (The borderline case: $W_d^1 \not\hookrightarrow L^\infty$)**.** On $\Omega = B(0,1) \subset \mathbb{R}^d$, consider $u(x) = \log\log \dfrac{2}{|x|}$. By the exercise,

$$\nabla u(x) = -\frac{1}{\log \frac{2}{|x|}}\, \frac{x}{|x|^2}, \qquad \int_\Omega |\nabla u|^d \,\mathrm{d}x = C_d \int_0^1 \frac{1}{\left(\log \frac{2}{r}\right)^d}\, \frac{\mathrm{d}r}{r} < \infty,$$

so $u \in W_d^1(\Omega)$, yet u is unbounded near 0. Hence $W_d^1(\Omega) \not\hookrightarrow L^\infty(\Omega)$ (the Sobolev number $\mathrm{sob}(W_d^1) = 1 - d/d = 0$ is the borderline where L^∞ fails). By superposition, taking $\sum_i \alpha_i u(\cdot - x_i)$ with x_i dense and $\alpha_i \to 0$ fast, one can construct W_d^1 functions that are infinite on a prescribed dense set.

Example 1.7 (The fundamental solution)**.** The functions

$$\Phi(x) = \begin{cases} -\frac{1}{2\pi} \log|x|\,, & d = 2, \\ \frac{1}{4\pi|x|}\,, & d = 3, \end{cases}$$

satisfy $\langle -\Delta\Phi, \varphi\rangle = \varphi(0)$ for all $\varphi \in C_c^\infty(\mathbb{R}^d)$, i.e. $-\Delta\Phi = \delta_0$ weakly, where δ_0 is the Dirac mass at the origin. Note $\Phi \in W^1_{p,\mathrm{loc}}(\mathbb{R}^d)$ for $p < d/(d-1)$, but $\Phi \notin H^1_{\mathrm{loc}}(\mathbb{R}^d)$ for $d \geq 2$.

1.6 Sobolev spaces and the Sobolev number

Definition 1.8 (Sobolev space)**.** For $1 \leq p \leq \infty$ and $m \in \mathbb{N}_0$,

$$W_p^m(\Omega) = \bigl\{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega)\ \forall\, |\alpha| \leq m\bigr\},$$

with norm and seminorm

$$\|u\|_{W_p^m} = \Bigl(\sum_{|\alpha|\leq m} \|D^\alpha u\|_{L^p}^p\Bigr)^{1/p}, \qquad |u|_{W_p^m} = \Bigl(\sum_{|\alpha|= m} \|D^\alpha u\|_{L^p}^p\Bigr)^{1/p}, \qquad \text{for } 1 \leq p < \infty,$$

$$\|u\|_{W_\infty^m} = \max_{|\alpha|\leq m} \|D^\alpha u\|_{L^\infty}\,, \qquad |u|_{W_\infty^m} = \max_{|\alpha|= m} \|D^\alpha u\|_{L^\infty}\,.$$

We write $H^m = W_2^m$ and define $H_0^1(\Omega)$ as the closure of $C_c^\infty(\Omega)$ in H^1.

Exercise 1.9. Show that $W_p^m(\Omega)$ is a *Banach space*. *Hint:* embed it isometrically as a closed subspace of $\bigoplus_{|\alpha|\leq m} L^p(\Omega)$ via $u \mapsto (D^\alpha u)_{|\alpha|\leq m}$, and use that $L^p(\Omega)$ is complete.

Remark 1.10 ($H^m(\Omega)$ is a Hilbert space)*.* For $p = 2$ the norm arises from the inner product $\langle u, v\rangle_{H^m} = \sum_{|\alpha|\leq m} \int_\Omega D^\alpha u\, D^\alpha v\,\mathrm{d}x$, making $H^m(\Omega)$ a Hilbert space. In particular, $H_0^1(\Omega)$ carries the inner product $\langle u, v\rangle_{H_0^1} = \int_\Omega \nabla u \cdot \nabla v\,\mathrm{d}x$, which is equivalent to the full H^1 inner product by the Poincaré–Friedrichs inequality (Lemma 1.17, proved in Section 1.9).

Well-posedness of the Poisson problem in $H_0^1(\Omega)$. We now have all the pieces to make the existence argument of Section 1 rigorous. The weak formulation (3) set in $H_0^1(\Omega)$ reads: find $u \in H_0^1(\Omega)$ such that

$$\mathscr{B}[u, v] = \int_\Omega \nabla u \cdot \nabla v\,\mathrm{d}x = \int_\Omega f\, v\,\mathrm{d}x =: F(v) \qquad \forall\, v \in H_0^1(\Omega). \tag{7}$$

The left-hand side is the H_0^1 inner product $\langle u, v\rangle_{H_0^1} = \mathscr{B}[u,v]$; that $\mathscr{B}[\cdot,\cdot]$ really is an inner product on $H_0^1(\Omega)$ rests on the zero-trace Poincaré–Friedrichs inequality, stated and proved later as Lemma 1.17 in Section 1.9, which forces $\mathscr{B}[v,v] = \|\nabla v\|_{L^2}^2 = 0 \Rightarrow v = 0$. The right-hand side defines a bounded linear functional: by Cauchy–Schwarz and the same Poincaré–Friedrichs inequality $\|v\|_{L^2} \le C_\Omega \|\nabla v\|_{L^2}$,

$$|F(v)| \le \|f\|_{L^2} \|v\|_{L^2} \le C_\Omega \|f\|_{L^2} \|\nabla v\|_{L^2} = C_\Omega \|f\|_{L^2} \|v\|_{H_0^1},$$

so $F \in (H_0^1(\Omega))^*$. The Riesz Representation Theorem 1.1 (with $\mathbb{V} = H_0^1(\Omega)$) therefore guarantees a *unique* solution $u \in H_0^1(\Omega)$, satisfying the stability bound

$$\|u\|_{H_0^1(\Omega)} = \|F\|_{(H_0^1)^*} \le C_\Omega \|f\|_{L^2(\Omega)}.$$

Sobolev Number. The single most useful bookkeeping device is the *Sobolev number*

$$\mathrm{sob}(W_p^m) = m - \frac{d}{p}. \tag{8}$$

It appears in every scaling argument. Let $\widehat{x} = \lambda x$ be a dilation $\Omega \to \widehat{\Omega}$ and set $\widehat{v}(\widehat{x}) = v(x)$. Then $D^\alpha v(x) = \lambda^{|\alpha|} D^\alpha \widehat{v}(\widehat{x})$, and changing variables ($\mathrm{d}\widehat{x} = \lambda^d \,\mathrm{d}x$),

$$\|D^\alpha v\|_{L^p(\Omega)} = \lambda^{|\alpha| - d/p} \|D^\alpha \widehat{v}\|_{L^p(\widehat{\Omega})}.$$

At the top order $|\alpha| = m$ the exponent is exactly $m - d/p = \mathrm{sob}(W_p^m)$: this is the power of λ that $|\cdot|_{W_p^m}$ acquires under dilation. A bounded, scale-invariant embedding $W_p^m \hookrightarrow W_q^k$ is therefore only possible when $\mathrm{sob}(W_p^m) \ge \mathrm{sob}(W_q^k)$.

1.7 Sobolev embeddings

Motivation: a family of radial powers. On the unit ball $B \subset \mathbb{R}^d$ take the radial power functions

$$v_\gamma(x) = |x|^\gamma, \qquad \gamma \in \mathbb{R}.$$

Their homogeneity turns every norm into an elementary radial integral, so we can see exactly how much integrability survives a given amount of differentiability. Writing $\mathrm{d}x = \omega_{d-1} r^{d-1}\,\mathrm{d}r$ in polar coordinates (ω_{d-1} the surface area of the unit sphere),

$$\int_B |v_\gamma|^q \,\mathrm{d}x = \omega_{d-1} \int_0^1 r^{\gamma q + d - 1}\,\mathrm{d}r < \infty \iff \gamma q + d > 0 \iff \gamma > -\frac{d}{q},$$

so $v_\gamma \in L^q(B) \iff \gamma > -d/q$. The gradient is $\nabla v_\gamma(x) = \gamma\, |x|^{\gamma-2} x$, hence $|\nabla v_\gamma| = |\gamma|\, |x|^{\gamma-1}$, and the same computation with the exponent lowered by one gives

$$\nabla v_\gamma \in L^p(B) \iff (\gamma - 1)p + d > 0 \iff \gamma > 1 - \frac{d}{p}.$$

The companion requirement $v_\gamma \in L^p$ (that is, $\gamma > -d/p$) is weaker, and for these γ the pointwise gradient really is the weak gradient—the singularity at the origin is too mild to leave a distributional mass there. Hence

$$v_\gamma \in W_p^1(B) \iff \gamma > 1 - \frac{d}{p}.$$

Now compare the two thresholds. If $v_\gamma \in W_p^1(B)$ then $\gamma > 1 - d/p$, and this *forces* $v_\gamma \in L^q(B)$ exactly when every such γ also clears the bar $-d/q$, i.e. when

$$1 - \frac{d}{p} \;\geq\; -\frac{d}{q} \qquad \Longleftrightarrow \qquad \frac{1}{q} \;\geq\; \frac{1}{p} - \frac{1}{d} \qquad \Longleftrightarrow \qquad q \leq \frac{dp}{d-p} \quad (p < d).$$

This is precisely the Sobolev-number condition $\mathrm{sob}(W_p^1) \geq \mathrm{sob}(W_q^0)$ of (8) (recall $L^q = W_q^0$, so $\mathrm{sob}(W_q^0) = -d/q$), now read off from concrete functions rather than from a dilation. The examples also show that the condition is *sharp*: as soon as $1 - d/p < -d/q$ (that is, $q > dp/(d-p)$), any γ in the nonempty range $1 - d/p < \gamma \leq -d/q$ yields a function $v_\gamma \in W_p^1(B)$ that is *not* in $L^q(B)$, so no embedding can hold. The theorem below says that this necessary scaling relation is also sufficient, on any bounded Lipschitz domain.

Theorem 1.11 (Sobolev embedding)**.** *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, $m > k \geq 0$, $1 \leq p, q \leq \infty$. Then, if* $\mathrm{sob}(W_p^m) > \mathrm{sob}(W_q^k)$*, ($m - d/p > k - d/q$) or* $\mathrm{sob}(W_p^m) = \mathrm{sob}(W_q^k)$ *and $q < \infty$,*

$$W_p^m(\Omega) \hookrightarrow W_q^k(\Omega).$$

Also, $\|v\|_{W_q^k} \leq C\, \|v\|_{W_p^m}$ for all $v \in W_p^m(\Omega)$, and the embedding is compact *when the inequality is strict and $m > k$.*

Examples. *1. $W_p^1 \hookrightarrow L^q$ (gaining integrability).* Take $m = 1$, $k = 0$: the embedding $W_p^1 \hookrightarrow L^q$ holds for $1 - d/p \geq -d/q$, i.e. $q \leq dp/(d-p)$ when $p < d$. For $d = 2$ and $p = 2$: $H^1(\Omega) \hookrightarrow L^q(\Omega)$ for all $1 \leq q < \infty$ (any finite exponent).

2. The borderline sob $= 0$ *fails for L^∞.* For $p = d$ we have $\mathrm{sob}(W_d^1) = 0$. Example 1.6 above shows $u(x) = \log\log\frac{2}{|x|} \in W_d^1(B_1) \setminus L^\infty(B_1)$, confirming $W_d^1(\Omega) \not\hookrightarrow L^\infty(\Omega)$. (The embedding does hold into L^q for all $q < \infty$.)

3. Morrey embedding into Hölder spaces. If $\mathrm{sob}(W_p^1) = 1 - d/p > 0$ (i.e. $p > d$), then $W_p^1(\Omega) \hookrightarrow C^{0,\alpha}(\overline{\Omega})$ with $\alpha = 1 - d/p$.

4. Why Lipschitz? Embeddings can fail on non-Lipschitz domains. Consider the cusp $\Omega = \{(x,y) \in \mathbb{R}^2 : |y| < x^\gamma,\ 0 < x < 1\}$, $\gamma > 1$, which is not Lipschitz. For $p > 2$ and small $\varepsilon > 0$ with $p < \gamma + 1 - \varepsilon$, the function $v(x,y) = x^{-\varepsilon/p}$ satisfies $v \in W_p^1(\Omega)$ yet $v \notin L^\infty(\Omega)$, so the embedding $W_p^1 \hookrightarrow L^\infty$ fails.

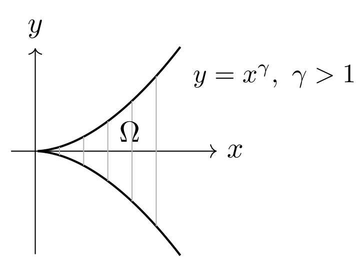


Figure 2: A cusp domain ($\gamma > 1$): not Lipschitz. The Sobolev embedding $W_p^1 \hookrightarrow L^\infty$ ($p > 2$) fails on this domain.

Remark 1.12 (Lipschitz functions). On a Lipschitz domain, $W_\infty^1(\Omega) = C^{0,1}(\overline{\Omega})$: the two conditions "$v$ and ∇v are essentially bounded" and "v is Lipschitz" are equivalent.

Remark 1.13 (nodal interpolation). Embeddings into C^0 are what make nodal interpolation (Section 2) well defined, since they let us evaluate functions at points. In particular for $d = 2$: $H^2 \hookrightarrow C^0$ since $\mathrm{sob}(H^2) = 2 - 1 = 1 > 0$, and the borderline space W_1^2 also embeds since $\mathrm{sob}(W_1^2) = 2 - 2 = 0$ (this case is outside the scope of Theorem 1.11, but holds because the case $p = 1$ is also special).

1.8 Density

Theorem 1.14 (density on Lipschitz domains)**.** *If $\Omega \subset \mathbb{R}^d$ is Lipschitz and $1 \le p < \infty$, then $C^\infty(\overline{\Omega})$ is dense in $W_p^m(\Omega)$: for every $u \in W_p^m(\Omega)$ there exists $u_n \in C^\infty(\overline{\Omega})$ with $\|u - u_n\|_{W_p^m(\Omega)} \to 0$.*

Theorem 1.15 (density on open domains)**.** *If $\Omega \subset \mathbb{R}^d$ is open and $1 \le p < \infty$, then $C^\infty(\Omega) \cap W_p^m(\Omega)$ is dense in $W_p^m(\Omega)$.*

Remark 1.16. Both theorems fail for $p = \infty$: the uniform limit of continuous functions is continuous, but $C^0(\overline{\Omega}) \subsetneq L^\infty(\Omega)$.

1.9 Poincaré–Friedrichs inequalities

Two inequalities of Poincaré type recur throughout these notes. One controls a function by its gradient when the function has zero boundary trace; it is what makes the energy form an inner product on $H_0^1(\Omega)$ and underlies the well-posedness argument of Section 1.6. The other applies when the function has zero mean; it is the analytic engine of the interpolation theory of Section 2. Both express the same idea: modulo the kernel of the gradient (the constants), the full W_p^1 norm is controlled by the gradient alone.

Lemma 1.17 (Poincaré–Friedrichs inequality, zero trace)**.** *Let $\Omega \subset \mathbb{R}^d$ be bounded and $1 \le p < \infty$. There is a constant C_Ω, depending only on* $\operatorname{diam}\Omega$ *and p, such that*

$$\|v\|_{L^p(\Omega)} \le C_\Omega \, \|\nabla v\|_{L^p(\Omega)} \qquad \text{for all } v \in W_{p,0}^1(\Omega),$$

where $W_{p,0}^1(\Omega)$ is the closure of $C_c^\infty(\Omega)$ in $W_p^1(\Omega)$ (so $W_{2,0}^1 = H_0^1$). Consequently $|v|_{W_p^1} = \|\nabla v\|_{L^p}$ is a norm on $W_{p,0}^1(\Omega)$, equivalent to the full W_p^1 norm.

Proof. By density it suffices to prove the bound for $v \in C_c^\infty(\Omega)$; the general case follows by passing to the limit. Since Ω is bounded we may translate so that $\Omega \subset \{x : 0 < x_1 < L\}$ with $L = \operatorname{diam}\Omega$. Extending v by zero outside Ω and using the fundamental theorem of calculus in the x_1 direction,

$$v(x) = \int_0^{x_1} \partial_1 v(t, x') \, \mathrm{d}t, \qquad x = (x_1, x').$$

For $p > 1$, Hölder's inequality gives

$$|v(x)|^p \le x_1^{p-1} \int_0^L |\partial_1 v(t, x')|^p \, \mathrm{d}t \le L^{p-1} \int_0^L |\partial_1 v(t, x')|^p \, \mathrm{d}t$$

(for $p = 1$ the factor x_1^{p-1} is simply absent). Integrating first in $x_1 \in (0, L)$ and then in x',

$$\|v\|_{L^p(\Omega)}^p \le L^p \int_\Omega |\partial_1 v|^p \, \mathrm{d}x \le L^p \, \|\nabla v\|_{L^p(\Omega)}^p ,$$

which is the claim with $C_\Omega = \operatorname{diam}\Omega$. The norm equivalence then follows from $\|\nabla v\|_{L^p} \le \|v\|_{W_p^1} \le (1 + C_\Omega) \, \|\nabla v\|_{L^p}$. $\square$

The energy inner product. Lemma 1.17 is exactly what makes the Dirichlet form $\mathscr{B}[u,v] = \int_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x$ an *inner product* on $H_0^1(\Omega)$: bilinearity and symmetry are immediate, and positive definiteness is the implication $\mathscr{B}[v,v] = \|\nabla v\|_{L^2}^2 = 0 \Rightarrow v = 0$, which is precisely the case $C_\Omega \cdot 0 \geq \|v\|_{L^2}$ of the lemma. This is the fact invoked (by forward reference) in the well-posedness argument of Section 1, where it also supplies the bound $\|v\|_{L^2} \leq C_\Omega \|\nabla v\|_{L^2}$ used to show F is bounded.

Lemma 1.18 (Poincaré-Friedrichs inequality, zero mean)**.** *Let $B \subset \mathbb{R}^d$ be a bounded, connected, Lipschitz domain and $1 \leq p \leq \infty$. There is a constant $C_P = C_P(B,p)$ such that every $g \in W_p^1(B)$ with vanishing mean ($\int_B g \,\mathrm{d}x = 0$) satisfies*

$$\|g\|_{L^p(B)} \leq C_P \; |g|_{W_p^1(B)} = C_P \; \|\nabla g\|_{L^p(B)} \, .$$

Proof. Argue by contradiction. If no such C_P existed, there would be a sequence $g_n \in W_p^1(B)$ with

$$\int_B g_n \,\mathrm{d}x = 0, \qquad \|g_n\|_{L^p(B)} = 1, \qquad |g_n|_{W_p^1(B)} \to 0.$$

The g_n are then bounded in $W_p^1(B)$, so by the Rellich–Kondrachov theorem (the embedding $W_p^1(B) \hookrightarrow L^p(B)$ is compact for a Lipschitz B) a subsequence, not relabeled, converges in $L^p(B)$ to some g. Since $\nabla g_n \to 0$ in $L^p(B)$ as well, the sequence is Cauchy in $W_p^1(B)$ and the limit obeys $\nabla g = 0$; hence g is constant on the connected set B. Passing to the limit in $\int_B g_n \,\mathrm{d}x = 0$ gives $\int_B g \,\mathrm{d}x = 0$, so that constant is zero, i.e. $g \equiv 0$. This contradicts $\|g\|_{L^p(B)} = \lim \|g_n\|_{L^p(B)} = 1$. $\square$

Remark 1.19 (two faces of one idea). Each lemma removes the kernel of the gradient by a normalization: a boundary condition in Lemma 1.17, a zero-mean condition in Lemma 1.18. The zero-trace form drives coercivity and well-posedness (Section 1); the zero-mean form drives polynomial approximation and the interpolation estimates (Section 2). The contradiction proof of Lemma 1.18 applies verbatim to Lemma 1.17 (replacing the zero-mean constraint by the zero-trace one); we gave the direct, constant-explicit argument for the latter because it also yields $C_\Omega = \operatorname{diam} \Omega$.

1.10 Extension*

* *This subsection is supplementary and may be skipped on first reading: the extension theorem is not invoked anywhere else in this short course. It is included because extension operators are a fundamental tool in the broader theory of Sobolev spaces—underlying trace theorems, embeddings on general domains, and interpolation of operators.*

Theorem 1.20 (extension)**.** *Let $\Omega \subset \mathbb{R}^d$ be Lipschitz. For every $1 \leq p \leq \infty$ and $m \geq 1$ there exists a bounded linear* extension operator

$$E \colon W_p^m(\Omega) \longrightarrow W_p^m(\mathbb{R}^d)$$

such that $Eu|_\Omega = u$ and $\|Eu\|_{W_p^m(\mathbb{R}^d)} \leq C(\Omega) \; \|u\|_{W_p^m(\Omega)}$.

The proof (Calderón; see Evans for the $k = 1$, $\partial\Omega \in C^1$ case) proceeds by *flattening and reflection*: near each boundary point flatten $\partial\Omega$ to a hyperplane, reflect u across it with coefficients chosen to match the first $m-1$ derivatives, then patch the local extensions with a partition of unity.

Remark 1.21 (necessity of Lipschitz). Extension fails on cusp domains. If $E \colon W_p^1(\Omega) \to W_p^1(\mathbb{R}^d)$ existed for the cusp above, the Sobolev embedding on the surrounding ball would give $W_p^1(\Omega) \hookrightarrow L^\infty(\Omega)$ for $p > 2$, contradicting Example 4.

1.11 Nonsymmetric elliptic problems and the Lax–Milgram theorem

The Riesz representation theorem exploits symmetry: $\mathscr{B}[\cdot,\cdot]$ is itself an inner product on $H_0^1(\Omega)$, and the solution u is simply the Riesz representative of $F = \langle f, \cdot\rangle$. For a *nonsymmetric* problem — e.g. one with a convection term $\boldsymbol{b}\cdot\nabla u$ or a nondivergence-form coefficient matrix $A \neq A^T$ — the bilinear form is no longer symmetric and this argument breaks down. The correct replacement is the following.

Theorem 1.22 (Lax–Milgram)**.** *Let $\mathbb{V}$ be a Hilbert space with norm $\|\cdot\|_{\mathbb{V}}$, let $\mathscr{B}[\cdot,\cdot] : \mathbb{V}\times\mathbb{V} \to \mathbb{R}$ be a bilinear form satisfying*

$$\begin{aligned} |\mathscr{B}[u,v]| &\leq M\ \|u\|_{\mathbb{V}}\,\|v\|_{\mathbb{V}}\,, && \textit{(continuity)}\\ \mathscr{B}[v,v] &\geq \alpha\ \|v\|_{\mathbb{V}}^2\,, && \textit{(coercivity)} \end{aligned}$$

for constants $M, \alpha > 0$ independent of $u, v \in \mathbb{V}$. Let $F : V \to \mathbb{R}$ be a bounded linear functional with $|F(v)| \leq \Lambda\ \|v\|_{\mathbb{V}}$. Then there exists a unique *$u \in \mathbb{V}$ such that*

$$\mathscr{B}[u,v] = F(v) \qquad \forall\, v \in \mathbb{V},$$

and the solution satisfies the stability bound *$\|u\|_{\mathbb{V}} \leq \Lambda/\alpha$.*

Proof sketch. Fix $u \in \mathbb{V}$. Since $v \mapsto \mathscr{B}[u,v]$ is a bounded linear functional, the Riesz theorem gives a unique $Au \in \mathbb{V}$ with $\mathscr{B}[u,v] = \langle Au, v\rangle_{\mathbb{V}}$ for all v; the map $A : \mathbb{V} \to \mathbb{V}$ is linear and bounded ($\|Au\|_{\mathbb{V}} \leq M\,\|u\|_{\mathbb{V}}$). The equation $\mathscr{B}[u,v] = F(v)$ becomes $Au = u_F$ (the Riesz representative of F). Coercivity gives $\alpha\,\|u\|_{\mathbb{V}}^2 \leq \mathscr{B}[u,u] = \langle Au, u\rangle_{\mathbb{V}} \leq \|Au\|_{\mathbb{V}}\,\|u\|_{\mathbb{V}}$, so $\|Au\|_{\mathbb{V}} \geq \alpha\,\|u\|_{\mathbb{V}}$: A is bounded below and hence injective with closed range. One checks surjectivity (e.g. by a Banach fixed-point iteration $u \leftarrow u - \rho(Au - u_F)$ for $\rho > 0$ small enough), yielding the result. ☐

Remark 1.23. When $\mathscr{B}[\cdot,\cdot]$ is symmetric it defines an inner product equivalent to $\langle\cdot,\cdot\rangle_{\mathbb{V}}$, and A is the identity (up to scaling), so Lax–Milgram reduces to the Riesz theorem. Thus Riesz is the symmetric special case of Lax–Milgram, and for the model Poisson problem of this Section the simpler Riesz theorem already suffices.

A general elliptic model problem. Consider the *nonsymmetric second-order elliptic problem*

$$\begin{cases} -\operatorname{div}(A\nabla u) + \boldsymbol{b}\cdot\nabla u + cu = f & \text{in } \Omega,\\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{9}$$

where $A \in L^\infty(\Omega;\mathbb{R}^{d\times d})$ is symmetric and uniformly positive definite, $\boldsymbol{b} \in L^\infty(\Omega;\mathbb{R}^d)$, $c \in L^\infty(\Omega)$, and $f \in L^2(\Omega)$. Set $V = H_0^1(\Omega)$ with norm $\|v\|_{\mathbb{V}} = |v|_{H^1} = \|\nabla v\|_{L^2}$ (equivalent by Poincaré). The bilinear form is

$$\mathscr{B}[u,v] = \int_\Omega A\nabla u\cdot\nabla v\,\mathrm{d}x + \int_\Omega (\boldsymbol{b}\cdot\nabla u)\,v\,\mathrm{d}x + \int_\Omega cu\,v\,\mathrm{d}x.$$

Continuity is immediate from Hölder and the L^∞ bounds. *Coercivity* uses the uniform positive definiteness $A(x)\xi\cdot\xi \geq \lambda\,|\xi|^2$ a.e. and the antisymmetric structure of the convection term: integration by parts gives

$$\int_\Omega (\boldsymbol{b}\cdot\nabla v)\,v\,\mathrm{d}x = -\tfrac{1}{2}\int_\Omega (\operatorname{div}\boldsymbol{b})\,v^2\,\mathrm{d}x,$$

so $\mathscr{B}[v,v] \geq \lambda\,\|\nabla v\|_{L^2}^2 + \int_\Omega (c - \tfrac{1}{2}\operatorname{div}\boldsymbol{b})\,v^2\,\mathrm{d}x \geq \lambda\,\|\nabla v\|_{L^2}^2$ provided $c - \tfrac{1}{2}\operatorname{div}\boldsymbol{b} \geq 0$ a.e. Hence the Lax–Milgram theorem applies, giving a unique $u \in H_0^1(\Omega)$ solving (9) weakly.

Remark 1.24. When $\boldsymbol{b} = 0$ and $c = 0$ the bilinear form reduces to $\int_\Omega A\nabla u \cdot \nabla v$, which is symmetric (since $A = A^T$), and Riesz applies. As soon as $\boldsymbol{b} \neq 0$ the convection term $\int_\Omega (\boldsymbol{b} \cdot \nabla u)\, v$ is not symmetric in u and v, so $\mathscr{B}[u, v] \neq \mathscr{B}[v, u]$ in general and Riesz is no longer available — Lax–Milgram is the right tool. Finite element discretization of (9) proceeds by the Galerkin method exactly as in Section 2; Céa's lemma holds with the ratio M/α (not equal to 1 as in the symmetric case), and the approximation theory is unchanged.

2 Discretization

2.1 Galerkin formulation

The Galerkin method replaces the infinite-dimensional space $\mathbb{V}$ in the weak problem (3) by a finite-dimensional subspace $\mathbb{V}_{\mathscr{T}} \subset \mathbb{V}$, where $\mathscr{T}$ will later denote a mesh over Ω: find $U \in \mathbb{V}_{\mathscr{T}}$ such that

$$\mathscr{B}[U, V] = \langle f, V\rangle \qquad \forall\, V \in \mathbb{V}_{\mathscr{T}}. \tag{10}$$

We point out that (10) reduces to an algebraic linear system, which can be solved by direct or iterative methods – a topic we will not consider in these lectures. To see this, let $\{\phi_j\}_{j=1}^N$ be a basis of $\mathbb{V}_{\mathscr{T}}$, write $U = \sum_{j=1}^N u_j\phi_j$ and set

$$\mathbf{U} = (u_j)_{j=1}^N, \quad \mathbf{F} = (\langle f, \phi_i\rangle)_{i=1}^N, \quad \mathbf{K} = (\mathscr{B}[\phi_j, \phi_i])_{i,j=1}^N.$$

Then, (10) reads equivalently: find $\mathbf{U} \in \mathbb{R}^N$ such that

$$\mathbf{K}\mathbf{U} = \mathbf{F}$$

Exercise 2.1. Show that $\mathbf{K}$ is positive definite and is symmetric if $\mathscr{B}$ is symmetric. Therefore, a method s choice is the preconditioned conjugate gradient method.

Subtracting (10) from (3) gives *Galerkin orthogonality*

$$\mathscr{B}[u - U, V] = 0 \qquad \forall\, V \in \mathbb{V}_{\mathscr{T}}. \tag{11}$$

For the symmetric, coercive form $\mathscr{B}[\cdot, \cdot]$ this says U is the $\mathscr{B}[\cdot, \cdot]$-orthogonal projection of u onto $\mathbb{V}_{\mathscr{T}}$, i.e. the best approximation in the energy norm $|\!|\!|v|\!|\!| = \mathscr{B}[v, v]^{1/2}$. In general, *Céa's lemma* provides the *quasi-best approximation* estimate

$$|u - U|_{H_0^1(\Omega)} \le \frac{M}{\alpha} \inf_{V\in\mathbb{V}_{\mathscr{T}}} |u - V|_{H_0^1(\Omega)}\,, \tag{12}$$

where M and α are the continuity and coercivity constants of $\mathscr{B}$. Therefore, the discretization error is controlled by the *approximation error*, which we estimate next via interpolation.

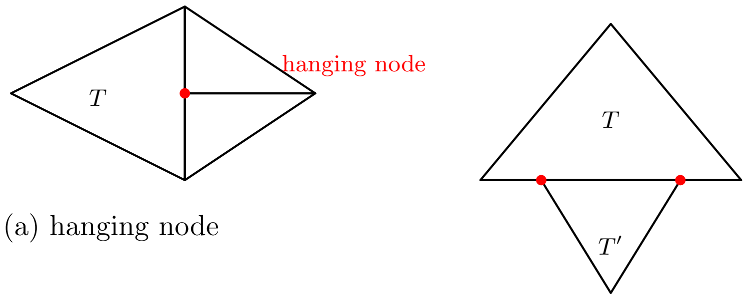
2.2 Triangulations

Let $\Omega \subset \mathbb{R}^d$ be a bounded polyhedral domain. We focus on $d = 2$. i.e. on polygonal domains.

Definition 2.2 (conforming triangulation)**.** A *conforming triangulation* (mesh or partition) $\mathscr{T}$ of Ω is a finite collection of closed triangles such that

1. $\overline{\Omega} = \bigcup_{T\in\mathscr{T}} T$;
2. the interior of any two distinct triangles is disjoint;
3. any two triangles $T, T' \in \mathscr{T}$ share either the empty set, a common vertex, or a common complete edge (no *hanging nodes*).

The third condition ensures that the triangulation is consistent: edges are not partially shared. Figure 3 shows two configurations it rules out. We write $h_T = |T|^{1/2}$ and $h = \max_{T\in\mathscr{T}} h_T$ for the global mesh size.



(a) hanging node

(b) partially shared edge

Figure 3: Two situations *excluded* by conformity. (a) A vertex of the two right-hand triangles falls in the interior of an edge of T, creating a *hanging node*. (b) The triangles T, T' share only *part* of an edge, so their intersection is neither a common vertex nor a common complete edge.

Definition 2.3 (shape regularity)**.** A triangulation $\mathscr{T}$ is *shape-regular* (or *non-degenerate*) with constant $\sigma > 0$ if

$$\frac{\operatorname{diam}(T)}{\rho_T} \leq \sigma \qquad \forall\, T \in \mathscr{T},$$

where ρ_T is the diameter of the largest ball inscribed in T. A *family* $\{\mathscr{T}_h\}_h$ of triangulations is shape-regular if σ can be chosen independently of h. In this case, $h_T = |T|^{1/2}$ is equivalent to the diameter of T.

Shape regularity prevents triangles from becoming flat (large angles approaching π) or needle-like (small angles approaching 0). It is the minimal geometric assumption under which interpolation error constants are bounded uniformly in h.

2.3 Piecewise linear finite elements ($\mathscr{P}_1$)

Fix a triangle T with vertices z_1, z_2, z_3. The *barycentric coordinates* $\lambda_1, \lambda_2, \lambda_3$ on T are the unique affine functions satisfying $\lambda_i(z_j) = \delta_{ij}$ and $\lambda_1 + \lambda_2 + \lambda_3 = 1$; they are non-negative on T. The space of *piecewise linear polynomials* on T is

$$\mathscr{P}_1(T) = \operatorname{span}\{\lambda_1, \lambda_2, \lambda_3\}, \qquad \dim \mathscr{P}_1(T) = 3.$$

Any $p \in \mathscr{P}_1(T)$ is uniquely determined by its values at the three vertices

$$p = \sum_{i=1}^{3} p(z_i)\, \lambda_i,$$

so the vertex evaluations are the degrees of freedom. The functions $\lambda_1, \lambda_2, \lambda_3$ are themselves the *nodal (hat) basis* of $\mathscr{P}_1(T)$.

The *global piecewise-linear space* is

$$\mathbb{V}^1_{\mathscr{T}} = \left\{ V \in C^0(\bar{\Omega}) : V|_T \in \mathscr{P}_1(T)\ \forall\, T \in \mathscr{T} \right\}.$$

It has one degree of freedom per mesh node (vertex) $z \in \mathscr{N}$, and basis (hat) functions $\{\phi_z\}_{z \in \mathscr{N}}$: ϕ_z is the unique function of $\mathbb{V}^1_{\mathscr{T}}$ with $\phi_z(z) = 1$ and $\phi_z(z') = 0$ at every other node z'. Its support

$$\omega_z := \operatorname{supp} \phi_z$$

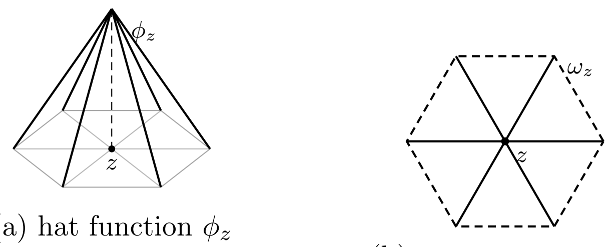
is the patch of triangles surrounding z, from now on called the *star* of z (Figure 4); the stars ω_z will be the natural localization patches in the a posteriori theory of Section 5. Every $V \in \mathbb{V}^1_{\mathscr{T}}$ has the nodal representation

$$V = \sum_{z \in \mathscr{N}} V(z)\, \phi_z.$$

The H^1_0-conforming subspace of $\mathbb{V}^1_{\mathscr{T}}$ is

$$\mathscr{V}_{\mathscr{T}} = \mathbb{V}^1_{\mathscr{T}} \cap H^1_0(\Omega) = \left\{ V \in \mathbb{V}^1_{\mathscr{T}} : V = 0 \text{ on } \partial\Omega \right\},$$

and its nodal basis is $\{\phi_z\}_{z \in \mathscr{N}^0}$ where $\mathscr{N}^0$ denote the interior nodes of $\mathscr{T}$.



(a) hat function ϕ_z

(b) star $\omega_z = \operatorname{supp} \phi_z$

Figure 4: The piecewise-linear nodal basis (hat) function ϕ_z of an interior node z (a), and its support — the star ω_z of triangles containing z (b, boundary dashed, interior skeleton solid). ϕ_z equals 1 at z, 0 at all other nodes, and is linear on each triangle of ω_z.

Continuity across edges (whence H^1-conformity) holds because a $\mathscr{P}_1$ function on a shared edge is determined by its two endpoint values, which are matched by construction. The basis functions $\{\phi_z\}_{z \in \mathscr{N}}$ of $\mathbb{V}^1_{\mathscr{T}}$ satisfy the *partition of unity* property

$$\sum_{z \in \mathscr{N}} \phi_z = 1. \tag{13}$$

2.4 Higher-order Lagrange elements ($\mathscr{P}_k$)

Quadratic case $k = 2$. The space of polynomials of total degree at most 2 on a triangle T is

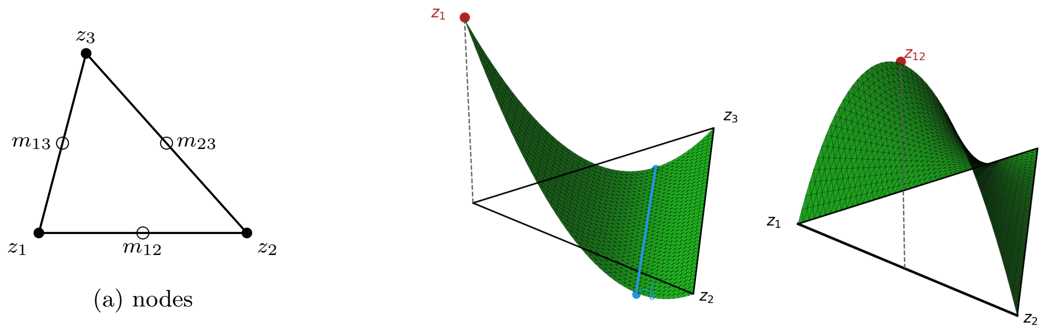
$$\mathscr{P}_2(T) = \Big\{ p = \sum_{|\alpha| \le 2} c_\alpha x^\alpha : c_\alpha \in \mathbb{R} \Big\}, \qquad \dim \mathscr{P}_2(T) = 6.$$

The natural degrees of freedom are point evaluations at the three vertices z_1, z_2, z_3 and the three edge midpoints $m_{ij} = \frac{1}{2}(z_i + z_j)$ (see Figure 5). In barycentric coordinates the nodal basis functions are

$$\underbrace{N_i = \lambda_i\,(2\lambda_i - 1)}_{\text{vertex } z_i}, \qquad \underbrace{N_{ij} = 4\,\lambda_i \lambda_j}_{\text{midpoint } m_{ij}} \quad (i \ne j).$$

The Kronecker property is checked directly: $N_i(z_i) = 1 \cdot (2-1) = 1$, while N_i vanishes at z_j ($\lambda_i = 0$) and at every midpoint m_{ij} ($\lambda_i = \frac{1}{2} \Rightarrow 2\lambda_i - 1 = 0$); and $N_{ij}(m_{ij}) = 4 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1$ while N_{ij} vanishes at all other nodes. Note that a vertex function N_i dips negative near the opposite edge, whereas an edge function N_{ij} is an *edge bubble* peaking at m_{ij}.
General case $k \ge 1$ The space of polynomials of total degree at most k on a triangle T is

$$\mathscr{P}_k(T) = \Big\{ p = \sum_{|\alpha| \le k} c_\alpha x^\alpha : c_\alpha \in \mathbb{R} \Big\}, \qquad \dim \mathscr{P}_k(T) = \binom{k+2}{2} = \frac{(k+1)(k+2)}{2}.$$



(a) nodes

Figure 5: The six $\mathscr{P}_2$ nodes on a triangle (a): vertices z_1, z_2, z_3 (filled) carry the vertex functions $N_i = \lambda_i(2\lambda_i - 1)$, whose graph (b) equals 1 at its own vertex and dips negative near the opposite edge; edge midpoints m_{ij} (open) carry the edge bubbles $N_{ij} = 4\lambda_i\lambda_j$, whose graph (c) peaks at the midpoint and vanishes on ∂T.

The natural degrees of freedom are point evaluations at the *principal lattice nodes* of order k,

$$\mathscr{L}_k(T) = \Big\{ \tfrac{i}{k} z_1 + \tfrac{j}{k} z_2 + \tfrac{\ell}{k} z_3 : i, j, \ell \geq 0,\ i + j + \ell = k \Big\}, \qquad \# \mathscr{L}_k(T) = \dim \mathscr{P}_k(T).$$

These nodes are *unisolvent* on $\mathscr{P}_k(T)$: a polynomial $p \in \mathscr{P}_k(T)$ vanishing at all lattice nodes must be zero. The corresponding nodal basis $\{\phi_\alpha\}$ satisfies $\phi_\alpha(x_\beta) = \delta_{\alpha\beta}$ for $x_\alpha, x_\beta \in \mathscr{L}_k(T)$, and can be written explicitly in terms of barycentric coordinates; see Figure 6.

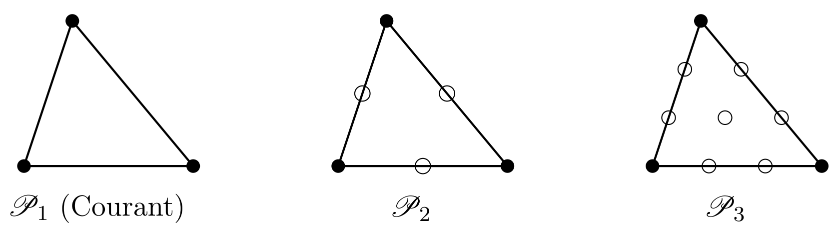


Figure 6: Principal lattice nodes for $\mathscr{P}_1$, $\mathscr{P}_2$, $\mathscr{P}_3$ on a triangle. Filled circles: vertices (shared by all orders). Open circles: edge and interior Lagrange nodes.

The global $\mathscr{P}_k$ space and its H^1_0-conforming subspace are

$$\begin{aligned} \mathbb{V}^k_{\mathscr{T}} &= \left\{ V \in C^0(\bar{\Omega}) : V|_T \in \mathscr{P}_k(T)\ \forall T \in \mathscr{T} \right\}, \\ \mathscr{V}^k_{\mathscr{T}} &= \mathbb{V}^k_{\mathscr{T}} \cap H^1_0(\Omega). \end{aligned}$$

Global continuity of functions in $\mathscr{V}^k_{\mathscr{T}}$ follows because a $\mathscr{P}_k$ polynomial on a shared edge is determined by the $k+1$ edge lattice nodes, which are shared between the two adjacent triangles. For $k = 1$ this recovers the $\mathscr{P}_1$ space of the previous subsection.

2.5 Interpolation

Lagrange interpolation. For $v \in C^0_0(\overline{\Omega})$, we define the *nodal (Lagrange) interpolant* $I_{\mathscr{T}} v \in \mathbb{V}_{\mathscr{T}}$ by matching v at all nodes:

$$I_{\mathscr{T}} v := \sum_{z \in \mathscr{N}_h^0} v(z) \phi_z, \tag{14}$$

where $\mathscr{N}_h^0$ is the set of interior nodes. The requirement $v \in C^0$ holds by the embeddings of Section 1 when $\operatorname{sob}(W_p^m) > 0$, e.g. $H^2 \hookrightarrow C^0$ for $d = 2, 3$:

$$\operatorname{sob}(H^2) = 2 - \frac{d}{2} > 0.$$

The cornerstone local estimate compares v with a good *polynomial* approximation on a reference element and then scales back. The polynomial we use is the Verfürth projection, which we introduce first; its key approximation property is a *generalized Poincaré inequality* that will do the analytic work in the proof.

The Verfürth projection. On a bounded, connected Lipschitz domain B (below, the reference element $\hat{T}$), the *Verfürth projection* [21] $P_{k,B}v \in \mathscr{P}_k(B)$ of a function $v \in W_p^{k+1}(B)$ is defined by the moment conditions

$$\int_B D^\alpha\big(v - P_{k,B}v\big)\,\mathrm{d}x = 0 \qquad \text{for all } |\alpha| \le k. \tag{15}$$

Construction and uniqueness. Expanding $q = P_{k,B}v$ about the barycenter x_0 of B,

$$q(x) = \sum_{|\alpha|\le k} c_\alpha \frac{(x - x_0)^\alpha}{\alpha!},$$

its coefficients $(c_\alpha)_{|\alpha|\le k}$ are determined by the equally many conditions (15), which are *triangular* in the order $|\alpha|$. In fact, since

$$\int_B D^\beta q\,\mathrm{d}x = c_\beta\,|B| \;+ \sum_{\alpha>\beta,\ |\alpha|\le k} c_\alpha \int_B \frac{(x-x_0)^{\alpha-\beta}}{(\alpha-\beta)!}\,\mathrm{d}x,$$

the top-order coefficients ($|\beta| = k$, empty sum) are determined first by $c_\beta = \fint_B D^\beta v\,\mathrm{d}x$, and descending in $|\beta|$ fixes the rest uniquely. Computing

$$D^\beta q(x) = \sum_{\alpha\ge\beta,\ |\alpha|\le k} c_\alpha \frac{(x-x_0)^{\alpha-\beta}}{(\alpha-\beta)!}$$

exhibits the commuting-projection property $D^\beta P_{k,B}v = P_{k-|\beta|,B}(D^\beta v)$; in particular $D^{k+1}q = 0$ since $q \in \mathscr{P}_k$. Propose as exercise.

The reason $P_{k,B}$ is the right comparison polynomial is its approximation property, which is nothing but the following higher-order Poincaré inequality: for $k = 0$ this is exactly the zero-mean Poincaré inequality of Lemma 1.18; the general case follows by iterating Lemma 1.18.

Lemma 2.4 (generalized Poincaré inequality)**.** *Let B be a bounded, connected Lipschitz domain, $1 \le p \le \infty$ and $k + 1 > d/p$. Then, there exists a constant $C = C(B, k, p)$ such that*

$$|v - P_{k,B}v|_{W_p^m(B)} \;\le\; C\,|v|_{W_p^{k+1}(B)} \qquad \textit{for all } 0 \le m \le k.$$

Proof. Write $q = P_{k,B}v$, so $D^{k+1}q = 0$ and, by (15), $\int_B D^\alpha(v-q)\,\mathrm{d}x = 0$ for all $|\alpha| \le k$. Fix a multi-index α with $|\alpha| = k$. Since $D^\alpha(v-q)$ has zero mean, Lemma 1.18 gives

$$\|D^\alpha(v-q)\|_{L^p(B)} \le C\,|D^\alpha(v-q)|_{W_p^1(B)} \le C\,\|D^{k+1}(v-q)\|_{L^p(B)} = C\,|v|_{W_p^{k+1}(B)},$$

the last equality using $D^{k+1}q = 0$. Descending to $|\alpha| = k-1$, the zero-mean condition again applies and $|D^\alpha(v-q)|_{W_p^1(B)} \le \|D^k(v-q)\|_{L^p(B)}$, which was just bounded; iterating down to $|\alpha| = 0$ yields the assertion. $\square$

Theorem 2.5 (local estimate for Lagrange interpolation)**.** *For a shape-regular element T and $0 \le m \le k+1$ with $k+1 > d/p$,*

$$|v - I_{\mathscr{T}} v|_{W_p^m(T)} \le C\, h_T^{k+1-m}\, |v|_{W_p^{k+1}(T)}\,. \tag{16}$$

In particular, for $\mathscr{P}_1$ and $p = 2$, we have

$$\|v - I_{\mathscr{T}} v\|_{L^2(T)} + h_T\, \|\nabla(v - I_{\mathscr{T}} v)\|_{L^2(T)} \le C\, h_T^2\, |v|_{H^2(T)}\,.$$

Proof. The argument has two steps.

Step 1: Invariance. We work on the reference element $\hat{T}$, with $\hat{I}$ being the Lagrange interpolation operator in $\hat{T}$ and $q = P_{k,\hat{T}} v$. Because $\hat{I}$ *reproduces* $\mathscr{P}_k$ exactly (i.e. $\hat{I} q = q$ for $q \in \mathscr{P}_k$),

$$v - \hat{I} v = (v - q) - \hat{I}(v - q) \quad \forall\, q \in \mathscr{P}_k.$$

For the interpolated term, boundedness of $\hat{I} : C^0(\hat{T}) \to \mathscr{P}_k(\hat{T})$, the embedding $W_p^{k+1}(\hat{T}) \hookrightarrow C^0(\hat{T})$, and norm equivalence on the finite-dimensional space $\mathscr{P}_k(\hat{T})$ give

$$|\hat{I}(v-q)|_{W_p^m(\hat{T})} \lesssim \|\hat{I}(v-q)\|_{C^0(\hat{T})} \lesssim \|v - q\|_{C^0(\hat{T})} \lesssim \|v - q\|_{W_p^{k+1}(\hat{T})}.$$

By Lemma 2.4, both $|v - q|_{W_p^m(\hat{T})}$ and the full norm $\|v - q\|_{W_p^{k+1}(\hat{T})}$ are controlled by $|v|_{W_p^{k+1}(\hat{T})}$ (for the top order use $D^{k+1} q = 0$). Hence

$$|v - \hat{I} v|_{W_p^m(\hat{T})} \le |v - q|_{W_p^m(\hat{T})} + |\hat{I}(v-q)|_{W_p^m(\hat{T})} \le C\, |v|_{W_p^{k+1}(\hat{T})}. \tag{17}$$

Step 2: Scaling. Let $F_T : \hat{T} \to T$ be the affine bijection, $F_T(\hat{x}) = B\hat{x} + b$, with $\|B\| \approx h_T$, $\|B^{-1}\| \approx h_T^{-1}$, and $|\det B| \approx |T| \approx h_T^d$. For any function w on T set $\hat{w} = w \circ F_T$ on $\hat{T}$. By the chain rule $|D^j \hat{w}(\hat{x})| \approx h_T^j |D^j w(F_T(\hat{x}))|$, so a change of variables ($\mathrm{d}x = |\det B|\, \mathrm{d}\hat{x}$) gives

$$\|D^j \hat{w}\|_{L^p(\hat{T})} \approx h_T^j\, h_T^{-d/p}\, \|D^j w\|_{L^p(T)},$$

i.e. $\|D^j w\|_{L^p(T)} \approx h_T^{d/p - j} \|D^j \hat{w}\|_{L^p(\hat{T})}$. This is exactly the statement that $\mathrm{sob}(W_p^j) = j - d/p$ governs the scaling. Applying this with $w = v - I_T v$ (and noting $\widehat{v - I_T v} = \hat{v} - \hat{I}\hat{v}$) at orders m and $k+1$:

$$\begin{aligned} |v - I_T v|_{W_p^m(T)} &\lesssim h_T^{d/p - m}\, |\hat{v} - \hat{I}\hat{v}|_{W_p^m(\hat{T})} \le C\, h_T^{d/p - m}\, |\hat{v}|_{W_p^{k+1}(\hat{T})} \qquad \text{(by (17))} \\ &\lesssim C\, h_T^{d/p - m} \cdot h_T^{-(d/p - (k+1))}\, |v|_{W_p^{k+1}(T)}. \end{aligned}$$

Collecting exponents $(d/p - m) - (d/p - (k+1)) = k + 1 - m$ gives the asserted estimate. □

Quasi-interpolation. In contrast to (14), for functions without point values, such as functions in $H^1(\Omega)$ for $d \ge 2$, we need to define nodal values by averaging. This leads to *quasi-interpolation*, which we briefly discuss now for polynomial degree $k = 1$. For each triangle $T \in \mathscr{T}$, generated by vertices $\{z_i\}_{i=1}^3$, the *dual functions* $\{\lambda_i^*\}_{i=1}^3 \subset \mathscr{P}_1(T)$ to the barycentric coordinates $\{\lambda_i\}_{i=1}^3$ are given by

$$\lambda_i^* = \frac{9}{|T|}\lambda_i - \frac{3}{|T|}\sum_{j \ne i} \lambda_j \qquad \forall\, 1 \le i \le 3,$$

and satisfy the bi-orthogonality relation $\int_T \lambda_i^* \lambda_j = \delta_{ij}$; the numerators 3 and 9 correspond to $1+d$ and $(1+d)^2$ for dimension $d=2$. The *Courant dual basis* ϕ_z^* of the space of piecewise linear discontinuous functions over $\mathscr{T}$ is given by

$$\phi_z^* = \frac{1}{\nu_z} \sum_{T \ni z} (\lambda_z^T)^* \chi_T \qquad \forall\, z \in \mathscr{N},$$

where $\nu_z \in \mathbb{N}$ is the valence of z (number of elements of $\mathscr{T}$ containing z) and χ_T is the characteristic function of T. These functions have the same support ω_z as the nodal basis ϕ_z and satisfy the global bi-orthogonality relation

$$\int_\Omega \phi_z^* \phi_y = \delta_{zy} \qquad \forall\, z, y \in \mathscr{N}.$$

We now introduce the operator $I_{\mathscr{T}} : L^1(\Omega) \to \mathbb{V}^1_{\mathscr{T}}$ due to Scott and Zhang [8, 18], from now on called quasi-interpolation operator:

$$I_{\mathscr{T}} v := \sum_{z \in \mathscr{N}} \langle v, \phi_z^* \rangle \phi_z. \tag{18}$$

By construction, this operator is invariant over $\mathbb{V}^1_{\mathscr{T}}$, namely

$$I_{\mathscr{T}} v = v \quad \forall\, v \in \mathbb{V}^1_{\mathscr{T}}. \tag{19}$$

In addition, since the averaging process giving rise to the values of $I_{\mathscr{T}} v$ for each element $T \in \mathscr{T}$ takes place in the neighborhood ω_T (union of starts containing T), we also deduce the local invariance

$$I_{\mathscr{T}} v|_T = v \quad \forall v \in \mathscr{P}_1(\omega_T) \tag{20}$$

as well as the local stability estimate for any $1 \le q \le \infty$

$$\|I_{\mathscr{T}} v\|_{L^q(T)} \lesssim \|v\|_{L^q(\omega_T)}. \tag{21}$$

Theorem 2.6 (local estimate for quasi-interpolation)**.** *If $0 < s \le 2$ (integer) is a regularity index and $1 \le p \le \infty$ is an integrability index, and $v \in W_p^s(\omega_T)$ for $T \in \mathscr{T}$, then*

$$|v - I_{\mathscr{T}} v|_{W_q^t(T)} \lesssim h_T^{\operatorname{sob}(W_p^s) - \operatorname{sob}(W_q^t)} |v|_{W_p^s(\omega_T)}, \tag{22}$$

provided $0 \le t \le s$, $1 \le q \le \infty$ are such that $\operatorname{sob}(W_p^s) > \operatorname{sob}(W_q^t)$.

Proof. In view of (20), we may write

$$v - I_{\mathscr{T}} v|_T = (v - w) - I_{\mathscr{T}}(v - w)|_T.$$

This combined with (21) are the key ingredients in the proof of Theorem 2.5. Essentially the same two-step proof applies now and gives (22). □

Before finishing this section it is worth realizing that, even though (22) is not fully local as (16), it still yields global estimates because of the finite overlapping properties of the sets ω_T for all $T \in \mathscr{T}$.

2.6 A priori error estimates

We now apply the interpolation theory developed in Section 2.5 to quantity the Galerkin error.

Theorem 2.7 (a priori energy estimate)**.** *Let the bilinear form $\mathscr{B}$ be continuous and coercive in $H_0^1(\Omega)$. If $u \in H^{k+1}(\Omega)$ and $\mathbb{V}_{\mathscr{T}}$ uses $\mathscr{P}_k$ Lagrange elements on a shape-regular mesh $\mathscr{T}$, with $k \geq 1$, then*

$$|u - U|_{H_0^1(\Omega)} \leq C\, h^k\ |u|_{H^{k+1}(\Omega)}\,. \tag{23}$$

Proof. Since $H^{k+1}(\Omega) \hookrightarrow C^0(\Omega)$, we may invoke Theorem 2.5 (local estimate for Lagrange interpolation) for $d = 2, 3$. Otherwise, we resort to Theorem 2.6 (local estimate for quasi-interpolation) in case u is not continuous. In either case, we sum (16) or (22) over $\mathscr{T}$ to deduce

$$|u - I_{\mathscr{T}} u|_{H^1(\Omega)} \leq C h^k |u|_{H^{k+1}(\Omega)},$$

where $h = \max_{T \in \mathscr{T}} h_T$. We finally apply the quasi-best approximation estimate (12) (Céa's lemma) to derive the desired estimate (23) $\square$

Remark 2.8 (curse of low regularity). Estimate (23) requires $u \in H^{k+1}(\Omega)$. On a polygon with a reentrant corner for $d = 2$ the solution behaves like $u \sim r^\gamma$ with $\frac{1}{2} < \gamma < 1$, so $u \in H^{1+s}(\Omega)$ only for $0 < s < \gamma$. Since $H^{1+s}(\Omega) \hookrightarrow C^0(\Omega)$ for $d = 2$, (16) yields

$$|u - U|_{H_0^1(\Omega)} \leq C\, h^s\ |u|_{H^{s+1}(\Omega)}\,.$$

for quasi-uniform meshes $\mathscr{T}$. Equivalently, in terms of degrees of freedom $N = \#\mathscr{T}$ this gives the suboptimal rate $h^s \approx N^{-s/2}$. Recovering the optimal rate $N^{-k/2}$ is exactly the motivation for *adaptivity*. This is discussed from Section 4 onward.

We conclude this section with L^2-error estimate. We might expect from Section 2.5 that the rate gains one extra order in L^2. This is precisely what the following duality argument due to Aubin-Nitsche delivers provided the underlying dual problem is H^2-regular.

Theorem 2.9 (a priori L^2-error estimate)**.** *Let the dual problem associated with the bilinear form $\mathscr{B}$ be H^2-regular, namely*

$$\phi \in H_0^1(\Omega): \qquad \mathscr{B}[v, \phi] = (\ell, v) \quad \forall\, v \in H_0^1(\Omega) \tag{24}$$

satisfies

$$\|\phi\|_{H^2(\Omega)} \leq C \|\ell\|_{L^2(\Omega)}$$

for a constant $C = C(\Omega)$. Then

$$\|u - U\|_{L^2(\Omega)} \leq C\, h^{k+1}\, |u|_{H^{k+1}(\Omega)}\,. \tag{25}$$

Proof. Let $\phi \in H_0^1(\Omega) \cap H^2(\Omega)$ be the solution (24) with $\ell = u - U$. Then

$$\|\phi\|_{H^2(\Omega)} \leq C \|u - U\|_{L^2(\Omega)}.$$

Exploiting the definition (24) and Galerkin orthogonality yields

$$\|u - U\|_{L^2(\Omega)}^2 = (\ell, u - U) = \mathscr{B}[u - U, \phi] = \mathscr{B}[u - U, \phi - I_{\mathscr{T}}\phi] \leq M |u - U|_{H^1(\Omega)} |\phi - I_{\mathscr{T}}\phi|_{H^1(\Omega)}$$

The inequality (25) finally follows from combining Theorem 2.7 with either Theorem 2.5 or 2.6. $\square$

Remark 2.10 (lack of H^2-regularity). The estimate (25) hinges on H^2-regularity of the dual problem (24). For a polygonal domain with a reentrant corner, this is an unrealistic requirement. Instead, in view of Remark 2.8, we could expect that $\phi \in H^{s+1}(\Omega)$ with $0 < s < \gamma$ and

$$\|\phi\|_{H^{s+1}(\Omega)} \leq C\|u - U\|_{L^2(\Omega)}.$$

The same proof of Theorem 2.9 gives

$$\|u - U\|_{L^2(\Omega)} \leq Ch^{2s}|u|_{H^{s+1}(\Omega)}.$$

On the one hand, this estimate seems sub-optimal, because for the interpolation error (Theorem 2.6) we have order h^{1+s}. On the other hand, though, it turns out that this estimate is optimal as the following argument by Schatz demostrates. If $\mathscr{B}$ is the Dirichlet form associated with the Laplacian Δ, then

$$\|\nabla(u - U)\|_{L^2(\Omega)}^2 = \mathscr{B}[u - U, u - U] = \mathscr{B}[u - U, u] = -\int_\Omega (u - U)f \leq \|u - U\|_{L^2(\Omega)}\|f\|_{L^2(\Omega)}.$$

3 Computational Implementation on a Fixed Mesh

We now turn the Galerkin method of Section 2 into a working program for $\mathscr{P}_1$ elements, following a minimal Matlab/Octave code. The mathematics — weak formulation (Section 1), hat functions and the Galerkin problem (Section 2) — is not repeated; the emphasis is on *how* each object is actually computed. Five tasks organize the code:

- Representing the mesh.
- Assembling the element matrices and load vector.
- Imposing boundary conditions.
- Solving the linear system.
- Verifying the result against the a priori theory of Section 2.

3.1 The model problem

The code solves a mild generalization of the Poisson problem of Section 1,

$$-\operatorname{div}(a\nabla u) + c\,u = f \quad \text{in } \Omega, \qquad u = g_D \quad \text{on } \Gamma_D, \qquad a\,\partial_n u = g_N \quad \text{on } \Gamma_N, \tag{26}$$

with $\partial\Omega = \Gamma_D \cup \Gamma_N$, diffusion coefficient (scalar) $a > 0$ and reaction $c \geq 0$. Both coefficients are allowed to be *piecewise constant* on the mesh — possibly discontinuous from one element to the next, as in the Kellogg benchmark below — and the code (both the fixed-mesh driver `fem.m` of this Section and the adaptive driver `assemble_and_solve.m` of Section 6) accepts a and c as ordinary Matlab function handles, $a(x)$ and $c(x)$, rather than hardcoded constants. Testing with functions that vanish on Γ_D and integrating by parts, the Neumann datum enters the right-hand side, and the weak form reads: find $u \in H^1(\Omega)$ with $u = g_D$ on Γ_D such that

$$\int_\Omega a\,\nabla u\cdot\nabla v + \int_\Omega c\,u\,v = \int_\Omega f\,v + \int_{\Gamma_N} g_N\,v \qquad \forall\, v \in H^1_{0,\Gamma_D}(\Omega), \tag{27}$$

$H^1_{0,\Gamma_D} = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}$. Well-posedness is the Lax–Milgram theory of Section 1; the $\mathscr{P}_1$ Galerkin problem replaces H^1 by the space $\mathbb{V}^1_{\mathscr{T}}$ of Section 2.

3.2 From the weak form to a linear system

Section 2 built the nodal basis $\{\phi_z\}_{z\in\mathscr{N}}$ of $\mathbb{V}^1_{\mathscr{T}}$: every $V \in \mathbb{V}^1_{\mathscr{T}}$ is $V = \sum_{z\in\mathscr{N}} V(z)\,\phi_z$, so its $N_p = \#\mathscr{N}$ nodal values are exactly its degrees of freedom. Write the Galerkin solution as

$$U = \sum_{j=1}^{N_p} U_j\,\phi_{z_j}, \qquad U_j := U(z_j),$$

and test the weak form (27) against $v = \phi_{z_i}$, one node at a time. For a node $z_i \notin \Gamma_D$ this gives one linear equation in the unknowns $U_1, \ldots, U_{N_p}$,

$$\sum_{j=1}^{N_p} U_j\,\mathscr{B}[\phi_{z_j}, \phi_{z_i}] \;=\; \int_\Omega f\,\phi_{z_i} + \int_{\Gamma_N} g_N\,\phi_{z_i}, \qquad \mathscr{B}[u,v] := \int_\Omega a\nabla u\cdot\nabla v + \int_\Omega c\,u\,v,$$

the bilinear form of (27). For a Dirichlet node $z_i \in \Gamma_D$ there is no equation to test, only the imposed value $U_i = g_D(z_i)$. Collecting all N_p conditions gives a square linear system

$$\mathbf{A}\,\mathbf{U} = \mathbf{b}, \qquad \mathbf{U} = (U_1, \ldots, U_{N_p})^T, \tag{28}$$

with entries

$$A_{ij} = \begin{cases} \mathscr{B}\big[\phi_{z_j}, \phi_{z_i}\big], & z_i \notin \Gamma_D, \\ \delta_{ij}, & z_i \in \Gamma_D, \end{cases} \qquad b_i = \begin{cases} \displaystyle\int_\Omega f\,\phi_{z_i} + \int_{\Gamma_N} g_N\,\phi_{z_i}, & z_i \notin \Gamma_D, \\ g_D(z_i), & z_i \in \Gamma_D. \end{cases}$$

Solving (28) for $\mathbf{U}$ and reading off $U = \sum_j U_j \phi_{z_j}$ *is* the Galerkin solution: everything from here on is about how to build $\mathbf{A}$ and $\mathbf{b}$ efficiently, and how to solve (28).

3.3 Decoupling over elements

Building $\mathbf{A}$ and $\mathbf{b}$ entry by entry from the formulas above looks expensive: A_{ij} is an integral over all of Ω, and there are up to N_p^2 of them. Two structural facts make it tractable instead.

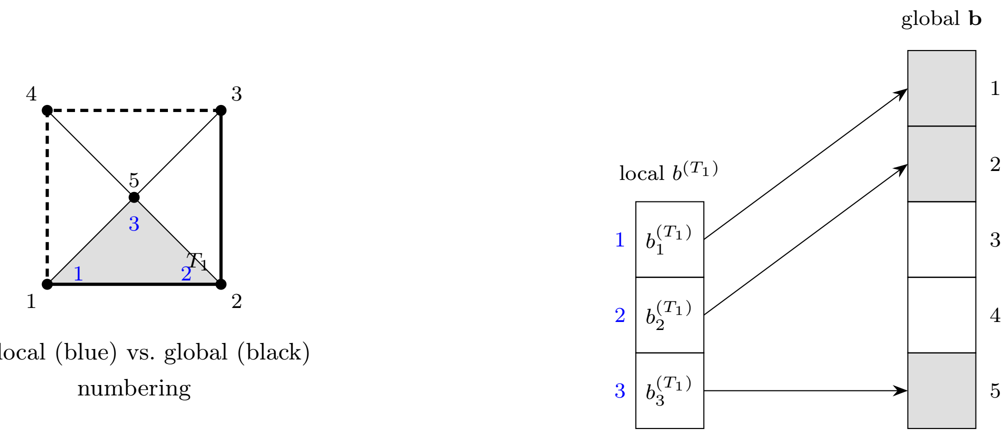
First, every integral over Ω splits additively over the triangulation, $\int_\Omega(\cdot) = \sum_{T\in\mathscr{T}} \int_T(\cdot)$, because the triangles tile Ω and overlap only on sets of measure zero. Second, and more importantly, the hat function ϕ_z vanishes outside its star ω_z (Section 2), so on a *single* triangle T only the three hat functions of its own vertices are nonzero; every other ϕ_z contributes exactly zero to $\int_T$. Consequently:

- $A_{ij} \neq 0$ only if z_i, z_j are vertices of a common triangle — a handful of neighbours per node on a typical mesh — so $\mathbf{A}$ is sparse;
- each triangle T, visited once, contributes to only $3 \times 3 = 9$ entries of $\mathbf{A}$ and 3 entries of $\mathbf{b}$: those indexed by its own three vertices.

This is what we mean by *decoupling over elements*: instead of assembling $\mathbf{A}$ and $\mathbf{b}$ column by column from global integrals, we *loop over triangles*; for each T we compute a small, dense 3×3 *element matrix* and 3-vector *element load*, using only the local data of T; and we *scatter-add* these few numbers into the (much larger) global $\mathbf{A}$ and $\mathbf{b}$, at the rows/columns given by T's three vertices. No triangle ever needs to know about any other triangle, and the global system is touched only through this scatter-add.

Figure 7 illustrates the idea on the element T_1 of the running example of Figure 8 below, whose vertices are the global nodes $1, 2, 5$. Panel (a) shows the *local* numbering $1, 2, 3$ used inside the element computation, alongside the *global* numbering of the mesh: local vertex k of T_1 *is* global vertex $(1, 2, 5)_k$. Panel (b) shows the element load $b^{(T_1)} \in \mathbb{R}^3$, computed from T_1 alone, scattered into positions $1, 2, 5$ of the global $\mathbf{b} \in \mathbb{R}^{N_p}$ (here $N_p = 5$); the untouched entries $3, 4$ receive contributions only from the other triangles that share those vertices. The element *matrix* $A^{(T_1)} \in \mathbb{R}^{3\times 3}$ scatters the same way, into the 3×3 block of $\mathbf{A}$ indexed by $(1, 2, 5) \times (1, 2, 5)$.

The rest of this Section makes this precise: we first fix the local-to-global correspondence — the ordered list of vertices of each T — as data read from a file, then show how to compute the element matrix and load from the equation's data, and finally close the loop with the scatter-add itself and the imposition of boundary conditions.



(a) local (blue) vs. global (black) numbering

(b) scatter-add of the element load into the global vector

Figure 7: Decoupling over elements, on the element T_1 (vertices $1, 2, 5$) of the running example of Figure 8. (a) Local numbering $1, 2, 3$ (blue) assigned inside the element computation versus the global mesh numbering (black); here local $1, 2, 3$ are global $1, 2, 5$. (b) The 3-entry element load $b^{(T_1)}$, computed from T_1 in isolation, is scattered into the three matching positions of the global load vector $\mathbf{b}$ — the other entries are left for other elements to fill in. The element matrix $A^{(T_1)}$ scatters the same way into a 3×3 block of $\mathbf{A}$.

3.4 Mesh representation

The local-to-global correspondence used in Figure 7 — for each triangle, the ordered list of its three vertices — is exactly the piece of data the code needs in order to know what to scatter where. It, together with the vertex coordinates and the boundary partition into Γ_D, Γ_N, is stored in four plain-text files:

- `vertex_coordinates.txt` — one line per vertex, its (x, y) coordinates;
- `elem_vertices.txt` — one line per triangle, the three global vertex indices (counterclockwise);
- `dirichlet.txt` — the indices of the vertices lying on Γ_D;
- `neumann.txt` — one line per boundary segment on Γ_N, its two vertex indices.

Figure 8 shows the convention on a four-triangle mesh of the unit square. In the program the first two files are read into arrays `coord` ($N_p \times 2$) and `elem` ($N_t \times 3$); the global $\mathscr{P}_1$ basis $\{\phi_i\}_{i=1}^{N_p}$ is indexed by vertices, one hat function per vertex.

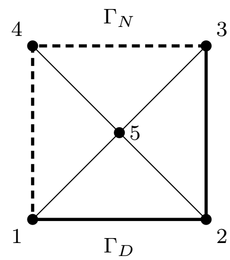
3.5 Reference element and the affine map

Every element computation is pulled back to the reference triangle $\widehat{T} = \{\hat{x} : \hat{x}_1, \hat{x}_2 \geq 0,\ \hat{x}_1 + \hat{x}_2 \leq 1\}$, with vertices $\hat{v}^1 = (0, 0)$, $\hat{v}^2 = (1, 0)$, $\hat{v}^3 = (0, 1)$ and reference hat functions

$$\hat{\phi}_1 = 1 - \hat{x}_1 - \hat{x}_2, \quad \hat{\phi}_2 = \hat{x}_1, \quad \hat{\phi}_3 = \hat{x}_2, \qquad \hat{\nabla}\hat{\phi}_1 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \ \hat{\nabla}\hat{\phi}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \ \hat{\nabla}\hat{\phi}_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A triangle T with vertices v^1, v^2, v^3 is the image of $\widehat{T}$ under the affine map (Figure 9)

$$F_T(\hat{x}) = v^1 + B\,\hat{x}, \qquad B = [\,v^2 - v^1 \quad v^3 - v^1\,] = DF_T, \qquad |T| = \tfrac{1}{2}\,|\det B|\,.$$




```
vertex_coordinates.txt
0.0   0.0
1.0   0.0
1.0   1.0
0.0   1.0
0.5   0.5
```

```
elem_vertices.txt
 1   2   5
 2   3   5
 3   4   5
 4   1   5
```

```
dirichlet.txt
1
2
3
```

```
neumann.txt
 3   4
 4   1
```

Figure 8: Mesh representation on a four-triangle mesh of the unit square (vertex 5 at the centre). Solid edges lie on Γ_D (Dirichlet vertices $1, 2, 3$), dashed edges on Γ_N (segments 3–4 and 4–1). The four text files store the geometry and the boundary partition.

The local hat function $\hat{\phi}_i \circ F_T^{-1}$ coincides with the global ϕ_i on T, and by the chain rule the physical and reference gradients are related by

$$\hat{\nabla}\hat{\phi}_i = B^T \nabla\phi_i, \qquad \text{equivalently} \qquad \nabla\phi_i = B^{-T}\,\hat{\nabla}\hat{\phi}_i. \tag{29}$$

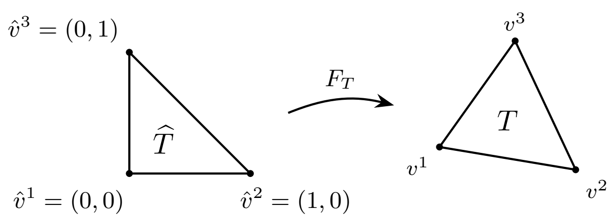


Figure 9: The affine map $F_T(\hat{x}) = v^1 + B\hat{x}$ sends the reference triangle $\widehat{T}$ onto the element T; its Jacobian $B = [\, v^2 - v^1 \;\; v^3 - v^1 \,]$ carries all the geometry, with $|T| = \frac{1}{2}\,|\det B|$ and $\nabla\phi_i = B^{-T}\hat{\nabla}\hat{\phi}_i$.

3.6 Element matrices and load vector

By §3.3, on a single triangle T only the three hat functions ϕ_i associated with its own vertices are nonzero, so all T can contribute to $\mathbf{A}, \mathbf{b}$ is a 3×3 *element (stiffness–mass) matrix*

$$A^{(T)} = \big(A_{ij}^{(T)}\big)_{i,j=1}^{3}, \qquad A_{ij}^{(T)} = \int_T a\,\nabla\phi_i \cdot \nabla\phi_j + \int_T c\,\phi_i\phi_j,$$

together with a 3-vector $b_i^{(T)} = \int_T f\,\phi_i$ (plus a boundary term on Neumann edges), scattered into $\mathbf{A}, \mathbf{b}$ through the vertex numbering (§3.7). It suffices, then, to know how to build these 3×3 local matrices. For convenience, we collect the reference gradients as the columns of $\hat{G} = [\,\hat{\nabla}\hat{\phi}_1 \;\; \hat{\nabla}\hat{\phi}_2 \;\; \hat{\nabla}\hat{\phi}_3\,] \in \mathbb{R}^{2\times 3}$.

Stiffness. Since B — hence each $\nabla\phi_i$ — is constant on T, (29) gives

$$A_{ij}^{(T)} = \int_T a\,\nabla\phi_i \cdot \nabla\phi_j = a\;|T|\;(\hat{\nabla}\hat{\phi}_i)^T B^{-1}B^{-T}(\hat{\nabla}\hat{\phi}_j), \qquad \text{i.e.} \qquad A^{(T)} = a\;|T|\;\hat{G}^T\big(B^{-1}B^{-T}\big)\hat{G}.$$

Mass (reaction). With the exact reference mass matrix,

$$M_{ij}^{(T)} = \int_T c\,\phi_i\phi_j = c\;|T|\;\widehat{M}, \qquad \widehat{M} = \frac{1}{12}\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

Remark 3.1 (evaluating a piecewise-constant coefficient). Both formulas pull a and c out of the element integral, which is only legitimate if a, c are literally constant on T. Given $a(x)$ and $c(x)$ as function handles, the code enforces this by evaluating each *once per element*, at its barycenter $x_T = \frac{1}{3}(v^1 + v^2 + v^3)$: $a|_T := a(x_T)$, $c|_T := c(x_T)$. This is *exact* whenever the data are genuinely piecewise constant *and* every discontinuity of a (or c) lies on the mesh skeleton — which is how the Kellogg benchmark's coefficient jump (Table 3) is handled: since N is even, the jump lines $x_1 = 0$ and $x_2 = 0$ coincide with mesh edges, so no element straddles the interface and barycenter evaluation never sees the wrong value. For a smoothly varying a, evaluating at the barycenter is simply the midpoint-rule approximation used for f above, and inherits the same accuracy.

Load. The right-hand side is evaluated with the three-point *midpoint rule*, exact for quadratics,

$$\int_T g \,\mathrm{d}x \;\approx\; \frac{|T|}{3}\big(g(m_{12}) + g(m_{23}) + g(m_{31})\big), \qquad m_{ab} = \tfrac{1}{2}(v^a + v^b),$$

where m_{ab} are the edge midpoints. As each ϕ_i equals $\frac{1}{2}$ at the midpoints of its two adjacent edges and 0 at the opposite one,

$$b^{(T)} = \frac{|T|}{3}\begin{pmatrix} \frac{1}{2}\, f(m_{12}) + \frac{1}{2}\, f(m_{31}) \\ \frac{1}{2}\, f(m_{12}) + \frac{1}{2}\, f(m_{23}) \\ \frac{1}{2}\, f(m_{23}) + \frac{1}{2}\, f(m_{31}) \end{pmatrix}.$$

Exercise 3.2 (convection: the non-symmetric case)**.** Extend the code to the full convection–diffusion–reaction problem

$$-\operatorname{div}(a\nabla u) + \boldsymbol{b}\cdot\nabla u + c\,u = f,$$

with a given velocity field $\boldsymbol{b}$ (piecewise constant on the mesh). Show that it adds to every element the *convection matrix*

$$C^{(T)}_{ij} = \int_T \phi_i\,(\boldsymbol{b}\cdot\nabla\phi_j)\,\mathrm{d}x = \frac{|T|}{3}\,\boldsymbol{b}\cdot\nabla\phi_j \qquad (i, j = 1, 2, 3),$$

using that $\nabla\phi_j = B^{-T}\hat{\nabla}\hat{\phi}_j$ is constant on T and $\int_T \phi_i \,\mathrm{d}x = |T|/3$; in particular the three *rows* of $C^{(T)}$ coincide. Add `Cloc = (area/3)*ones(3,1)*(bvec(:)'*Grad)` — with `Grad = B'\Ghat` the physical gradients as columns — to `Aloc` in the assembly loop. Because $C^{(T)}$ is *not* symmetric, the global matrix loses symmetry: the remark on Cholesky/conjugate gradients below no longer applies, and one uses a general sparse solver (the backslash still works; iteratively, GMRES or BiCGStab). Verify that for a convection-dominated problem ($|\boldsymbol{b}|\ h \gg a$) the Galerkin solution develops spurious oscillations, whose cure (e.g. SUPG stabilization) lies beyond the scope of these notes.

Remark 3.3 (quadratic elements). Moving from $\mathscr{P}_1$ to $\mathscr{P}_2$ changes remarkably little in the overall structure; only four ingredients need attention. *(i) Degrees of freedom.* A quadratic function on T is fixed by its values at the three vertices *and* the three edge midpoints, so each element now carries six local unknowns and the element matrices become 6×6. Globally $N_p = \#\{\text{vertices}\} + \#\{\text{edges}\}$, which forces one extra bookkeeping step: build a list of the mesh edges (each interior edge shared by two triangles) and assign each a global index, so that every triangle knows the six global numbers of its vertex and midpoint nodes. *(ii) Reference basis.* The three hat functions are replaced by the six quadratic Lagrange shape functions $\hat{\phi}_i(\hat{x}, \hat{y})$ on the reference triangle; the affine map and its Jacobian B are unchanged, and physical gradients are still recovered by $\nabla\phi_i = B^{-T}\hat{\nabla}\hat{\phi}_i$. *(iii) Quadrature.* This is the essential difference: the reference gradients are now *linear* in $(\hat{x}, \hat{y})$ rather than constant, so $\hat{G}$ must be evaluated at each quadrature point instead of once per element, and the integrands are of higher degree — $a\,\nabla\phi_i\cdot\nabla\phi_j$ is quadratic and $c\,\phi_i\phi_j$ is quartic when the data are constant.

The 3-point midpoint rule (exact only for quadratics) no longer integrates the mass and load terms exactly; one replaces it by a rule exact for degree 4, e.g. a symmetric 6- or 7-point Gauss rule on the triangle, and accumulates $A^{(T)}$, $M^{(T)}$, $b^{(T)}$ as weighted sums over its points. *(iv) Boundary and assembly.* The scatter-add loop is verbatim, only with the six-node local-to-global map. A Neumann edge now involves its two endpoints and its midpoint, and Simpson's rule integrates the resulting quadratic datum exactly; Dirichlet values are prescribed at the boundary vertices and at the boundary edge midpoints alike. The pay-off is the higher order: for smooth u one gains a full power of h, $|u - U|_{H^1} = O(h^2)$ and $\|u - U\|_{L^2} = O(h^3)$.

3.7 Global assembly

The global sparse matrix $\mathbf{A}$ ($N_p \times N_p$) and load vector $\mathbf{b}$ are formed by adding each element's contribution into the rows and columns indexed by its global vertices (*scatter-add*). This is the heart of `fem.m`:

```
grd_bas_fcts = [-1 -1; 1 0; 0 1]'; % reference gradients (columns)
A = sparse(n_vertices, n_vertices);
fh = zeros(n_vertices, 1);
for el = 1 : n_elem
    v_elem = elem_vertices(el, :); % global indices of the 3 vertices
    v1 = vertex_coordinates(v_elem(1),:)'; % vertex coordinates (columns)
    v2 = vertex_coordinates(v_elem(2),:)';
    v3 = vertex_coordinates(v_elem(3),:)';
    m12 = (v1+v2)/2; m23 = (v2+v3)/2; m31 = (v3+v1)/2;
    bary = (v1+v2+v3)/3; % barycenter of the element
    f12 = fc_f(m12); f23 = fc_f(m23); f31 = fc_f(m31);
    B = [ v2-v1 v3-v1 ]; % Jacobian [v2-v1 v3-v1]
    el_area = abs(det(B))*0.5; Binv = inv(B);
    el_mat = coeff_a(bary) * grd_bas_fcts' * (Binv*Binv') * grd_bas_fcts * el_area ...
           + coeff_c(bary) * el_area * [1/6 1/12 1/12; 1/12 1/6 1/12; 1/12 1/12 1/6];
    f_el = [ (f12+f31)*0.5; (f12+f23)*0.5; (f23+f31)*0.5 ] * (el_area/3);
    A(v_elem, v_elem) = A(v_elem, v_elem) + el_mat; % scatter-add
    fh(v_elem) = fh(v_elem) + f_el;
end
```

Here `coeff_a` and `coeff_c` are function handles (Remark 3.1); a constant-coefficient problem simply sets `coeff_a = @(x) 1.0;` etc. The matrix $\mathbf{A}$ is sparse, symmetric, positive (semi-)definite, with $O(N_p)$ nonzeros because each vertex couples only to its mesh neighbours.

3.8 Boundary conditions

Neumann. Each segment $s \subset \Gamma_N$ with endpoints z_a, z_b contributes $\int_s g_N \phi_i$ to its two endpoints. Using Simpson's rule (with $m = \frac{1}{2}(z_a + z_b)$), $\int_s g \approx \frac{|s|}{6}\big(g(z_a) + 4g(m) + g(z_b)\big)$, and ϕ_a, ϕ_b being linear along s,

```
n_neumann_segments = size(neumann, 1);
for i = 1:n_neumann_segments
    v_seg = neumann(i, :);
    v1 = vertex_coordinates(v_seg(1),:); v2 = vertex_coordinates(v_seg(2),:);
    m = (v1+v2)/2; segment_length = norm(v2-v1);
    g1 = fc_gN(v1); g2 = fc_gN(v2); gm = fc_gN(m);
    f_seg = [ g1+2*gm ; 2*gm+g2 ] * segment_length/6;
    fh(v_seg) = fh(v_seg) + f_seg;
```

```
end
```

Dirichlet. The values $U_i = g_D(z_i)$ at Dirichlet vertices are imposed by *row replacement*: overwrite row i of $\mathbf{A}$ by e_i^T and set $b_i = g_D(z_i)$.

```
for i = 1:length(dirichlet)
    diri = dirichlet(i);
    A(diri,:) = zeros(1, n_vertices);
    A(diri,diri) = 1;
    fh(diri) = fc_gD( vertex_coordinates(diri,:) );
end
uh = A \ fh; % sparse solve
```

Remark 3.4 (symmetry). Row replacement is the simplest imposition but breaks the symmetry of $\mathbf{A}$. A symmetric alternative eliminates the Dirichlet unknowns (restrict to the free nodes and move the known columns to the right-hand side, $\mathbf{b}_{\text{free}} -= \mathbf{A}_{\text{free},D}\,\mathbf{g}$), letting one use a sparse Cholesky factorization or conjugate gradients.

3.9 Verification: error and convergence

Correctness is checked with a *manufactured* solution: choose u, set $f = -\operatorname{div}(a\nabla u) + cu$ and $g_D = u|_{\Gamma_D}$, solve, and measure the error. The energy (H^1-seminorm) and L^2 errors are computed element by element with the same midpoint rule; on each T the discrete gradient is the constant vector $\nabla U|_T = B^{-T}\hat{G}\,U|_T$:

```
B = [ v2-v1 v3-v1 ]; area_el = abs(det(B))/2;
graduef = B' \ (grd_bas_fcts * uef(v_elem)); % constant grad of U on the element
dif12 = graduef - grad_u(m12); % (and dif23, dif31 at m23, m31)
integral_el = (dif12'*dif12 + dif23'*dif23 + dif31'*dif31)/3 * area_el;
integral = integral + integral_el; % H1 seminorm error (sqrt at the end)
```

By the a priori estimate of Section 2 (Theorem 2.7) a smooth u on a quasi-uniform mesh should give $|u-U|_{H^1} = O(h) = O(N_p^{-1/2})$ and, by the Aubin–Nitsche duality, $\|u-U\|_{L^2} = O(h^2) = O(N_p^{-1})$. Table 1 and Figure 10, for $u(x) = e^{-10|x|^2}$ on the unit square with $a = 1$, $c = 0$ and a sequence of uniform meshes, confirm exactly these rates.

N	N_p	$\lvert u-U\rvert_{H^1}$	rate	$\lVert u-U\rVert_{L^2}$	rate
4	25	3.93×10^{-1}	—	3.29×10^{-2}	—
8	81	2.19×10^{-1}	0.85	9.88×10^{-3}	1.73
16	289	1.12×10^{-1}	0.96	2.58×10^{-3}	1.94
32	1089	5.64×10^{-2}	0.99	6.51×10^{-4}	1.98
64	4225	2.83×10^{-2}	1.00	1.63×10^{-4}	2.00
128	16641	1.41×10^{-2}	1.00	4.09×10^{-5}	2.00

Table 1: Convergence for the smooth manufactured solution $u = e^{-10|x|^2}$ on uniform meshes of the unit square. The observed orders in $h = 1/N$ (1 for the energy error, 2 for L^2) match Theorem 2.7.

The effect of a reentrant corner. Running the *same* program on the L-shaped domain $\Omega = (-1,1)^2 \setminus \big([0,1]\times[-1,0]\big)$ (`gen_mesh_L_shape`), with the classical reentrant-corner solution

$$u(r,\theta) = r^{2/3}\sin\!\big(\tfrac{2}{3}\theta\big), \qquad \theta \in [0, \tfrac{3\pi}{2}],$$

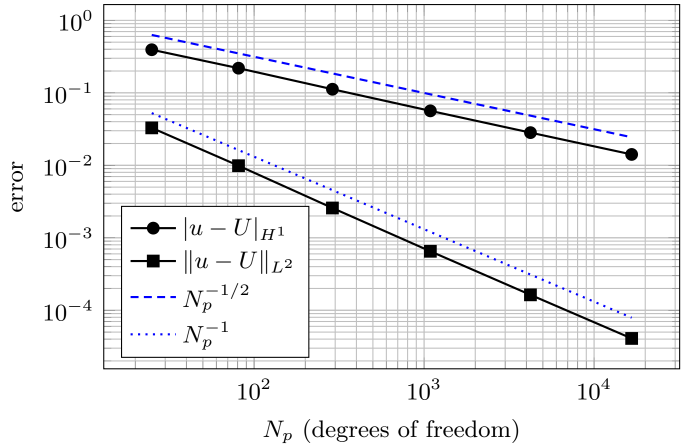


Figure 10: Energy and L^2 errors versus the number of degrees of freedom N_p, for the data of Table 1. The slopes match the reference lines $N_p^{-1/2}$ $(= O(h))$ and N_p^{-1} $(= O(h^2))$.

exposes the limitation of uniform meshes. This u is harmonic (so $f = 0$) and vanishes on the two edges meeting at the corner, but $u \in H^{1+2/3-\varepsilon}$ only, so Theorem 2.7 no longer applies at full order. Table 2 and Figure 11 report the measured errors: the energy error decays like $h^{2/3} = N_p^{-1/3}$ instead of $N_p^{-1/2}$. More strikingly, the L^2 error decays like $h^{4/3} = N_p^{-2/3}$ — *not* the $h^2 = N_p^{-1}$ one might expect. The reason is that the Aubin–Nitsche duality gains only one extra factor $h^{2/3}$ over the energy error, not h: the dual problem inherits the *same* corner singularity and is no more regular than the primal one, so the two singular factors multiply to $h^{2/3} \cdot h^{2/3} = h^{4/3}$.

N	N_p	$\|u-U\|_{H^1}$	rate	$\\|u-U\\|_{L^2}$	rate
4	65	1.82×10^{-1}	—	1.84×10^{-2}	—
8	225	1.17×10^{-1}	0.64	7.24×10^{-3}	1.34
16	833	7.46×10^{-2}	0.65	2.87×10^{-3}	1.34
32	3201	4.74×10^{-2}	0.65	1.14×10^{-3}	1.33
64	12545	3.01×10^{-2}	0.66	4.55×10^{-4}	1.33
128	49665	1.90×10^{-2}	0.66	1.81×10^{-4}	1.33

Table 2: L-shaped domain with the singular solution $u = r^{2/3}\sin(\frac{2}{3}\theta)$ on uniform meshes. The observed orders in $h = 1/N$ are $\frac{2}{3}$ (energy) and $\frac{4}{3}$ (L^2), i.e. $|u-U|_{H^1} = O(N_p^{-1/3})$ and $\|u-U\|_{L^2} = O(N_p^{-2/3})$ — both suboptimal.

The Kellogg benchmark: a coefficient jump. A second, complementary stress test keeps the domain perfectly smooth and instead makes the *coefficient* a singular. On the square $\Omega = (-1,1)^2$ (`gen_mesh_rectangle` with $N = M$), take the piecewise-constant, checkerboard diffusion coefficient

$$a(x) = \begin{cases} 161.4476387975881, & x_1x_2 > 0 \ \ \text{(quadrants I and III)}, \\ 1, & x_1x_2 < 0 \ \ \text{(quadrants II and IV)}, \end{cases} \qquad c = 0, \qquad f = 0,$$

jumping by more than two orders of magnitude across the two coordinate axes, and take $g_D = u|_{\partial\Omega}$ from the classical singular solution of Kellogg [12]

$$u(r,\phi) = r^{\gamma}\,\mu(\phi), \qquad \gamma = 0.1,$$

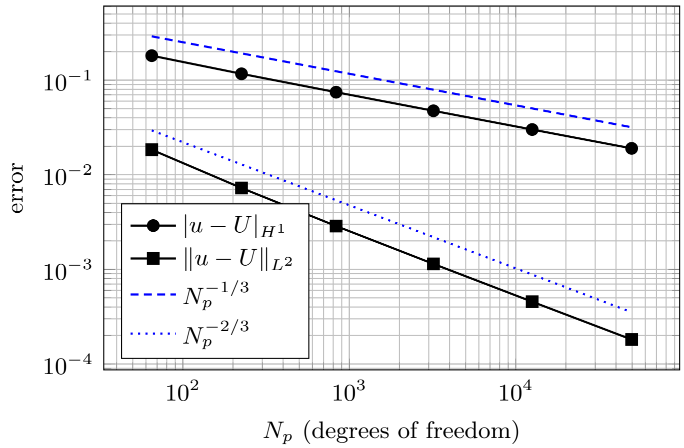


Figure 11: L-shaped domain: energy and L^2 errors versus N_p, for the data of Table 2. The slopes match the reference lines $N_p^{-1/3}$ and $N_p^{-2/3}$; both are below the smooth-case slopes of Figure 10.

where μ is a piecewise-trigonometric function, one formula per quadrant, chosen so that u and the conormal flux $a\,\partial_\nu u$ match across each axis (`kellogg_u_exact.m`, `kellogg_grad_u_exact.m`). Unlike the L-shaped example, Ω has *no* geometric singularity at all — the mesh is uniform triangles on a square — and u is singular purely because a is discontinuous: the transmission conditions across $x_1 = 0$ and $x_2 = 0$ force $u \in H^{1+\gamma-\varepsilon}(\Omega)$ only, with $\gamma = 0.1$ chosen deliberately far below the $\gamma = 2/3$ of the reentrant corner. Because $N = M$ is always even, every grid line of `gen_mesh_rectangle` passes through the origin, so the jump in a never crosses the interior of a triangle, and evaluating a once per element (at the barycenter, as `fem.m` does) is exact.

Table 3 and Figure 12 report the measured errors on a sequence of uniform meshes. The energy error is nearly *flat*: over this whole range it decays like only $N_p^{-0.014}$, far short of even the modest asymptotic rate $N_p^{-\gamma/2} = N_p^{-0.05}$ that Remark 2.8 would suggest — a signature of the very long pre-asymptotic regime that makes this benchmark so demanding in practice. The L^2 error fares somewhat better ($N_p^{-0.21}$ overall) but is likewise far from the optimal $N_p^{-1/2}$ line also plotted for reference. Uniform refinement is, for practical purposes, useless on this problem, which is exactly why the Kellogg benchmark is a standard test of whether an adaptive algorithm can recover a usable rate in the presence of a genuinely discontinuous coefficient A — the setting targeted by the a posteriori theory of Section 5 and the discontinuous-coefficient extension of Section 9.

N	N_p	$\lvert u-U\rvert_{H^1}$	rate	$\|u-U\|_{L^2}$	rate
4	25	1.42×10^{-1}	—	3.18×10^{-2}	—
8	81	1.39×10^{-1}	0.02	2.20×10^{-2}	0.53
16	289	1.38×10^{-1}	0.02	1.61×10^{-2}	0.45
32	1089	1.35×10^{-1}	0.02	1.23×10^{-2}	0.39
64	4225	1.33×10^{-1}	0.03	9.74×10^{-3}	0.34
128	16641	1.30×10^{-1}	0.03	7.88×10^{-3}	0.31

Table 3: Kellogg checkerboard-coefficient benchmark on $\Omega = (-1,1)^2$, $\gamma = 0.1$, on uniform meshes. Rates are the observed order in $h = 2/N$ between successive rows (i.e. rate $= \log_2(\mathrm{err}_{\mathrm{prev}}/\mathrm{err})$). Both errors decay far more slowly than in Table 1 or even Table 2, reflecting the severity of the coefficient jump.

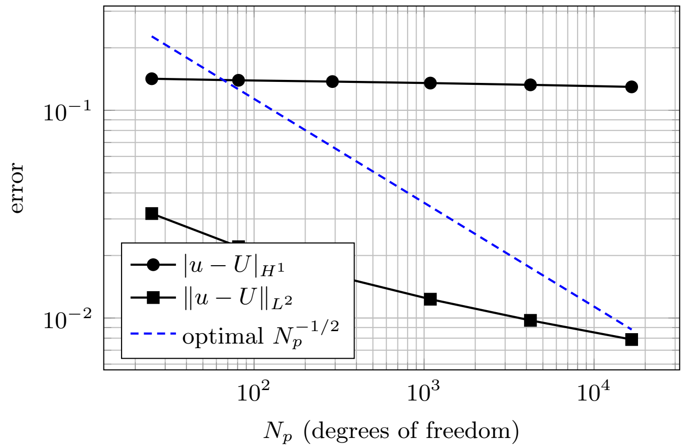


Figure 12: Kellogg benchmark: energy and L^2 errors versus N_p, for the data of Table 3, together with the reference slope $N_p^{-1/2}$ that a smooth problem (or an optimally adapted mesh) would achieve. Uniform refinement falls dramatically short of this line — far more so than in Figure 11 — because the singularity here comes from the coefficient, not the geometry.

Recovering the optimal rate $N_p^{-1/2}$ by *local* mesh refinement — without knowing the location or strength of the singularity (geometric, as in the L-shaped domain, or coefficient-driven, as in the Kellogg benchmark) in advance — is exactly the motivation for the adaptive methods of Sections 4–9.

3.10 Source code

The complete Matlab/Octave code and the verification scripts are available in the companion repository

https://github.com/morinpedro/fixed-mesh-fem

The driver `fem.m` is shown below in condensed form (mesh input and the Neumann assembly elided) to convey the overall structure; the complete files live in the repository.

```
% fem.m -- solve -div(a grad u) + c u = f with mixed boundary conditions
% (full version, with mesh input and Neumann assembly, in the repository)

% ---- problem data ----
coeff_a = @(x) 1.0; coeff_c = @(x) 0.0; % function handles: a(x), c(x)
fc_f = inline('20*exp(-10*norm(x)^2)*(2-20*norm(x)^2)', 'x');
fc_gD = inline('exp(-10*norm(x)^2)', 'x');

% ---- read mesh: elem_vertices, vertex_coordinates, dirichlet, neumann ----
% ... load the four text files ...
n_vertices = size(vertex_coordinates,1); n_elem = size(elem_vertices,1);
A = sparse(n_vertices, n_vertices); fh = zeros(n_vertices,1);
grd_bas_fcts = [ -1 -1 ; 1 0 ; 0 1 ]'; % reference gradients (columns)

% ---- assembly: loop over elements ----
for el = 1 : n_elem
    v_elem = elem_vertices(el, :);
```

```
    v1 = vertex_coordinates(v_elem(1),:)'; v2 = vertex_coordinates(v_elem(2),:)';
    v3 = vertex_coordinates(v_elem(3),:)';
    m12 = (v1+v2)/2; m23 = (v2+v3)/2; m31 = (v3+v1)/2;
    bary = (v1+v2+v3)/3; % barycenter, where a,c are evaluated
    f12 = fc_f(m12); f23 = fc_f(m23); f31 = fc_f(m31);
    B = [ v2-v1 v3-v1 ]; el_area = abs(det(B))*0.5; Binv = inv(B);
    f_el = [ (f12+f31)*0.5 ; (f12+f23)*0.5 ; (f23+f31)*0.5 ] * (el_area/3);
    el_mat = coeff_a(bary) * grd_bas_fcts' * (Binv*Binv') * grd_bas_fcts * el_area ...
          + coeff_c(bary) * el_area * [1/6 1/12 1/12; 1/12 1/6 1/12; 1/12 1/12 1/6];
    fh(v_elem) = fh(v_elem) + f_el;
    A(v_elem, v_elem) = A(v_elem, v_elem) + el_mat;
end

% ... Neumann segments added to fh with Simpson's rule ...

% ---- Dirichlet by row replacement, then solve ----
for i = 1 : length(dirichlet)
    diri = dirichlet(i);
    A(diri,:) = 0; A(diri,diri) = 1; fh(diri) = fc_gD(vertex_coordinates(diri,:));
end
uh = A \ fh;
trimesh(elem_vertices, vertex_coordinates(:,1), vertex_coordinates(:,2), uh);
```

The repository contains the following files.

- fem.m — the driver above: assembles and solves $-\operatorname{div}(a\nabla u) + c\,u = f$ with mixed boundary conditions.
- gen_mesh_rectangle.m — uniform triangulation of a rectangle (writes the four mesh files).
- gen_mesh_L_shape.m — uniform triangulation of the L-shaped domain.
- H1_err.m — H^1-seminorm error against a known solution (midpoint quadrature).
- L2_err.m — L^2 error against a known solution.
- fem_convergence.m — Convergence study on the unit square (Table 1).
- fem_convergence_L_shape.m — Convergence study on the L-shaped domain (Table 2).
- kellogg_u_exact.m, kellogg_grad_u_exact.m — the Kellogg singular solution u and ∇u [12], evaluated quadrant by quadrant.
- fem_convergence_Kellogg.m — Convergence study for the checkerboard-coefficient Kellogg benchmark (Table 3).
- fem_convergence.py, fem_convergence_L_shape.py — Python/NumPy ports of fem_convergence.m and fem_convergence_L_shape.m, for readers without Matlab/Octave.

4 Adaptive Approximation and Mesh Refinement

4.1 Why adapt? Error equidistribution

On a quasi-uniform mesh the convergence rate is limited by the *global* Sobolev regularity of u. Adaptivity instead tailors the mesh to u, via a nonlinear process. We consider first the one-dimensional case with $\Omega = (0,1)$ and assume the function to approximate v is absolutely continuous.

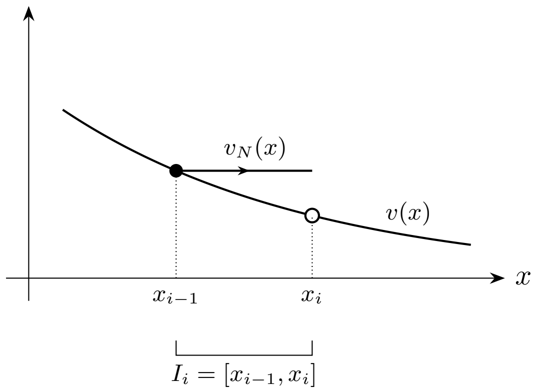
- *Uniform mesh: smooth data.* If $v \in W^1_\infty(\Omega)$ and $\mathscr{T}$ is uniform, $h_i = h = 1/N$, then integrating $|v'|$ over I_i gives $|v(x) - v_N(x)| \le h\, \|v'\|_{L^\infty(\Omega)}$, so
$$\|v - v_N\|_{L^\infty(\Omega)} \;\le\; \frac{1}{N}\, |v|_{W^1_\infty(\Omega)}\,. \tag{30}$$

- *Graded mesh: rough data.* Now let v be merely in $W^1_1(\Omega)$, possibly with an unbounded derivative. A uniform mesh no longer delivers (30), but the range of the monotone *error-accumulation function*
$$\phi(x) = \int_0^x \big|v'(s)\big|\, \mathrm{d}s, \qquad \phi(0) = 0, \quad \phi(1) = |v|_{W^1_1(\Omega)}\,,$$
can be cut into N equal pieces.

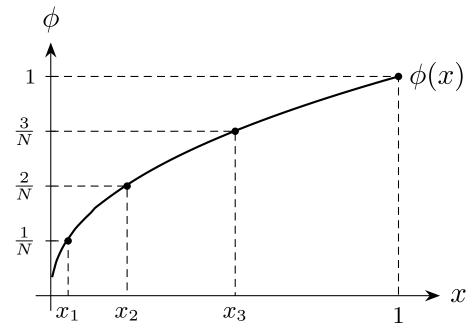
  As a side note, for $v(x) = x^\gamma$ with $0 < \gamma < 1$ we have $v'(x) = \gamma x^{\gamma-1} \notin L^\infty(0,1)$, so the uniform bound (30) is unavailable; yet $v \in W^1_1(\Omega)$, and $\phi(x) = \int_0^x \gamma s^{\gamma-1}\,\mathrm{d}s = x^\gamma = v(x)$. The error-accumulation function is the solution itself, and its singular behaviour at the origin is exactly what makes a graded mesh profitable.

  Choosing the nodes x_i so that
$$\phi(x_i) = \frac{i}{N}\, |v|_{W^1_1(\Omega)} \qquad (0 \le i \le N)$$
— that is, cutting the *range* of ϕ into N equal pieces (Figure 13(b)) — *equalizes* the local errors: for $x \in I_i$,
$$|v(x) - v_N(x)| \le \int_{x_{i-1}}^{x_i} \big|v'\big|\, \mathrm{d}s = \phi(x_i) - \phi(x_{i-1}) = \frac{1}{N}\, |v|_{W^1_1(\Omega)}\,,$$
whence
$$\|v - v_N\|_{L^\infty(\Omega)} \;\le\; \frac{1}{N}\, |v|_{W^1_1(\Omega)}\,. \tag{31}$$

The graded mesh thus recovers the *same* $O(1/N)$ rate for the rougher class W^1_1 as the uniform mesh gives for W^1_∞. The construction did exactly one thing — it placed the nodes so that every subinterval carries the same portion $1/N$ of the total variation, i.e. it *equidistributed* the local error — and this automatically clusters small elements where $|v'|$ is large (near the singularity of x^γ). We will see that this property hinges on the following relation between the regularity space W^1_1 and the approximation space L^∞:
$$\mathrm{sob}(W^1_1) = 1 - \frac{1}{1} \ge 0 - \frac{1}{\infty} = \mathrm{sob}(L^\infty).$$



(a) piecewise-constant v_N



(b) equidistributing the range of ϕ

Figure 13: The one-dimensional mechanism. (a) Piecewise-constant approximation v_N on an interval I_i; the error $v - v_N$ is controlled by $\int_{I_i} |v'|$. (b) Grading by equidistribution: cutting the range of the monotone error-accumulation function $\phi(x) = \int_0^x |v'|$ into N equal pieces places the nodes x_i (via $\phi(x_i) = i/N$) so that every subinterval carries the same error $1/N$; the nodes cluster where $|v'|$ is large.

Heuristics for the two-dimensional case. The starting point is the local interpolation estimate written in a form that carries a *vanishing power of h_T*. In 2D one has, for $p \geq 1$, that $W_p^2(\Omega) \hookrightarrow W_1^2(\Omega) \hookrightarrow C^0(\bar{\Omega})$, so the Lagrange interpolant $I_{\mathscr{T}} u$ is well defined and

$$\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(T)} \;\leq\; C\; |u|_{W_1^2(T)} \;=:\; e_{\mathscr{T}}(u, T) \qquad \text{for all } T \in \mathscr{T}; \tag{32}$$

The structure of (32) is what makes the discrete argument below valid.

A discrete minimization problem. Collect the surrogate local errors into the vector $\mathbf{e} = \big(e_{\mathscr{T}}(u,T)\big)_{T \in \mathscr{T}} \in \mathbb{R}^N$, $N = \#\mathscr{T}$, and consider the upper bound of the total (squared) energy error

$$E_{\mathscr{T}}(u)^2 := \sum_{T \in \mathscr{T}} e_{\mathscr{T}}(u,T)^2.$$

Given the structure of (32), we compute the ℓ^1-norm of $\mathbf{e}$ in terms of the W_1^2-norm of u as follows:

$$\sum_{T \in \mathscr{T}} e_{\mathscr{T}}(u,T) = C|u|_{W_1^2(\Omega)}. \tag{33}$$

It is worth realizing that the choice of ℓ^1-summability corresponds to the accumulation of $W_1^2(T)$ over $\mathscr{T}$, thereby reflecting equality of the Sobolev numbers $\mathrm{sob}(W_1^2) = \mathrm{sob}(H^1)$ in (32). This ties the surrogate local errors to the global regularity through the *constraint* (33). The key question now is: among all ways of distributing the fixed quantity $\sum_{T \in \mathscr{T}} e_{\mathscr{T}}(u,T)$ over N elements, which one minimizes $E_{\mathscr{T}}(u)^2$? To simplify this calculation, we idealize $e_{\mathscr{T}}(u,T)$ to range over all nonnegative reals, namely we ignore that shape regularity couples element neighbors and restricts such a range, and form the Lagrangian in terms of $\mathbf{e}$ and the Lagrange multiplier λ

$$\mathscr{L}[\mathbf{e}, \lambda] = \sum_{T \in \mathscr{T}} e_{\mathscr{T}}(u,T)^2 - \lambda\Big(\sum_{T \in \mathscr{T}} e_{\mathscr{T}}(u,T) - C\; |u|_{W_1^2(\Omega)} \Big).$$

The optimality condition $\partial\mathscr{L}/\partial e_{\mathscr{T}}(u,T) = 2\, e_{\mathscr{T}}(u,T) - \lambda = 0$ forces

$$e_{\mathscr{T}}(u,T) = \frac{\lambda}{2} \qquad \forall T \in \mathscr{T},$$

that is, the local error is the *same on every element*. We record the conclusion:

$$\textbf{A graded mesh is quasi-optimal if the local error is equidistributed.} \tag{34}$$

The resulting rate. Equidistribution yields $e_{\mathscr{T}}(u,T) = \frac{\lambda}{2}$, whence

$$E_{\mathscr{T}}(u)^2 = \frac{\lambda^2}{4}\, N, \qquad C\; |u|_{W_1^2(\Omega)} = \frac{\lambda}{2}\, N,$$

and eliminating λ gives the optimal decay rate

$$\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(\Omega)} \;\le\; C\, |u|_{W_1^2(\Omega)}\, N^{-1/2} \;\le C|\Omega|^{\frac{1}{q}} |u|_{W_p^2(\Omega)}\, N^{-1/2}, \tag{35}$$

where $q = \frac{p}{p-1}$ for $p > 1$. This rate $N^{-1/2}$ is the same as that delivered by uniform refinement for $u \in H^2(\Omega)$, but now requiring only $u \in W_p^2(\Omega)$, a much larger space. This is a genuine instance of nonlinear approximation, in which the mesh $\mathscr{T}$ is designed for the particular u at hand.

Example 4.1 (W_p^2-regularity at a reentrant corner)**.** Let $u(r,\theta) \sim r^{\gamma}$ with $0 < \gamma < 1$, the typical corner singularity. Then $D^2 u \sim r^{\gamma-2} \notin L^2$, so $u \notin H^2(\Omega)$; but

$$\left\|D^2 u\right\|_{L^p(\Omega)} = c \int_0^1 r^{p(\gamma-2)} \cdot r\, dr = c\int_0^1 r^{p(\gamma-2)+1}\, dr < \infty \implies u \in W_p^2(\Omega),$$

provided $1 \le p < p_0 = \frac{2}{2-\gamma}$. Hence $|u|_{W_p^2(\Omega)} < \infty$, and by (35) a mesh that equidistributes the local energy error — necessarily *graded* toward the corner — achieves $\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(\Omega)} \lesssim |u|_{W_p^2(\Omega)}\, N^{-1/2}$, the optimal 2D rate despite $u \notin H^2(\Omega)$.

Remark 4.2 (discrete vs. continuous equidistribution). The classical route to the same conclusion is *continuous* and reads as follows: introduce a mesh-density $h(x)$, express the number of degrees of freedom $\#\mathscr{T} \approx \int_\Omega h^{-2}\, \mathrm{d}x$ and the total energy error $E_{\mathscr{T}}(u)^p = \int_\Omega h^{2(p-1)} \left|D^2 u\right|^p \mathrm{d}x$, $p > 1$, and minimize $\#\mathscr{T}$ subject to the constraint $E_{\mathscr{T}}(u) = \varepsilon$ via a Lagrangian. The resulting pointwise Euler–Lagrange equation

$$h(x)^{2p}\, \left|D^2 u(x)\right|^p = \text{const} \qquad \text{(Babuška–Rheinboldt for } d = 1;\ \text{Nochetto–Veeser for } d = 2)$$

expresses the same elementwise equidistribution because $|T| \approx h(x)^2$ for $x \in T$. The discrete formulation above is closer to what an adaptive code actually manipulates — element indicators on a partition — and it needs no mesh-density idealization and no positive power of h_T in (32).

Remark 4.3 (fractional regularity). We may wonder what dictates the exponent $\frac{1}{2}$ in (35), which is unrelated to the integrability $p \ge 1$. The denominator 2 is the dimension of Ω and the numerator 1 is the difference of differentiability indices between the regularity space W_p^2 and the approximation space H^1. Since we are dealing with piecewise linear polynomial approximation, namely $k = 1$, Theorem 2.5 shows that the regularity space cannot have differentiability index larger than 2. But approximation theory does not prevent fractional differentiability $1 < s < 2$, which we now explore formally. Inequality (32) is still valid

$$\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(T)} \le C|u|_{W_p^s(\omega_T)} = e_{\mathscr{T}}(u,T) \quad \forall T \in \mathscr{T},$$

where ω_T is a discrete neighborhood of T (i.e. union of stars that contain T), provided

$$\mathrm{sob}(W_p^s) - \mathrm{sob}(H^1) = s - \frac{2}{p} = 0.$$

Since u may not be continuous in Ω, $I_{\mathscr{T}}$ stands now for a quasi-interpolation operator defined for integrable functions over Ω. Accumulating the quantities $e_{\mathscr{T}}(u,T)$ in ℓ^p yields the constraint

$$\sum_{T\in\mathscr{T}} e_{\mathscr{T}}(u,T)^p \approx |u|^p_{W_p^s(\Omega)},$$

and a similar calculation to that above leads to $e_{\mathscr{T}}(u,T) = \Lambda$ constant for all $T \in \mathscr{T}$. Therefore,

$$E_{\mathscr{T}}(u)^2 = \Lambda^2 N, \qquad \sum_{T\in\mathscr{T}} e_{\mathscr{T}}(u,T)^p = \Lambda^p N,$$

whence $\Lambda \le C|u|_{W_p^s(\Omega)} N^{-1/p}$. Altogether, this results in

$$\|\nabla(u - I_{\mathscr{T}}u)\|_{L^2(\Omega)} \le C|u|_{W_p^s(\Omega)} N^{-\frac{s-1}{2}},$$

We see that the exponent of N has a numerator $s-1$, which entails the difference between the regularity indices of the regularity space $W_p^s(\Omega)$ and the approximation space $H^1(\Omega)$, as asserted. Obviously, this inequality remains valid if $\mathrm{sob}(W_p^s) = s - \frac{2}{p} > 0$.

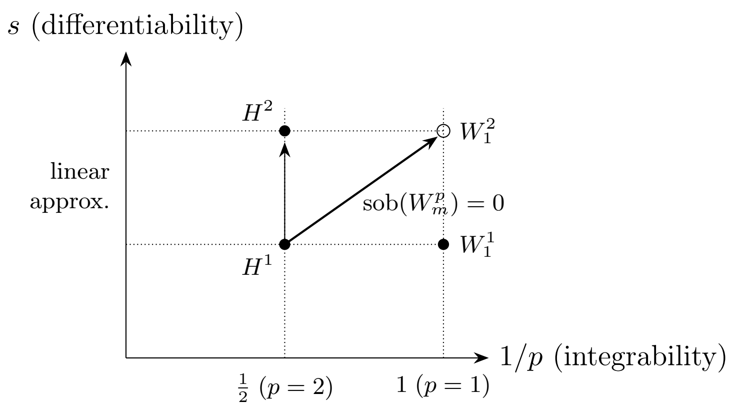


Figure 14: DeVore diagram. The Sobolev line $\mathrm{sob}(W_p^s) = s - 2/p = 0$ (i.e. $\mathrm{sob}(H^1)$) governs nonlinear approximation: graded meshes reach W_1^2 (open circle) at the same $N^{-1/2}$ rate that uniform meshes reach H^2 (filled circle).

Remark 4.4 (DeVore diagram). The above discussion can be summarized in the diagram of Figure 14. Given a differentiability index $1 < s \le 2$, in order to achieve the optimal convergence rate $N^{-\frac{s-1}{2}}$ the regularity space $W_p^s(\Omega)$ must lie on or above the oblique Sobolev line

$$s - \frac{2}{p} = \mathrm{sob}(W_p^s) = \mathrm{sob}(H^1) = 0$$

connecting $H^1(\Omega)$ for $p = 2$ with $W_1^2(\Omega)$ for $p = 1$. *Linear approximation theory* increases s from 1 to 2 keeping $p = 2$, thereby reaching the regularity space $H^2(\Omega)$. In contrast, *nonlinear approximation theory* trades off integrability from $p = 2$ to $p = 1$ with differentiability from $s = 1$ to 2, thereby reaching the nonlinear Sobolev scale $W_1^2(\Omega)$. The latter delivers the optimal convergence rate $N^{-\frac{s-1}{2}}$ with regularity $H^s(\Omega)$, a much more restrictive regularity requirement on u.

The practical question is how to *construct* optimal graded meshes automatically, by local refinement, while preserving conformity and shape regularity. The answer is bisection.

4.2 The Bisection Method

The refinement engine later in Section 6 rests on a single operation: *bisection* of one element. To this end, each element T carries a distinguished *refinement edge* $E(T)$, opposite to a vertex $v(T)$ (see Figure 15). Bisecting T joins $v(T)$ to the midpoint of $E(T)$, producing two children T_1, T_2 that share the newly created vertex $v(T_1) = v(T_2)$; each child is in turn assigned its own refinement edge $E(T_1), E(T_2)$ opposite to the new vertex. The *bisection method* therefore consists of two ingredients: a *labeling* of the initial mesh $\mathscr{T}_0$ that fixes each element's refinement edge, and a *rule* that propagates the labels to the children. For $d = 2$ this is *newest-vertex bisection*, described below; for $d > 2$ one needs in addition the notions of element *type* and *vertex order* [20, 17].

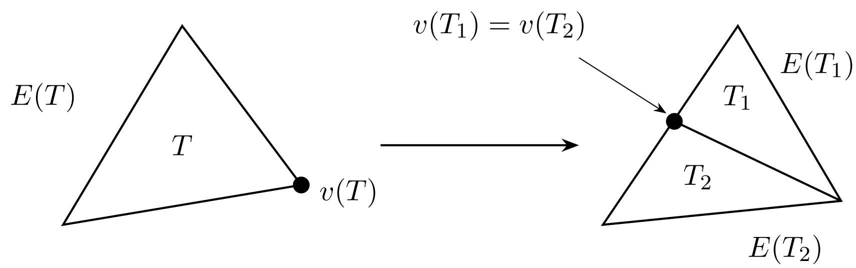


Figure 15: Triangle $T \in \mathscr{T}$ with vertex $v(T)$ and opposite refinement edge $E(T)$. The bisection rule for d = 2 consists of connecting $v(T)$ with the midpoint of $E(T)$, thereby giving rise to children T_1, T_2 with common vertex $v(T_1) = v(T_2)$, the newly created vertex, and opposite refinement edges $E(T_1), E(T_2)$.

Generation. The *generation* $g(T)$ of an element is the number of bisections needed to create it from its ancestor $T_0 \in \mathscr{T}_0$. Since each bisection halves the area, and $h_T = |T|^{1/2}$,

$$h_T = 2^{-g(T)/2}\, h_{T_0}, \qquad T_0 \in \mathscr{T}_0 \text{ the ancestor of } T. \tag{36}$$

We now assign the ordered edge label $(i, i+1, i+1)$ to each element $T \in \mathscr{T}$ of generation $g(T) = i$, the level-i edge being the refinement edge $E(T)$ (see Figure 16 (a)). Bisection splits the level-i edge and labels both new half-edges and the bisector $i+2$, leaving the other labels unchanged, so the two children T_1, T_2 carry labels $(i+1, i+2, i+2)$ (see Figure 16 (b)); the level-$(i+1)$ edges are the new refinement edges.

Shape regularity. The decisive geometric fact is that bisection never degrades element shape. This is because the previous labeling prevents consecutive bisections of the same edge, thereby inducing a rotation of edges.

Lemma 4.5 (shape regularity)**.** *Every partition generated from $\mathscr{T}_0$ by newest-vertex bisection satisfies a uniform minimal-angle condition: the ratio of element diameter to inscribed-ball diameter is bounded by a constant depending only on $\mathscr{T}_0$. Equivalently, the descendants of each $T_0 \in \mathscr{T}_0$ fall into finitely many similarity classes.*

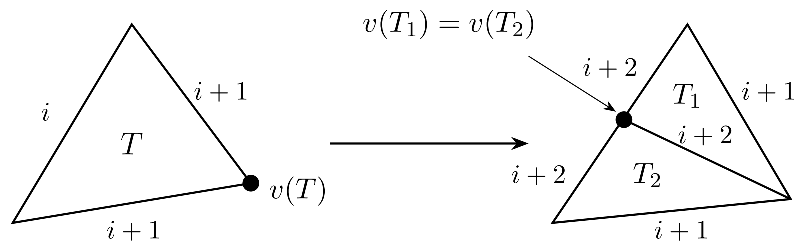


Figure 16: Newest-vertex bisection: triangle T of generation $g(T) = i$ carries labels $(i, i{+}1, i{+}1)$. T is split across its level-i refinement edge $E(T)$ (joining $v(T)$ to its midpoint) into children T_1, T_2 of generation $i+1$, sharing the new vertex and carrying labels $(i+1, i+2, i+2)$.

Write $\mathbb{T}$ for the set of all conforming refinements of $\mathscr{T}_0$ reachable by bisection, and $\mathscr{T}_* \geq \mathscr{T}$ when $\mathscr{T}_* \in \mathbb{T}$ refines $\mathscr{T} \in \mathbb{T}$. This partial order $(\mathbb{T}, \geq)$ is the arena of the whole theory: the loop Solve–Estimate–Mark–Refine of Section 5, realized as code in Section 6, produces an increasing sequence $\mathscr{T}_0 \leq \mathscr{T}_1 \leq \cdots \leq \mathscr{T}_k \leq \cdots$, and shape regularity keeps all the interpolation and a posteriori constants uniform along the way (see Figure 17).

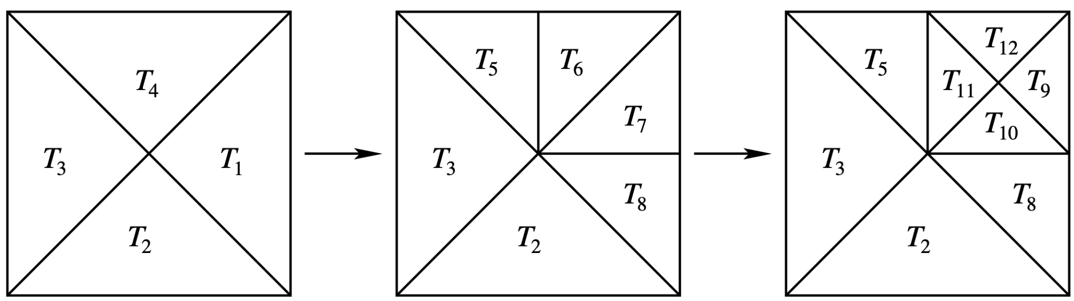


Figure 17: Sequence of bisection meshes $\{\mathscr{T}_k\}_{k=0}^{2}$ starting from the initial mesh $\mathscr{T}_0 = \{T_i\}_{i=1}^{4}$ with longest edges labeled for bisection. Mesh $\mathscr{T}_1$ is created from $\mathscr{T}_0$ upon bisecting T_1 and T_4, whereas mesh $\mathscr{T}_2$ arises from $\mathscr{T}_1$ upon refining T_6 and T_7. The bisection rule is described in Figure 16.

Compatible labeling. For the refinement to terminate and stay conforming, an element and its neighbor must *agree* on the refinement edge they share. In 2D this is secured by a *compatible initial labeling* of $\mathscr{T}_0$: label every edge 0 or 1 so that

$$\textit{every } T \in \mathscr{T}_0 \textit{ has exactly two edges labeled } 1 \textit{ and one labeled } 0, \tag{37}$$

the label-0 edge being its refinement edge (the generation-0 instance of the pattern $(i, i+1, i+1)$ above). When two elements share their refinement edge they form a *compatible patch* that can be bisected *together*, inserting one vertex and creating no hanging node. Such a labeling is not automatic, but it can always be arranged after refining $\mathscr{T}_0$: bisect every element of $\mathscr{T}_0$ twice and label the two newest edges 0 and the rest 1. For $d > 2$ the corresponding condition (on type and vertex order) is more delicate but still available [20, 17].

4.3 Complexity of Bisection

Given $\mathscr{T} \in \mathbb{T}$ and a marked set $\mathscr{M} \subset \mathscr{T}$, the module

$$\mathscr{T}_* = \textsc{Refine}\,(\mathscr{T}, \mathscr{M})$$

returns the smallest conforming refinement of $\mathscr{T}$ in which every $T \in \mathscr{M}$ has been bisected at least once. Bisecting a marked T across $E(T)$ generally leaves a *hanging node* on the neighbor across that edge; conformity is then restored by bisecting the neighbor as well, which may itself expose a new hanging node, and so on. The completion runs along a *chain* of elements linked by their refinement edges, each bisected in turn until a compatible patch is reached (the chain terminates at the boundary or at an already-compatible pair). Crucially, the elements of such a chain are bisected twice before the original edge is split, and the chain can be as long as the generation $g(T)$ of the marked element (see Figure 18).

Counting elements. Because one mark can trigger a completion chain, as depicted in Figure 18, the tempting local bound

$$\#\mathscr{T}_* - \#\mathscr{T} \;\leq\; \Lambda_0 \,\#\mathscr{M}$$

does *not* hold with a constant Λ_0 independent of the refinement level: the local cost may be as large as $g(T)$ for $T \in \mathscr{M}$, which grows without bound as refinement proceeds. The remedy is not to bound a single step but the *cumulative* cost over the whole history — the completion work then telescopes, and only the *total* number of marks survives. This extremely delicate property is due to Binev–Dahmen–DeVore for ($d = 2$) [1] and Stevenson for ($d \geq 3$) [20]; see the survey paper [17] for a full proof of Theorem 4.6.

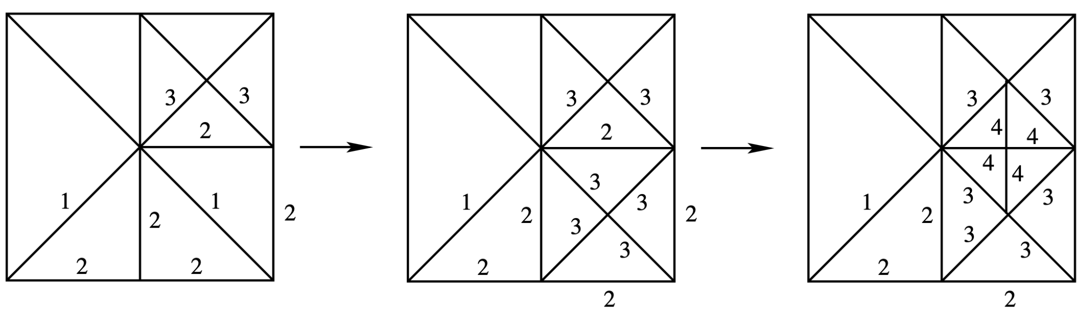


Figure 18: *Recursive refinement of element T_{10} of Figure 17.* Refining T_{10} is *not* local: to preserve conformity, completion must first refine an entire chain of elements linked by their refinement edges. Refining T_{10} forces the refinement chain $\{T_{10}, T_8, T_2\}$ in Figure 17, resolved in reverse — first T_2 (the bottom element, whose refinement edge lies on $\partial\Omega$, so it is a compatible patch on its own), then T_8 (lower right), and finally T_{10}. The three frames show the mesh after each bisection; every intermediate mesh is conforming, and the edge numbers are element generations. The number of forced bisections is the chain length $\#\{T_{10}, T_8, T_2\} = g(T_{10}) = 3$, and this length grows without bound as the mesh is refined — so no per-mark bound $\#\mathscr{T}_* - \#\mathscr{T} \leq \Lambda_0 \,\#\mathscr{M}$ can hold with Λ_0 independent of the refinement level.

Theorem 4.6 (complexity of bisection)**.** *Suppose $\mathscr{T}_0$ satisfies the compatible labeling (37) for $d = 2$ and the analogous type/vertex-order condition of Stevenson for $d > 2$, and let $\mathscr{M}_j \subset \mathscr{T}_j$ be the* *marked set at step j and $\mathscr{T}_{j+1} = \text{REFINE}(\mathscr{T}_j, \mathscr{M}_j)$. Then there exists a constant Λ_0, depending only on $\mathscr{T}_0$ and d, such that for all $k \geq 1$*

$$\#\mathscr{T}_k - \#\mathscr{T}_0 \;\leq\; \Lambda_0 \sum_{j=0}^{k-1} \#\mathscr{M}_j.$$

If each marked element is bisected a fixed number $b \geq 1$ of times (as with $b = 2$ in the code of Section 6), the bound persists with Λ_0 also depending on b.

This linear bound is the combinatorial backbone of the optimality theory: it converts a bound on the total number of *marks* into a bound on the number of *degrees of freedom*; see Section 9 for details. Without it the completion process — however local it looks element by element — could in principle overwhelm the savings of adaptivity; with it, the cost of keeping meshes conforming is provably a fixed multiple of the useful refined elements.

4.4 The Greedy Algorithm

Given $u \in H^1(\Omega) \cap W_p^2(\Omega)$, $p > 1$, we want to *construct* a near-optimal mesh by iterative bisection refinement. Define the local energy error

$$e_{\mathscr{T}}(T) = e_{\mathscr{T}}(u, T) := \|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(T)}.$$

Algorithm 4.7 (greedy adaptive approximation)**.** Given a tolerance $\delta > 0$ and an initial mesh $\mathscr{T}_0$:

> `function` GREEDY$(\mathscr{T}_0, \delta)$
> $\mathscr{T} := \mathscr{T}_0$
> **while** $\mathscr{M} := \{T \in \mathscr{T} : e_{\mathscr{T}}(T) > \delta\} \neq \emptyset$**:**
> $\mathscr{T} \leftarrow \text{REFINE}(\mathscr{T}, \mathscr{M})$
> **return** $\mathscr{T}$

Remark 4.8. This algorithm assumes we know u and can evaluate $e_{\mathscr{T}}(T)$. It serves as a theoretical benchmark for what adaptive methods can achieve; in practice $e_{\mathscr{T}}(T)$ is replaced by a computable estimator.

Theorem 4.9 (optimal greedy approximation)**.** *If $u \in H_0^1(\Omega) \cap W_p^2(\Omega)$ for some $p > 1$ and $d = 2$, the output $\mathscr{T}$ of* GREEDY$(\mathscr{T}_0, \delta)$ *satisfies*

1. $\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(\Omega)} \leq \delta\, (\#\mathscr{T})^{1/2}$;
2. $\#\mathscr{T} - \#\mathscr{T}_0 \lesssim \delta^{-1}\, |\Omega|^{1-1/p}\, |u|_{W_p^2(\Omega)}$.

In particular, with $N = \#\mathscr{T} \geq 2\#\mathscr{T}_0$,

$$\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(\Omega)} \lesssim |\Omega|^{1-1/p}\, |u|_{W_p^2(\Omega)}\, N^{-1/2}.$$

Proof. We proceed in five steps.

1. Local error bound. By Theorem 2.5 (local interpolation estimate), $e_{\mathscr{T}}(T) \leq C\, h_T^r\, |u|_{W_p^2(T)}$ with $r = \text{sob}(W_p^2) - \text{sob}(H^1) = 2(1 - 1/p) > 0$, so the algorithm terminates. At termination $e_{\mathscr{T}}(T) \leq \delta$ for all $T \in \mathscr{T}$, giving

$$\|\nabla(u - I_{\mathscr{T}} u)\|_{L^2(\Omega)}^2 = \sum_{T \in \mathscr{T}} e_{\mathscr{T}}(T)^2 \leq \delta^2\, \#\mathscr{T},$$

which is statement (1).

2. Counting elements of $\mathscr{T}$. Let $\mathscr{M}$ be the *total* set of all elements ever marked. By Theorem 4.6 (complexity of bisection) we have $\#\mathscr{T} - \#\mathscr{T}_0 \le \Lambda_0 \, \#\mathscr{M}$, so it suffices to bound $\#\mathscr{M}$. To do this, we partition $\mathscr{M}$ by element size: let P_j be the marked elements with area

$$2^{-(j+1)} \le |T| < 2^{-j}, \quad j \ge 0;$$

then $\#\mathscr{M} = \sum_{j\ge 0} \#P_j$. We now need to estimate $\#P_j$ for all $j \ge 0$.

3. Bounding $\#P_j$ (geometry). Elements in P_j are *disjoint* because if $T_1, T_2 \in P_j$ and their interiors intersect, then one of them is contained in the other, say $T_1 \subset T_2$, due to the bisection procedure; hence $|T_1| \le \frac{1}{2}|T_2|$, thereby contradicting the definition of P_j. Since each element in P_j has area $\ge 2^{-(j+1)}$, so

$$\#P_j \le |\Omega| \; 2^{j+1}. \tag{38}$$

4. Bounding $\#P_j$ (regularity). Every $T \in P_j$ was marked because $e_{\mathscr{T}}(T) > \delta$, and $h_T := |T|^{1/2} \le 2^{-j/2}$, so

$$\delta < e_{\mathscr{T}}(T) \lesssim h_T^r \, |u|_{W_p^2(T)} \lesssim 2^{-jr/2} \, |u|_{W_p^2(T)} \, .$$

Raising to the p-th power and summing over $T \in P_j$, we get $\delta^p \, \#P_j \lesssim 2^{-jrp/2} \, |u|^p_{W_p^2(\Omega)}$, whence

$$\#P_j \lesssim 2^{-jrp/2} \delta^{-p} \, |u|^p_{W_p^2(\Omega)} \, . \tag{39}$$

5. Crossover and summation. Bounds (38) and (39) are complementary; they cross at an index j_0 satisfying

$$2^{j_0} |\Omega| \simeq 2^{-\frac{j_0 rp}{2}} \delta^{-p} \, |u|^p_{W_p^2(\Omega)} \qquad \Longrightarrow \qquad 2^{j_0 (1 + \frac{rp}{2})} \simeq \delta^{-p} \, |u|^p_{W_p^2(\Omega)} \, |\Omega|^{-1}.$$

Since $1 + \frac{rp}{2} = p$, the crossover happens when

$$2^{j_0} \approx \delta^{-1} \, |u|_{W_p^2(\Omega)} \, |\Omega|^{-1/p} \, .$$

Using the geometric bounds for $j \le j_0$ and $j > j_0$, both geometric series relate to j_0 as follows:

$$\sum_{j \le j_0} 2^j \approx 2^{j_0}, \quad \sum_{j > j_0} (2^{-rp/2})^j \approx (2^{-rp/2})^{j_0} = 2^{-(p-1) j_0}.$$

This gives

$$2^{j_0} \approx \left(\delta^{-1} \, |u|_{W_p^2(\Omega)} \, |\Omega|^{-1/p} \right)^{p/(1+rp/2)},$$

whence

$$\#\mathscr{M} \lesssim |\Omega| \; 2^{j_0} \approx \delta^{-1} \, |u|_{W_p^2(\Omega)} \; |\Omega|^{1-1/p} \, ,$$

because $1 + \frac{rp}{2} = p$. This is statement (2) and concludes the proof. □

5 A posteriori error analysis

Adaptivity needs *computable* error information. We derive an estimator $\mathscr{E}_{\mathscr{T}}(U)$ depending only on the data (f, A, Ω) and the computed solution U, and show it is *two-sided equivalent* to the true energy error. The upper bound (*reliability*) prevents under-refinement; the lower bound (*efficiency*), valid up to a data *oscillation* term, prevents over-refinement and underpins optimality.

Model problem and residual

Consider

$$\begin{cases} -\operatorname{div}(A(x)\nabla u) = f & \text{in } \Omega \subset \mathbb{R}^d, \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

with weak solution $u \in \mathbb{V} = H_0^1(\Omega)$ and Galerkin solution $U \in \mathbb{V}$ (C^0 piecewise linears over a generic mesh $\mathscr{T}$ that has been obtained by a finite number of bisections applied to a given macro triangulation $\mathscr{T}_0$). The bilinear form $\mathscr{B}[u, v] = \int_\Omega A\nabla u \cdot \nabla v$ is coercive ($\mathscr{B}[v, v] \geq \alpha\, |v|_{H^1(\Omega)}^2$) and bounded ($|\mathscr{B}[u, v]| \leq M\, |u|_{H^1(\Omega)}\, |v|_{H^1(\Omega)}$) with respect to the $H^1(\Omega)$-seminorm $|v|_{H^1(\Omega)} = \|\nabla v\|_{L^2(\Omega)}$, which is itself a norm in $H_0^1(\Omega)$ due to Poincaré.

Goal. Estimate $|u - U|_{H^1(\Omega)}$ purely in terms of computable quantities depending on f, A, Ω, and the Galerkin solution U:

$$|u - U|_{H^1(\Omega)} \simeq \mathscr{E}_{\mathscr{T}}(U),$$

where $\mathscr{E}_{\mathscr{T}}(U)$ is the *a posteriori error estimator.*

We denote with $H^{-1}(\Omega)$ the dual space of $H_0^1(\Omega)$, with the dual norm

$$\|F\|_{H^{-1}(\Omega)} = \sup_{v \in \mathbb{V}\setminus\{0\}} \frac{\langle F, v\rangle}{|v|_{H^1(\Omega)}}, \qquad F \in H^{-1}(\Omega),$$

and we define the *residual* $R \in H^{-1}(\Omega)$ by

$$\langle R, v\rangle = \langle f, v\rangle - \mathscr{B}[U, v] = \mathscr{B}[u - U, v] \qquad \forall\, v \in \mathbb{V} = H_0^1(\Omega).$$

5.1 Error–residual relation

The energy error is equivalent to the dual norm of the residual:

$$\alpha\, |u - U|_{H^1(\Omega)} \leq \|R\|_{H^{-1}(\Omega)} \leq M\, |u - U|_{H^1(\Omega)}\,.$$

Both bounds follow directly. Observe first that, testing against $v = u - U$,

$$\alpha\, |u - U|_{H^1(\Omega)}^2 \leq \mathscr{B}[u - U, u - U] = \langle R, u - U\rangle \leq \|R\|_{H^{-1}}\, |u - U|_{H^1(\Omega)}\,,$$

so that after dividing by $|u - U|_{H^1(\Omega)}$ we get $\alpha\, |u - U|_{H^1(\Omega)} \leq \|R\|_{H^{-1}(\Omega)}$. For the *right inequality,* we resort to the boundedness of $\mathscr{B}[\cdot, \cdot]$ inner product:

$$|\langle R, v\rangle| = |\mathscr{B}[u - U, v]| \leq M\, |u - U|_{H^1(\Omega)}\, |v|_{H^1(\Omega)}\,,$$

so $\|R\|_{H^{-1}} = \sup_{\mathbb{V}} |\langle R, v\rangle| \,/\, |v|_{H^1(\Omega)} \leq M\, |u - U|_{H^1(\Omega)}$.

Thus $\|u - U\|_{H^1(\Omega)} \simeq |u - U|_{H^1(\Omega)} \simeq \|R\|_{H^{-1}(\Omega)}$, and it only remains to bound $\|R\|_{H^{-1}(\Omega)}$ by computable quantities.

5.2 Residual representation

Assume $A(x) \in C^{0,1}(\bar{\Omega}; \mathscr{T}_0)$ (Lipschitz) and $f \in L^2(\Omega)$. Integrating by parts element by element,

$$\begin{aligned}\langle R, v\rangle &= \sum_{T\in\mathscr{T}} \int_T fv - \int_T A(x)\nabla U \cdot \nabla v \\ &= \sum_{T\in\mathscr{T}} \int_T \big[f + \operatorname{div}(A\nabla U)\big]v - \int_{\partial T} A\nabla U \cdot \nu_T\, v.\end{aligned}$$

After cancellation over interior edges, the boundary contributions survive only as *flux jumps*. For two elements T, T' sharing an interior edge s with $\nu_{T'} = -\nu_T$ (see Figure 19), define the *jump residual*

$$j_s := -\big(A_T\nabla U_T \cdot \nu_T + A_{T'}\nabla U_{T'} \cdot \nu_{T'}\big) = [\![\, A\nabla U \cdot \boldsymbol{n}\,]\!]_s,$$

and the *interior (element) residual*

$$r_T := f + \operatorname{div}(A\nabla U)\big|_T.$$

Then

$$\langle R, v\rangle = \sum_{T\in\mathscr{T}} \int_T r_T\, v + \sum_{s\in\Gamma} \int_s j_s\, v,$$

where Γ is the mesh skeleton. For $\mathscr{P}_1$ with constant A, $\operatorname{div}(A\nabla U) = 0$ on each T, so $r_T = f|_T$.

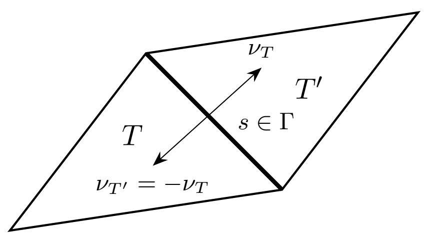


Figure 19: Two elements T, T' sharing interior edge s. Outward normals satisfy $\nu_{T'} = -\nu_T$; the flux jump $j_s = [\![\, A\nabla U \cdot \nu\,]\!]$ measures the discontinuity in the normal component of $A\nabla U$ across s.

Goal. We want to bound $\|R\|_{H^{-1}}$ without inverting any differential operator. The key is *Galerkin orthogonality*: since $\langle R, I_{\mathscr{T}}v\rangle = \mathscr{B}[u - U, I_{\mathscr{T}}v] = 0$, we may replace v by $v - I_{\mathscr{T}}v$ for *free*:

$$\langle R, v\rangle = \langle R, v - I_{\mathscr{T}}v\rangle \qquad \forall\, v \in \mathbb{V}.$$

Local interpolation estimates then localize the error to element patches.

5.3 Reliability upper bound

Theorem 5.1 (reliability)**.** *There is a constant C depending only on shape regularity such that*

$$|u - U|_{H_0^1(\Omega)} \le C\,\mathscr{E}_{\mathscr{T}}(U), \qquad \mathscr{E}_{\mathscr{T}}(U) = \Big(\sum_{T\in\mathscr{T}} \mathscr{E}^2_{\mathscr{T}}(U,T)\Big)^{1/2},$$

with the local element indicator $\mathscr{E}^2_{\mathscr{T}}(U,T) = h_T^2\,\|r_T\|^2_{L^2(T)} + h_T\,\|j\|^2_{L^2(\partial T)}$. *Due to the equivalence $|\cdot|_{H^1(\Omega)} \simeq |\!|\!|\cdot|\!|\!|_\Omega$ in $H^1_0(\Omega)$, there exists a constant C_1 which depends on shape regularity and the coefficient $A(x)$ of the equation, such that*

$$|\!|\!|u - U|\!|\!|_\Omega \le C_1\,\mathscr{E}_{\mathscr{T}}(U). \tag{40}$$

Proof 1 (quasi-interpolant / Clément–Scott–Zhang). By the error-residual equivalence it suffices to bound $\|R\|_{H^{-1}}$. By Galerkin orthogonality, $\langle R, v\rangle = \langle R, v - I_{\mathscr{T}} v\rangle$ for any $v \in \mathbb{V}$. Using the residual representation and Cauchy–Schwarz,

$$\begin{aligned}|\langle R, v\rangle| &= \Big|\sum_T \int_T r_T(v - I_{\mathscr{T}} v) + \sum_s \int_s j_s(v - I_{\mathscr{T}} v)\Big| \\ &\le \sum_T \|r_T\|_{L^2(T)} \|v - I_{\mathscr{T}} v\|_{L^2(T)} + \sum_s \|j_s\|_{L^2(s)} \|v - I_{\mathscr{T}} v\|_{L^2(s)} .\end{aligned}$$

The quasi-interpolant satisfies the local estimates

$$\|v - I_{\mathscr{T}} v\|_{L^2(T)} \lesssim h_T \|\nabla v\|_{L^2(\omega_T)} , \qquad \|v - I_{\mathscr{T}} v\|_{L^2(s)} \lesssim h_s^{1/2} \|\nabla v\|_{L^2(\omega_s)} ,$$

where, as before, ω_T is the union of stars containing T, and also ω_s is the union of stars containing s. Substituting and applying Cauchy–Schwarz again,

$$\begin{aligned}|\langle R, v\rangle| &\lesssim \underbrace{\Big(\sum_T h_T^2 \|r_T\|^2_{L^2(T)} + \sum_s h_s \|j_s\|^2_{L^2(s)}\Big)^{1/2}}_{=\mathscr{E}_{\mathscr{T}}(U)} \underbrace{\Big(\sum_T \|\nabla v\|^2_{L^2(\omega_T)}\Big)^{1/2}}_{\lesssim \|\nabla v\|_{L^2(\Omega)}} \\ &\lesssim \mathscr{E}_{\mathscr{T}}(U)\ \|\nabla v\|_{L^2(\Omega)} = \mathscr{E}_{\mathscr{T}}(U)\ |v|_{H^1(\Omega)} .\end{aligned}$$

where finite overlap of patches gives the last bound. Dividing by $|v|_{H^1(\Omega)}$ and taking the supremum yields the claim. □

Proof 2 (partition of unity). Let $\{\phi_z\}_{z\in\mathscr{N}}$ be the hat-function partition of unity, $\sum_z \phi_z = 1$. Since $\phi_z \in \mathbb{V}_{\mathscr{T}}$, Galerkin orthogonality gives $\langle R, \phi_z\rangle = 0$ for each interior node z, so

$$\langle R, v\rangle = \sum_z \langle R, v\phi_z\rangle = \sum_z \langle R, (v - \overline{v}_z)\phi_z\rangle , \qquad \overline{v}_z = \begin{cases} ⨍_{\omega_z} v & \text{if } z \notin \partial\Omega, \\ 0 & \text{if } z \in \partial\Omega, \end{cases} .$$

Expanding the residual on each star ω_z and applying the Poincaré inequalities

$$\|v - \overline{v}_z\|_{L^2(\omega_z)} \lesssim h_z \|\nabla v\|_{L^2(\omega_z)} , \qquad \|v - \overline{v}_z\|_{L^2(\partial\omega_z)} \lesssim h_z^{1/2} \|\nabla v\|_{L^2(\omega_z)} ,$$

and Cauchy–Schwarz over stars, we arrive at the same estimate as Proof 1. □

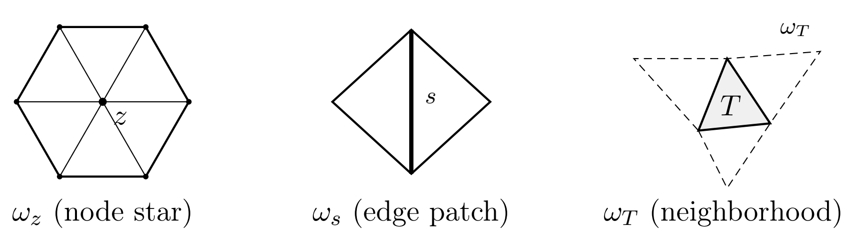


Figure 20: Local patches: node star ω_z (used in Proof 2), edge patch ω_s, and element neighborhood ω_T (used in Proof 1). Finite overlap of patches ensures the $\|\nabla v\|$ factors recombine globally.

Remark 5.2. $\mathscr{E}_{\mathscr{T}}(U)$ to zero forces the true error to zero. The element indicators $\mathscr{E}^2_{\mathscr{T}}(U, T)$ are exactly what MARK uses in Section 6 to decide which elements to refine.

5.4 Efficiency I. The simplest case of pw-constant data

Reliability prevents *under*-refinement. To prevent *over*-refinement — and to make adaptivity optimal — we need the converse: the estimator must not exceed the true error too much (*efficiency*). The price is a data *oscillation* term.

5.4.1 Element-bubble functions and the interior residual

Fix $T \in \mathscr{T}$ and let $r = r_T$ be the interior residual. Let $b_T = 27\lambda_1\lambda_2\lambda_3$ be the cubic *element bubble* ($b_T > 0$ in $\mathrm{int}(T)$, $b_T = 0$ on ∂T, $b_T(x_T) = 1$, with x_T the barycenter of T), so that $0 \le b_T \le 1$, $|\nabla b_T| \lesssim 1/h_T$ and

$$\int_T b_T \,\mathrm{d}x = \frac{27}{60}|T| \simeq |T| \simeq h_T^2 \quad \text{and} \quad \int_T b_T^2 \,\mathrm{d}x = \frac{27}{315}|T| \simeq |T| \simeq h_T^2.$$

For a *constant* residual r on T, we have

$$\begin{aligned}\|r\|_{L^2(T)}^2 &= \int_T r^2 = |T| r^2 \lesssim \int_T b_T r^2 = \langle r, r b_T\rangle \le \|r\|_{H^{-1}(T)} \, |r b_T|_{H^1(\Omega)} \\ &= \|r\|_{H^{-1}(T)} \, \|\nabla(r b_T)\|_{L^2(T)} = \|r\|_{H^{-1}(T)} \, \|r \nabla b_T\|_{L^2(T)} \lesssim \|r\|_{H^{-1}(T)} \; h_T^{-1} \, \|r\|_{L^2(T)}\,,\end{aligned}$$

where in the last inequality we have used that $|\nabla b_T| \lesssim h_T^{-1}$. Therefore, $h_T \,\|r\|_{L^2(T)} \lesssim \|r\|_{H^{-1}(T)}$. Since $\langle r_T, v\rangle = \langle R, v\rangle = \mathscr{B}[u - U, v]$ for $v \in H_0^1(T)$, we get $\|r_T\|_{H^{-1}(T)} \le \|\nabla(u-U)\|_{L^2(T)}$, so that

$$h_T \,\|r\|_{L^2(T)} \le \|\nabla(u-U)\|_{L^2(T)}\,,$$

which is the interior (computable) lower bound.

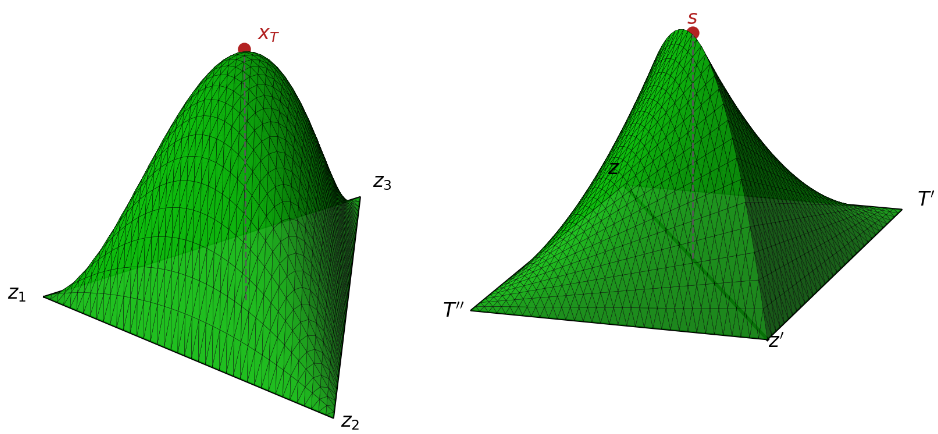


Figure 21: Left: the element bubble $b_T = 27\lambda_1\lambda_2\lambda_3$ on a triangle T with vertices z_1, z_2, z_3. Right: the edge bubble $b_s = 4\phi_z\phi_{z'}$ on the patch $\omega_s = T' \cup T''$ sharing the interelement side s with endpoints z, z'. Both vanish on the boundary of their support and attain their maximum at the marked point.

5.4.2 Edge-bubble functions and the jump residual

Let $b_s = 4\phi_z\phi_{z'}$ be the piecewise quadratic patch bubble function that is obtained as the product of the two hat functions corresponding to the vertices z, z' of the edge s, so that $0 \le b_s \le 1$, $|\nabla b_s| \lesssim 1/|s|$ and, for any $T \subset \omega_s$,

$$\int_T b_s \,\mathrm{d}x = \frac{1}{3}|T| \simeq h_T^2 \simeq |s|^2 \quad \text{and} \quad \int_T b_s^2 \,\mathrm{d}x = \frac{8}{45}|T| \simeq |T| \simeq h_T^2 \simeq |s|^2,$$

whence,

$$\int_{\omega_s} b_s \,\mathrm{d}x \simeq |s|^2 \simeq h_T^2 \quad \text{and} \quad \int_{\omega_s} b_s^2 \,\mathrm{d}x \simeq |\omega_s| \simeq |T| \simeq h_T^2,$$

and also,

$$\int_s b_s \,\mathrm{d}\sigma = \frac{2}{3}|s| \simeq |s| \simeq h_T \quad \text{and} \quad \int_s b_s^2 \,\mathrm{d}\sigma = \frac{8}{15}|s| \simeq |s| \simeq h_T.$$

If $j_s := j_{|s}$ is constant on the edge s,

$$\|j_s\|^2_{L^2(s)} = \int_s j_s^2 = |s| j_s^2 \lesssim \int_s j_s^2 b_s = \int_s [\![A\nabla(U-u)\cdot \boldsymbol{n}]\!]_s \, j_s b_s$$

If $\omega_s = T' \cup T''$ is the support of b_s,

$$\int_s [\![A\nabla(U-u)\cdot \boldsymbol{n}]\!]_s \, j_s b_s = \sum_{T\in\{T',T''\}} \int_{\partial T} A\nabla(U-u)\cdot \boldsymbol{n} j_s b_s \,\mathrm{d}\sigma = \sum_{T\in\{T',T''\}} \int_T \operatorname{div}[A\nabla(U-u) j_s b_s] \,\mathrm{d}x.$$

Now notice that

$$\begin{aligned}
\int_T \operatorname{div}[A\nabla(U-u) j_s b_s] &= \int_T \operatorname{div}[A\nabla(U-u)] j_s b_s + A\nabla(U-u)\cdot\nabla(j_s b_s) \\
&= \int_T r_T j_s b_s + A\nabla(U-u)\cdot\nabla(j_s b_s) \\
&\le \|r_T\|_{L^2(T)} \|j_s b_s\|_{L^2(T)} + \|\nabla(U-u)\|_{L^2(T)} \|j_s \nabla b_s\|_{L^2(T)} \\
&\le \|r_T\|_{L^2(T)} h_T |j_s| + \|\nabla(U-u)\|_{L^2(T)} |j_s|.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\int_s [\![A\nabla(U-u)\cdot \boldsymbol{n}]\!]_s \, j_s b_s &\lesssim \Big(h_s \|r\|_{L^2(\omega_s)} + \|\nabla(U-u)\|_{L^2(\omega_s)} \Big) |j_s| \\
&\lesssim \Big(h_s \|r\|_{L^2(\omega_s)} + \|\nabla(U-u)\|_{L^2(\omega_s)} \Big) h_s^{-1/2} \|j_s\|_{L^2(s)} ,
\end{aligned} \tag{41}$$

and consequently,

$$h_s^{1/2} \|j_s\|_{L^2(s)} \lesssim h_s \|r\|_{L^2(\omega_s)} + \|\nabla(U-u)\|_{L^2(\omega_s)} .$$

If we put together the estimates of the interior and jump residual we obtain the following lower bound:

$$h_T^2 \|r\|^2_{L^2(T)} + h_T \|j\|^2_{L^2(\partial T)} \lesssim |u-U|^2_{H^1(\omega_T)} , \tag{42}$$

where ω_T is the discrete neighborhood of T defined in Figure 20.

5.5 Oscillation and the full lower bound

The bubble argument is exact only when r is constant. For general data, one splits $r = \overline{r} + (r - \overline{r})$ with $\overline{r}_{|T} = \overline{r}_T := ⨍_T r$ and $j_s = \overline{j}_s + (j_s - \overline{j}_s)$, with $\overline{j}_{|s} := ⨍_s j_s$. The constant part obeys the equivalence above; the remainder contributes the local *oscillation terms*

$$\operatorname{osc}_{\mathscr{T}}(U,T)^2 = h_T^2 \|r - \overline{r}_T\|^2_{L^2(T)} + h_T \left\|j - \overline{j}\right\|^2_{L^2(\partial T)} , \tag{43}$$

which vanish for piecewise-constant data and are of higher order when f is smooth. The global oscillation term is given by

$$\operatorname{osc}^2_{\mathscr{T}}(U) := \sum_{T\in\mathscr{T}} \operatorname{osc}^2_{\mathscr{T}}(U,T). \tag{44}$$

Lemma 5.3 (local bound for interior residual)**.**

$$h_T \|r\|_{L^2(T)} \lesssim \|r\|_{H^{-1}(T)} + h_T \|r - \overline{r}_T\|_{L^2(T)} .$$

Proof. Write $\|h_T r\|_{L^2(T)} \le h_T \|\overline{r}_T\|_{L^2(T)} + h_T \|r - \overline{r}_T\|_{L^2(T)}$. For the first term, apply the constant-residual equivalence to $\overline{r}_T$: $h_T \|\overline{r}_T\|_{L^2(T)} \lesssim \|\overline{r}_T\|_{H^{-1}(T)} \lesssim \|r\|_{H^{-1}(T)} + \|r - \overline{r}_T\|_{H^{-1}(T)}$. It remains to bound $\|r - \overline{r}_T\|_{H^{-1}(T)}$. For any $v \in H^1_0(T)$,

$$|\langle r - \overline{r}_T, v\rangle| = |\langle r - \overline{r}_T, v - \overline{v}_T\rangle| \le \|r - \overline{r}_T\|_{L^2(T)} \|v - \overline{v}_T\|_{L^2(T)} \lesssim h_T \|r - \overline{r}_T\|_{L^2(T)} \|\nabla v\|_{L^2(T)} ,$$

by Poincaré, so $\|r - \overline{r}_T\|_{H^{-1}(T)} \lesssim h_T \|r - \overline{r}_T\|_{L^2(T)}$. Combining gives the lemma. □

The local bound for the jump residual is a bit more involved.

Lemma 5.4 (local bound for jump residual)**.**

$$h_s^{1/2} \|j\|_{L^2(s)} \lesssim |u - U|_{H^1(\omega_s)} + h_s^{1/2} \left\|j - \overline{j}_s\right\|_{L^2(s)} + \sum_{T \subset \omega_s} \|r\|_{H^{-1}(T)} .$$

Proof. We first observe that $h_s^{1/2} \|j\|_{L^2(s)} \le h_s^{1/2} \left\|\overline{j}_s\right\|_{L^2(s)} + h_s^{1/2} \left\|j - \overline{j}_s\right\|_{L^2(s)}$, and proceed to bound $h_s^{1/2} \left\|\overline{j}_s\right\|_{L^2(s)}$. We first do the following

$$\left\|\overline{j}_s\right\|^2_{L^2(s)} = \int_s \overline{j}_s^2 = |s| \overline{j}_s^2 \lesssim \int_s \overline{j}_s^2 b_s \lesssim \int_s j\, \overline{j}_s\, b_s + \int_s \overline{j}_s (j - \overline{j}_s) b_s.$$

We bound the last term on the right-hand side using Hölder as follows:

$$\int_s \overline{j}_s (j - \overline{j}_s) b_s \le \left\|\overline{j}_s\right\|_{L^2(s)} \left\|j - \overline{j}_s\right\|_{L^2(s)} .$$

The first term satisfies, due to (41),

$$\int_s j\, \overline{j}_s\, b_s = \int_s [\![A\nabla(U - u) \cdot \boldsymbol{n}]\!]_s\, \overline{j}_s b_s \lesssim \Big(h_s \|r\|_{L^2(\omega_s)} + \|\nabla(U - u)\|_{L^2(\omega_s)}\Big) h_s^{-1/2} \left\|\overline{j}_s\right\|_{L^2(s)} .$$

Combining we get

$$h_s^{1/2} \left\|\overline{j}_s\right\|_{L^2(s)} \lesssim \|\nabla(U - u)\|_{L^2(\omega_s)} + h_s \|r\|_{L^2(\omega_s)} + h_s^{1/2} \left\|j - \overline{j}_s\right\|_{L^2(s)}$$

□

Theorem 5.5 (efficiency / local lower bound)**.** *For each $T \in \mathscr{T}$, with ω_T the neighborhood of T,*

$$\mathscr{E}_{\mathscr{T}}(U, T) = \Big(h_T^2 \|r\|^2_{L^2(T)} + h_T \|j\|^2_{L^2(\partial T)}\Big)^{1/2} \;\lesssim\; \|\nabla(u - U)\|_{L^2(\omega_T)} + \operatorname{osc}_{\mathscr{T}}(U, \omega_T),$$

with $\operatorname{osc}^2_{\mathscr{T}}(U, \omega_T) = \sum_{T \subset \omega_T} \operatorname{osc}^2_{\mathscr{T}}(U, T)$. *Globally,*

$$C_2\, \mathscr{E}^2_{\mathscr{T}}(U) \le |\!|\!| u - U |\!|\!|^2_{\Omega} + \operatorname{osc}^2_{\mathscr{T}}(U), \tag{45}$$

with $\operatorname{osc}^2_{\mathscr{T}}(U) = \sum_{T \subset \mathscr{T}} \operatorname{osc}^2_{\mathscr{T}}(U, T)$.

Sketch. For the interior term: Lemma 5.3 gives $h_T \|r\|_{L^2(T)} \lesssim \|r\|_{H^{-1}(T)} + \operatorname{osc}_{\mathscr{T}}(U, \omega_T)$, and $\|r\|_{H^{-1}(T)} \le \|\nabla(u - U)\|_{L^2(T)}$ via the residual identity. Combining this estimate with the estimate from Lemma 5.4 we obtain the desired assertion. □

Remark 5.6 (two-sided control). Theorems 5.1 and 5.5 together give

$$C_2\,\mathscr{E}^2_{\mathscr{T}}(U) - \operatorname{osc}^2_{\mathscr{T}}(U) \;\le\; \|u-U\|^2_{H^1_0(\Omega)} \;\le\; C_1\,\mathscr{E}^2_{\mathscr{T}}(U).$$

The estimator is equivalent to the energy error up to oscillation. Since oscillation is reduced by refinement and is generically higher-order, $\mathscr{E}_{\mathscr{T}}(U)$ is a faithful, computable proxy for the energy error — the foundation for the convergence and optimality results of Sections 8–9.

Remark 5.7 (interior node property). When $\mathscr{T}_* \geq \mathscr{T}$ adds an interior node $z \in T$, the hat function $\phi_z \in \mathbb{V}(\mathscr{T}_*)$ satisfies $\int_T \phi_z \approx |T|$ and can replace the bubble b_T. Testing $\langle r_T, \phi_z\rangle$ and using Galerkin orthogonality with the new solution U_*,

$$\langle r_T, \phi_z\rangle = \mathscr{B}[u-U, \phi_z] = \mathscr{B}[U_* - U, \phi_z]\,,$$

one obtains

$$h_T\,\|r_T\|_{L^2(T)} \lesssim \|\nabla(U_* - U)\|_{L^2(\omega_T)} + \operatorname{osc}_{\mathscr{T}}(U, \omega_T). \tag{46}$$

This is the *interior node property*: refining T (adding an interior node) directly reduces the estimator, providing the key link between the marked set and the energy decrement used in the contraction proof of Section 8.

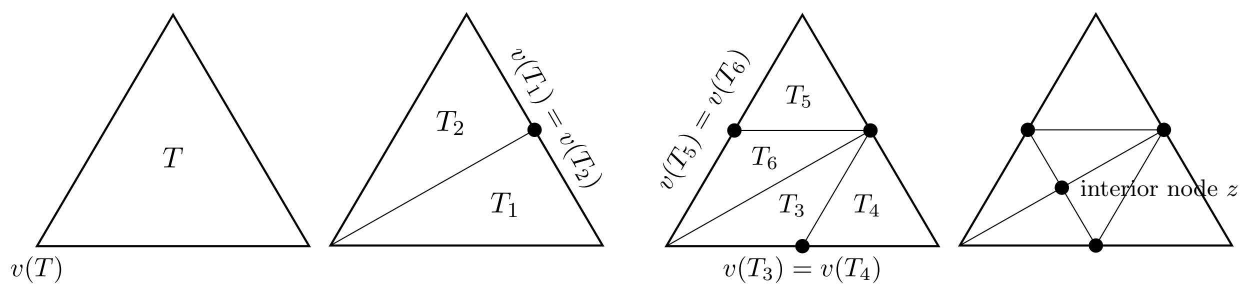


Figure 22: Interior node property: bisecting T (and neighbors for conformity) introduces an interior node $z \in T$. The associated hat function $\phi_z \in \mathbb{V}(\mathscr{T}_*)$ acts as a localized test function, showing that the marked elements contribute to the energy decrement.

Remark 5.8 (overestimation). The oscillation term $\operatorname{osc}_{\mathscr{T}}(U)$ cannot be removed from the lower bound (45), in other words the weighted L^2-residual estimator $\mathscr{E}_{\mathscr{T}}(U)$ may overestimate the error $|u-U|_{H^1_0(\Omega)}$. To see this, we consider the following *counterexample*: fix a mesh $\mathscr{T}$, a function $f \in H^{-1}(\Omega)\backslash L^2(\Omega)$, and a sequence of approximations $(f_k)_{k\in\mathbb{N}}$ in $L^2(\Omega)$ such that $\|f - f_k\|_{H^{-1}(\Omega)} \to 0$. The exact and Galerkin solutions $u_k \in H^1_0(\Omega)$ and $U_k \in \mathbb{V}_{\mathscr{T}}$ remain bounded but $\mathscr{E}_{\mathscr{T}}(U_n) \to \infty$ because $\|hf_k\|_{L^2(\Omega)} \to \infty$ as $k \to \infty$.

5.6 The adaptive loop

The estimator $\mathscr{E}_{\mathscr{T}}(U, T)$ just constructed is the last ingredient needed to state the adaptive method itself. Following the presentation of Nochetto [?], AFEM is the repeated application of four independent modules, each a *black box*: it consumes some data and returns some data, and the loop does not need to know how the box computes it internally, only what goes in and what comes out.

Module Solve. Given a conforming mesh $\mathscr{T}$ (obtained from the macro triangulation $\mathscr{T}_0$ by finitely many bisections),

$$U = \textsc{Solve}(\mathscr{T})$$

returns the Galerkin solution of Section 2: $U \in \mathbb{V}$ (C^0 piecewise linears over $\mathscr{T}$) with $\mathscr{B}[U,V] = \langle f, V\rangle$ for all $V \in \mathbb{V}$. Computationally, this is the assembly-and-solve of Section 3.

Module Estimate. Given $\mathscr{T}$ and $U = \textsc{Solve}(\mathscr{T})$,

$$\{\mathscr{E}_{\mathscr{T}}(U,T)\}_{T\in\mathscr{T}} = \textsc{Estimate}(U,\mathscr{T})$$

returns the local indicators of this Section (Theorem 5.1) — computable from f, A, and U alone, with no knowledge of the unknown u.

Module Mark. Given $\mathscr{T}$ and the indicators $\{\mathscr{E}_{\mathscr{T}}(U,T)\}_{T\in\mathscr{T}}$,

$$\mathscr{M} = \textsc{Mark}\big(\{\mathscr{E}_{\mathscr{T}}(U,T)\}_{T\in\mathscr{T}}, \mathscr{T}\big)$$

returns a subset $\mathscr{M} \subset \mathscr{T}$ of elements selected for refinement, on the strength of the indicators alone. Mark is the one module with genuine freedom of design; §6.4 fixes it to Dörfler (bulk) marking.

Module Refine. Given $\mathscr{T}$ and a marked set $\mathscr{M} \subset \mathscr{T}$,

$$\mathscr{T}_* = \textsc{Refine}(\mathscr{T}, \mathscr{M})$$

returns the smallest conforming refinement of $\mathscr{T}$ in which every $T \in \mathscr{M}$ has been bisected at least once (§4.2), so that $\mathscr{T}_* \geq \mathscr{T}$ in the partial order of Section 4; the completion procedure that restores conformity along the way is detailed in §6.5.

Algorithm 5.9 (AFEM)**.** Given an initial conforming mesh $\mathscr{T}_0$, set $k = 0$ and iterate

$U_k = \mathrm{SOLVE}(\mathscr{T}_k)$
$\{\mathscr{E}_k(U_k,T)\}_{T\in\mathscr{T}_k} = \mathrm{ESTIMATE}(U_k, \mathscr{T}_k)$
$\mathscr{M}_k = \mathrm{MARK}\big(\{\mathscr{E}_k(U_k,T)\}_{T\in\mathscr{T}_k}, \mathscr{T}_k\big)$
$\mathscr{T}_{k+1} = \mathrm{REFINE}(\mathscr{T}_k, \mathscr{M}_k);\ \ k \leftarrow k+1$

Remark 5.10 (what makes it *adaptive*). Every module above is specified purely by its input and output. What makes the loop *adaptive*, as opposed to *uniform* refinement (all of Section 3's examples), is that Mark and Refine route the mesh work to exactly the elements that Estimate flags as troublesome, instead of refining indiscriminately. Section 6 turns each module into a Matlab/Octave function and its driver `afem.m` into the loop above, literally; Sections 7–9 develop the properties of Solve–Estimate–Mark–Refine needed to prove that it converges, and at the optimal rate.

6 Implementation of AFEM

We now assemble the modules of the previous Sections into a running adaptive code. The reference implementation is a small Matlab/Octave program whose driver `afem.m` realizes the loop SOLVE → ESTIMATE → MARK → REFINE of §5.6 literally, one function per module. Rather than repeat the assembly of Section 3 or the black-box specification of each module (§5.6), we focus on what is genuinely new in the adaptive setting: the mesh data structure that makes local refinement and jump computations $O(1)$ per element, and the MARK and REFINE modules that have no counterpart on a fixed mesh.

6.1 The adaptive loop

This subsection shows how the four black-box modules of §5.6 are wired together in practice. The driver `[prob_data,adapt] = init_data()` returns the problem data and adaptivity parameters; `afem.m` then reads the initial mesh named by `prob_data.domain`, optionally applies a few uniform refinements to start from a reasonable resolution, and iterates the four modules until the global estimator falls below a tolerance, an iteration cap is reached, or the mesh exceeds a vertex-count cap (`adapt.max_vertices`, a safety net against runaway runs). Every module takes the data it needs as explicit arguments and returns what it updates — no global variables are involved at this level:

```
[prob_data, adapt] = init_data();
% ... read mesh.elem_vertices, elem_neighbours, elem_boundaries,
% vertex_coordinates from the prob_data.domain folder ...
uh = zeros(mesh.n_vertices,1); fh = zeros(mesh.n_vertices,1);
if (prob_data.initial_global_refinements) % optional uniform startup
  mesh.mark = prob_data.initial_global_refinements*ones(mesh.n_elem,1);
  [mesh,uh,fh] = refine_mesh(mesh,uh,fh);
end
iter_counter = 1;
while (1)
  [uh,fh] = assemble_and_solve(prob_data,mesh,uh,fh); % SOLVE (Section 3, on the current
       mesh)
  [est, mesh] = estimate(prob_data, adapt, mesh, uh); % ESTIMATE -> mesh.estimator,
      global est
  N_vert = mesh.n_vertices;
  if ((iter_counter >= adapt.max_iterations) || (est < adapt.tolerance) ...
      || (N_vert > adapt.max_vertices))
    break;
  else
    mesh = mark_elements(adapt, mesh); % MARK -> mesh.mark
    [mesh,uh,fh] = refine_mesh(mesh,uh,fh); % REFINE -> new conforming mesh
  end
  iter_counter = iter_counter + 1;
end
```

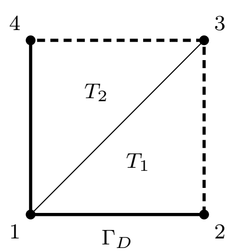
(The driver also records `H1_err`/`L2_err` and the estimator at every iteration and, optionally, plots the three side by side on a log-log scale; this bookkeeping is elided above since it does not affect the loop's logic.) The modules communicate through the `mesh` struct and the two nodal vectors `uh`, `fh`, passed and returned explicitly: ESTIMATE writes the per-element indicators into `mesh.estimator` and returns the updated `mesh` together with the scalar global estimator; MARK writes an integer refinement count into `mesh.mark`; REFINE rebuilds the connectivity and returns a new, larger `mesh`, interpolating the current solution `uh` (and `fh`) onto the finer mesh as it goes. Refine is the one exception to this argument-passing style: internally, `refine_mesh` and `refine_element` communicate through Matlab globals rather than return values, purely to avoid copying the whole mesh struct at every one of the many recursive calls a single refinement pass can make; `refine_mesh` still takes and returns `mesh`, `uh`, `fh` in the ordinary way, so the exception is invisible to the caller (see §6.5).

6.2 Mesh data structure for adaptivity

The four text files of Section 3 describe geometry and boundary vertices, which is enough to assemble on a *fixed* mesh. Adaptivity needs two more relations to be available in $O(1)$ per element: *who is my neighbour across each side* (for the jump term of the estimator and for propagating refinement), and *what kind of boundary each side carries*. The code therefore stores the triangulation in four arrays, all indexed by element:

- `elem_vertices` $(N_t \times 3)$ — the three global vertices v^1, v^2, v^3 of each triangle. The local ordering is part of the data: *side i is the edge opposite vertex i*, and side 3 (the edge v^1v^2, opposite the newest vertex v^3) is the designated *refinement edge* of Section 4.
- `elem_neighbours` $(N_t \times 3)$ — `elem_neighbours(el,i)` is the index of the triangle sharing side i with `el`, or 0 if that side lies on $\partial\Omega$.
- `elem_boundaries` $(N_t \times 3)$ — a flag per side: 0 for an interior side, > 0 for a Dirichlet side, < 0 for a Neumann side.
- `vertex_coordinates` $(N_p \times 2)$ — the (x, y) of each vertex, as before.

Figure 23 illustrates the convention on a two-triangle mesh. With `elem_neighbours` at hand the estimator reads a neighbouring gradient directly, and Refine finds the triangle on the other side of a refinement edge without any search; with `elem_boundaries` the Dirichlet/Neumann partition is reconstructed on the fly (`get_dirichlet_neumann`), so it need not be stored as separate vertex lists.



`elem_vertices`			
T_1:	3	1	2
T_2:	1	3	4

`elem_neighbours`			
T_1:	0	0	2
T_2:	0	0	1

`elem_boundaries`			
T_1:	$+1$	-1	0
T_2:	-1	$+1$	0

`vertex_coordinates`		
1:	0.0	0.0
2:	1.0	0.0
3:	1.0	1.0
4:	0.0	1.0

Figure 23: Element-based mesh data for adaptivity on two triangles $T_1 = (3, 1, 2)$, $T_2 = (1, 3, 4)$. Side i is opposite vertex i; the shared side is side 3 of both elements, so `elem_neighbours`$(T_1, 3) = T_2$ and vice versa. `elem_boundaries` flags each side (here $+1$ Dirichlet—solid line, -1 Neumann—dashed line, 0 interior). The refinement edge of each element is side 3 (the edge v^1v^2, opposite the newest vertex).

6.3 Solve and Estimate

Solve is exactly the fixed-mesh assembly of Section 3 (`assemble_and_solve`), applied to the current mesh: element stiffness and mass matrices via the reference map, midpoint quadrature for the load vector, Simpson's rule on Neumann segments, and Dirichlet rows replaced by the identity. Nothing changes but the mesh it runs on, so we do not repeat it here.

Estimate evaluates the residual indicator of Section 5, Theorem 5.1. For $\mathscr{P}_1$ elements and piecewise-constant coefficients the second derivative of U vanishes on each element, so the interior residual reduces to $r_T = b \cdot \nabla U + c\,U - f$, and the jump term collects the jumps of the normal flux $a\,\nabla U \cdot n$ across the three sides. As in `fem.m` (Remark 3.1), a and c are supplied as function handles and evaluated once per element at the barycenter, so the code first computes, for every element, the (constant) gradient of U together with $a|_T$ and $c|_T$, then loops again accumulating both estimator contributions:

```
grd_bas_fcts = [ -1 -1 ; 1 0 ; 0 1 ]'; % reference gradients
for el = 1:n_elem % constant grad U, and a|_T, c|_T, per element
  v = mesh.elem_vertices(el,:); B = [ V(v(2))-V(v(1)) , V(v(3))-V(v(1)) ];
  bary = (V(v(1))+V(v(2))+V(v(3)))/3;
  grduh(el,:) = ( B' \ (grd_bas_fcts*uh(v)) )';
  coeff_a(el) = prob_data.a(bary); coeff_c(el) = prob_data.c(bary);
end
for el = 1:n_elem
  ... el_area = abs(det(B))/2; h_T = sqrt(el_area);
  % interior residual r = b.gradU + c U - f, sampled at the three edge midpoints
  r12 = prob_data.b*grduh(el,:)' + coeff_c(el)*uh12 - f12; % (and r23, r31)
  % jump term: loop over the three sides
  jump_res2 = 0;
  for side = 1:3
    % ... tangential = (edge vector opposite vertex 'side'); midpoint = its midpoint ...
    normal = [tangential(2); -tangential(1)]/norm(tangential); % computed for every side
    if (mesh.elem_boundaries(el,side) == 0) % interior side
      neigh = mesh.elem_neighbours(el,side);
      vec = (coeff_a(el)*grduh(el,:) - coeff_a(neigh)*grduh(neigh,:)) * normal;
    elseif (mesh.elem_boundaries(el,side) < 0) % Neumann side
      vec = coeff_a(el)*grduh(el,:)*normal - prob_data.gN(midpoint);
    end % Dirichlet side: no jump
    jump_res2 = jump_res2 + vec*vec';
  end
  mesh.estimator(el) = sqrt( adapt.C(1)*h_T^2 * el_area*(r12^2+r23^2+r31^2)/3 ...
                        + adapt.C(2)*h_T * h_T*jump_res2 );
end
```

Remark 6.1 (the Neumann residual). On a Neumann side the code now evaluates *both* pieces pointwise, so that every occurrence of the coefficient in this listing reads `coeff_a(el)` rather than the bare function handle `prob_data.a`: it replaces the neighbouring flux $a_{\mathrm{neigh}}\nabla U_{\mathrm{neigh}} \cdot \boldsymbol{n}$ of the interior case by the prescribed datum g_N, sampled at the midpoint of that side, giving the boundary residual $a_{\mathrm{el}}\nabla U_{\mathrm{el}} \cdot \boldsymbol{n} - g_N$ — the natural boundary-residual analogue of the interior flux jump j_s of Section 5 (which is derived there only for the pure-Dirichlet problem, so this extension to Γ_N is not itself a proved theorem in these notes, but the standard one used in practice). Squaring and accumulating it into `jump_res2` exactly as an interior jump would be is what allows a single loop over the three sides to handle interior, Dirichlet, and Neumann edges uniformly.

This is precisely $\mathscr{E}_T^2 = C_1 h_T^2 \|r_T\|^2_{L^2(T)} + C_2 h_T \|J_T\|^2_{L^2(\partial T)}$ of Section 5, realized discretely: the interior integral uses the three-edge-midpoint quadrature, and each side of the piecewise-constant flux jump contributes $h_T \cdot |e|\,|J|^2 \sim h_T^2\,|J|^2$ (here $h_T = \sqrt{|T|}$). Two boundary rules are worth noting. On a Dirichlet side there is no jump, so it is skipped; on an interior side the code projects onto `normal` exactly the quantity the a posteriori theory calls for, $j_s = [\![A\nabla U \cdot \boldsymbol{n}]\!]_s$ of Section 5. *If* a is the same constant on both elements of the patch, the tangential derivative of U is continuous across the edge (both traces come from the same nodal values), so the jump of the normal flux equals the jump of the full gradient vector and the explicit projection could be skipped. The neighbour is read in $O(1)$ from `elem_neighbours`: this is the data structure of §6.2 paying off. In the L-shape example below $a = 1$, $b = 0$, $c = 0$ and $f = 0$, so the interior residual vanishes identically and the estimator is driven *entirely* by the gradient jumps — the discrete trace of the singularity at the reentrant corner.

6.4 Mark

Given the indicators $\{\mathscr{E}_T\}$, Mark selects the subset $\mathscr{M}$ of elements to be refined. The code implements three classical strategies through the single array `mesh.mark`, whose entries are the *number of times* each element is to be bisected (`adapt.n_refine`, typically 2; see Refine below).

```
function mesh = mark_elements(adapt, mesh)
mesh.mark = zeros(mesh.n_elem, 1);
switch adapt.strategy
case 'GR' % global (uniform) refinement: mark everything
  mesh.mark = adapt.n_refine * ones(size(mesh.estimator));
case 'MS' % maximum strategy: mark elements within a fraction of the largest
  mesh.mark(find(mesh.estimator > adapt.MS_gamma*mesh.max_est)) = adapt.n_refine;
case 'Doerfler' % Doerfler (bulk) strategy
  threshold = adapt.Doerfler_theta^2 * mesh.est_sum2;
  gamma = 1; est_sum2_marked = 0;
  while (est_sum2_marked < threshold)
    gamma = gamma - adapt.Doerfler_nu;
    f = find(mesh.estimator > gamma*mesh.max_est);
    mesh.mark(f) = adapt.n_refine;
    est_sum2_marked = sum(mesh.estimator(f).^2);
  end
end
```

The *maximum strategy* (`MS`) marks every element whose indicator exceeds a fixed fraction $\gamma \in (0, 1)$ of the largest one, $\mathscr{E}_T > \gamma \max_{T'} \mathscr{E}_{T'}$; it is cheap (one pass) and effective in practice. The `Doerfler` [11] strategy is *Dörfler* or *bulk* marking: it selects a set $\mathscr{M}$ that captures a fixed fraction of the total estimated energy,

$$\mathscr{E}(U, \mathscr{M}) \;\geq\; \theta\, \mathscr{E}(U, \mathscr{T}), \qquad 0 < \theta < 1, \qquad \mathscr{E}(U, \mathscr{M}) = \Big(\sum_{T \in \mathscr{M}} \mathscr{E}_T^2 \Big)^{1/2}, \tag{47}$$

which is the bulk criterion the optimality theory of Section 9 requires. The code realizes (47) directly: `threshold` is $\theta^2 \mathscr{E}(U, \mathscr{T})^2$, and the loop lowers the cutoff γ in steps of `adapt.Doerfler_nu` until the elements with $\mathscr{E}_T > \gamma \max_{T'} \mathscr{E}_{T'}$ accumulate at least that much of $\sum_T \mathscr{E}_T^2$ — so the parameter `adapt.Doerfler_theta` *is* θ of (47) directly, with no reparametrization. Setting `strategy = 'GR'` recovers uniform refinement and gives the baseline against which the adaptive runs are compared. The shipped default, `Doerfler_theta = 0.3`, is a conservative choice suited to watching the loop run interactively; §6.6 below uses a larger θ to reach the mesh sizes needed to see the asymptotic rate.

6.5 Refine: recursive bisection with completion

Refine turns the marked set into a new *conforming* mesh by newest-vertex bisection (Section 4, Figure 16). The driver `refine_mesh` pre-allocates the arrays and then repeatedly refines the first still-marked element until none remain:

```
first_marked = find(adapt_global_mesh.mark > 0, 1); % find first non_zero
while (first_marked)
  n_refined = n_refined + refine_element(first_marked);
  first_marked = find(adapt_global_mesh.mark > 0, 1);
end
```

(The pseudocode above writes `mesh` as if it were an ordinary variable in scope; as noted in §6, the actual `refine_mesh`/`refine_element` pair communicates through the globals `adapt_global_mesh`, `adapt_global_uh`, `adapt_global_fh` instead, purely so the recursion below does not copy the whole mesh struct on every call — `refine_mesh` itself still has the ordinary signature `[mesh,uh,fh,n_refined] = refine_mesh(mesh,uh,fh)`.)

All the work — and the reason adaptivity is subtle — is in `refine_element`. Bisecting a triangle across its refinement edge (side 3) inserts a new vertex at the edge midpoint. If the neighbour across that edge shares the *same* refinement edge, the two are a *compatible* patch and can be split together, producing a conforming mesh locally. If not, splitting `el` alone would leave a hanging node on the neighbour; the fix is to refine the neighbour *first*, recursively, until a compatible patch is exposed:

```
function n_refined = refine_element(el)
  neighbour = mesh.elem_neighbours(el, 3); % across the refinement edge
  if (neighbour > 0 && mesh.elem_neighbours(neighbour,3) ~= el) % not compatible
    refine_element(neighbour); % refine neighbour first
    neighbour = mesh.elem_neighbours(el, 3); % may have changed
  end
  % --- now (el, neighbour) is a compatible patch: bisect it ---
  new = new_vertex_at_midpoint(v1, v2); % v1,v2 = endpoints of side 3
  % el and its new child, with the NEW vertex as their vertex 3:
  mesh.elem_vertices(el,:) = [v3 v1 new];
  mesh.elem_vertices(child) = [v2 v3 new];
  % ... update elem_neighbours / elem_boundaries of the patch and its neighbours ...
  % ... interpolate uh, fh at 'new'; decrement mesh.mark on the children ...
end
```

Two points make this correct and efficient. First, placing the new vertex as vertex 3 of each child makes the opposite edge their refinement edge automatically, so the newest-vertex labeling of Section 4 is maintained without storing any explicit generation counter. Second, the recursion is exactly the *completion* step of Section 4: it terminates because a consistent initial labeling of $\mathscr{T}_0$ guarantees that following refinement edges cannot cycle, and by the Binev–Dahmen–DeVore/Stevenson bound (Theorem 4.6) the total number of elements it creates over the whole run is proportional to the total number of marks — so completion is never a bottleneck.

Finally, why `n_refine`$= 2$? Each call bisects the patch once and decrements the children's `mark` counter; an element marked twice is therefore bisected twice, which places a new node in the middle of its edges, halving the mesh-size h_T.

6.6 Validation: adaptive versus uniform refinement

6.6.1 L-shape domain

We close the loop on the benchmark that motivated adaptivity in Section 3: the L-shaped domain $\Omega = (-1,1)^2 \setminus \big([0,1]\times[-1,0]\big)$ with the singular harmonic solution $u(r,\theta) = r^{2/3}\sin(\frac{2}{3}\theta)$ (so $f = 0$, Dirichlet data $g_D = u$). On this problem uniform refinement stalls at the suboptimal energy rate $N_p^{-1/3}$ (Table 2), because the mesh wastes degrees of freedom far from the corner, where the solution is smooth. Running the adaptive loop above — ESTIMATE with the residual indicator, Dörfler/bulk marking at $\theta = 0.5$, and recursive bisection — concentrates refinement at the reentrant corner and recovers the *optimal* rate.

Table 4 and Figure 24 report a representative run, obtained from the shipped code. The energy error decays as $N_p^{-1/2}$, matching the smooth-case rate of Figure 10 and the optimal approximation rate of Section 4, even though the exact solution is only $H^{1+2/3-\varepsilon}$. This is the headline payoff of the whole method: adaptivity restores the convergence rate of a smooth problem by placing the mesh where the solution needs it, with no a priori knowledge of the singularity.

N_p	$\lvert u-U\rvert_{H^1}$	rate
124	8.859×10^{-2}	—
394	4.621×10^{-2}	0.56
1608	2.252×10^{-2}	0.51
6253	1.142×10^{-2}	0.50
13177	7.794×10^{-3}	0.51
31350	5.057×10^{-3}	0.50
47042	4.116×10^{-3}	0.51

Table 4: Adaptive run on the L-shape ($u = r^{2/3}\sin(\frac{2}{3}\theta)$), residual estimator with bulk marking $\theta = 0.5$. The observed order in N_p converges to $\frac{1}{2}$: the adaptive method recovers the *optimal* energy rate $N_p^{-1/2}$, versus the uniform $N_p^{-1/3}$ of Table 2.

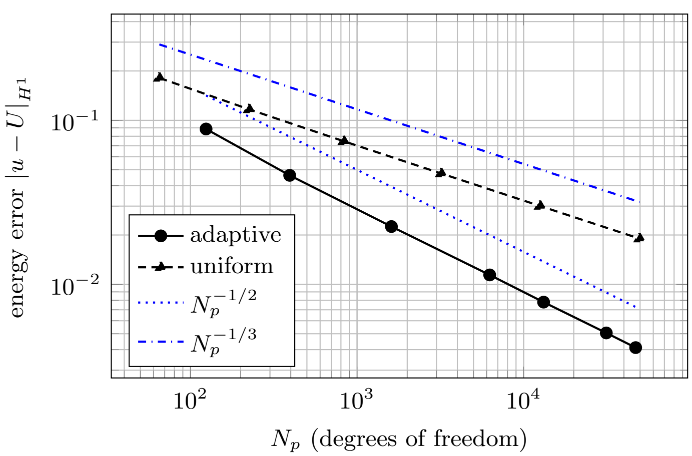


Figure 24: Energy error versus N_p on the L-shape: adaptive refinement (solid, data of Table 4) tracks the optimal reference slope $N_p^{-1/2}$, while uniform refinement (dashed, data of Table 2) is stuck at $N_p^{-1/3}$. The gap widens with N_p: at $N_p \approx 5\times10^4$ the adaptive error is more than four times smaller.

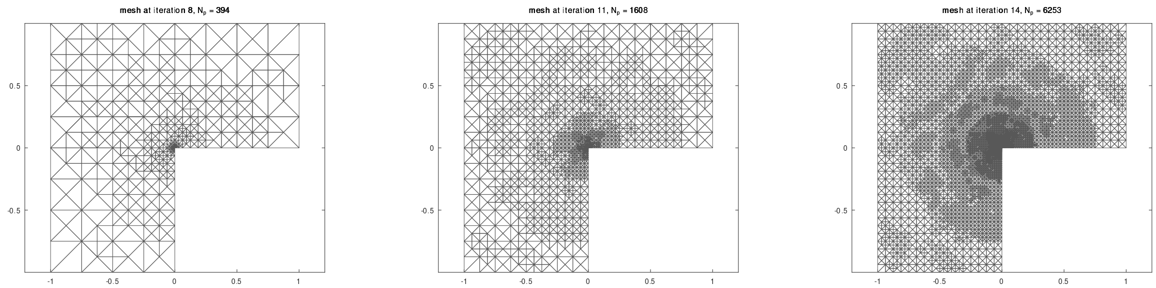


Figure 25: Adaptive meshes produced by `afem.m` on the L-shaped domain, at iterations 8, 11, and 14 of the run reported in Table 4 and Figure 24 ($N_p = 394, 1608, 6253$, respectively). Refinement concentrates almost exclusively at the reentrant corner $(0, 0)$, tracking the location of the singularity $u \sim r^{2/3}$ with no a priori knowledge of it; away from the corner the mesh stays close to its initial, coarse resolution throughout the run.

6.6.2 Example by Kellogg

Section 3.10 tested this same problem on a sequence of *uniform* meshes (Table 3, Figure 12) and found the energy error essentially flat, decaying at only $N_p^{-0.014}$: since $u \in H^{1+\gamma-\varepsilon}$ with $\gamma = 0.1$ *everywhere* on Ω, uniform refinement is, for practical purposes, useless. With two initial global refinements of `big_square_all_dirichlet` (Section 6.2) the starting mesh already conforms exactly to the jump of the coefficient $a(x)$ — every triangle lies entirely in one quadrant, so barycenter evaluation of a (Remark 3.1) is exact from the very first iteration — and this conformity is preserved automatically at every later step: bisection only ever adds a vertex at the midpoint of an existing edge, so an element that does not straddle an axis can never produce a child that does.

Because refinement here produces a *nested* sequence of conforming $\mathscr{P}_1$ spaces on which the discrete bilinear form $a_h(v, w) = \int_\Omega a \nabla v \cdot \nabla w$ is evaluated *exactly* (previous paragraph), Galerkin orthogonality guarantees that the a-weighted *energy* norm $|||u - U|||^2 = \int_\Omega a\, |\nabla(u - U)|^2$ can only decrease as the mesh is refined — this is just Céa's lemma applied at each step, and it holds regardless of how irregular the singularity is. It says nothing, however, about the *plain* (unweighted) seminorm $|u - U|_{H^1}$ used throughout this chapter's tables and figures for consistency with the rest of the notes: the two norms differ by a factor between 1 and $\sqrt{161.4476\ldots}$ depending on *where* the gradient error is concentrated, and since Dörfler marking is driven by the (energy-consistent) estimator rather than by $|\cdot|_{H^1}$ directly, refinement can — and here does — occasionally shift error mass from a low-a quadrant to a high-a one in a way that decreases the energy norm while increasing the plain seminorm. Table 5 lists both norms side by side: the energy column is monotone (up to solver round-off, $\sim 10^{-5}$ relative), while the H^1-seminorm column visibly is not, exactly as this argument predicts.

Table 5 and Figure 26 report a representative run (same Dörfler marking $\theta = 0.5$ and two initial global refinements as the L-shape run), pushed out to $N_p \approx 1.2 \times 10^6$ — comparable to the largest meshes used anywhere in these notes — to see whether the optimal rate $N_p^{-1/2}$ of Table 4 eventually emerges. It does not. The observed energy rate climbs from about 0.21 near $N_p = 385$ to a local peak of 0.27 near $N_p = 4491$, then *settles back down* to 0.20–0.23 and stays in that band for the remaining three orders of magnitude in N_p, showing no trend toward $\frac{1}{2}$. This is markedly faster than the uniform $N_p^{-0.014}$ of Table 3, but the plateau itself is an interesting finding.

N_p	$\lvert u-U\rvert_{H^1}$	$\lVert\lvert u-U\rvert\rVert$	rate ($\lVert\lvert\cdot\rvert\rVert$)
9	1.546×10^{-1}	1.561×10^{0}	—
385	1.329×10^{-1}	7.124×10^{-1}	0.21
967	1.238×10^{-1}	5.874×10^{-1}	0.21
1549	1.146×10^{-1}	5.264×10^{-1}	0.23
4491	9.255×10^{-2}	3.941×10^{-1}	0.27
17917	6.190×10^{-2}	2.817×10^{-1}	0.24
73769	4.977×10^{-2}	2.030×10^{-1}	0.23
296873	4.488×10^{-2}	1.488×10^{-1}	0.22
1189361	4.269×10^{-2}	1.124×10^{-1}	0.20

Table 5: Adaptive run on the Kellogg checkerboard-coefficient benchmark ($\gamma = 0.1$), residual estimator with bulk marking $\theta = 0.5$, pushed to $N_p \approx 1.2 \times 10^6$. $\lvert u-U\rvert_{H^1}$ is the plain (unweighted) gradient seminorm used elsewhere in this chapter; $\lVert\lvert u-U\rvert\rVert = (\int_\Omega a\,\lvert\nabla(u-U)\rvert^2)^{1/2}$ is the a-weighted energy norm, whose rate is listed since it is the quantity Galerkin orthogonality actually controls. Rows are sampled at the iterations between successive marking bursts (Figure 26 shows the full, non-monotone history in between). The energy rate never approaches the optimal $\frac{1}{2}$ of Table 4; after an initial rise it plateaus at 0.20–0.23 from $N_p = 17917$ onward, three orders of magnitude in N_p with no further improvement.

Does a smaller θ help? The theory behind AFEM's optimal-rate results (Sections 8–9) typically requires the Dörfler parameter θ to be *sufficiently small* in order for the optimal rate $N_p^{-1/2}$ to be guaranteed; $\theta = 0.5$ above is a common practical default, not a value singled out by the theory. It is natural to ask whether the plateau of Table 5 is an artifact of too large a θ, so Table 6 and Figure 27 repeat the identical run with $\theta = 0.2$ in place of $\theta = 0.5$ (everything else — $C_1 = C_2 = 1$, $n\_refine = 2$, two initial global refinements — held fixed), pushed to $N_p \approx 3.0 \times 10^5$ (the meshes produced by the smaller θ are more strongly graded, which makes the Jacobi–CG solve of Section 6.7 noticeably worse-conditioned at large N_p, so this run was stopped one order of magnitude short of the $\theta = 0.5$ run above rather than push the solver past what converges reliably in reasonable time). The observed energy rate is, if anything, marginally *higher* in the mid range (peaking near 0.29 around $N_p = 7075$, versus 0.27 for $\theta = 0.5$), but by the last few rows it has settled back into essentially the same 0.22–0.25 band. Halving θ does not close the gap to $\frac{1}{2}$: the plateau in Table 5 seems not to be an artifact of the marking parameter.

N_p	$\lvert u-U\rvert_{H^1}$	$\lVert\lvert u-U\rvert\rVert$	rate ($\lVert\lvert\cdot\rvert\rVert$)
9	1.546×10^{-1}	1.561×10^{0}	—
517	1.996×10^{-1}	6.784×10^{-1}	0.21
1043	1.411×10^{-1}	5.865×10^{-1}	0.21
2781	1.044×10^{-1}	4.722×10^{-1}	0.22
7075	8.931×10^{-2}	3.606×10^{-1}	0.29
18059	7.336×10^{-2}	2.783×10^{-1}	0.28
74801	5.593×10^{-2}	1.962×10^{-1}	0.25
299455	4.774×10^{-2}	1.441×10^{-1}	0.22

Table 6: Same benchmark and run as Table 5, with the Dörfler parameter lowered from $\theta = 0.5$ to $\theta = 0.2$, pushed to $N_p \approx 3.0 \times 10^5$. The energy rate again never approaches $\frac{1}{2}$ and settles into essentially the same 0.22–0.25 plateau as the $\theta = 0.5$ run, despite the smaller, theoretically more conservative marking fraction.

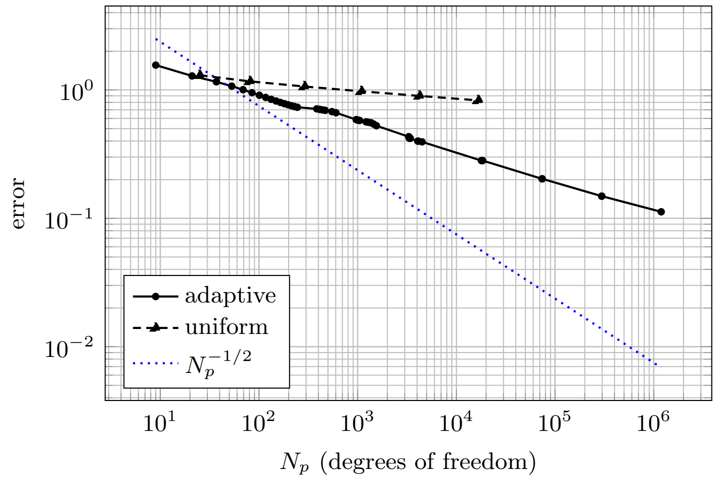


Figure 26: a-weighted energy error $\|u - U\| = (\int_\Omega a\,|\nabla(u-U)|^2)^{1/2}$ versus N_p on the Kellogg checkerboard problem, out to $N_p \approx 1.2 \times 10^6$: adaptive refinement (solid, same run as Table 5) versus uniform refinement (dashed, computed by the same `fem.m`/`gen_mesh_rectangle.m` pipeline as Table 3, with the plain H^1 seminorm there replaced by the energy norm here). In every other example in this chapter $a \equiv 1$, so the energy norm and the plain H^1 seminorm coincide exactly and the distinction is moot; only here, where a jumps by a factor of over 161, do the two differ enough to matter, and the energy norm is the one Galerkin orthogonality actually controls (it decreases essentially monotonically along the adaptive run, unlike the plain seminorm of Table 5). Neither curve tracks the optimal $N_p^{-1/2}$ reference slope (dotted): both bend away from it early and settle into a distinctly shallower, roughly parallel decay — the plateau documented in Table 5.

Is the plateau a quadrature artifact? Every energy-norm value reported above, in both Table 5 and Table 6, is computed with the same three-point edge-midpoint rule used throughout this chapter's `H1_err.m` (Section 3.10) — exact for quadratic polynomials, and normally more than adequate. Kellogg's exact gradient, however, blows up like $r^{\gamma-1} = r^{-0.9}$ as $r \to 0$, and on any element with a vertex at (or very near) the origin, two of that rule's three quadrature points — the midpoints of the two edges meeting at that vertex — sit at *half* the distance to the singularity of the third point, so the rule systematically *overestimates* the local gradient error precisely on the elements the adaptive loop keeps concentrating there. Two initial global refinements guarantee the origin is a mesh vertex at every level, so this is a permanent feature of the mesh sequence used throughout this subsubsection, not a rare edge case.

A second, independent issue compounds it: direct inspection of the mesh sequence shows the estimator never stops refining an ever-shrinking neighbourhood of the origin, where all four quadrant values of a meet — element areas there shrink by roughly a factor of two at *every* iteration, reaching about 10^{-24} by iteration 40, many orders of magnitude below double-precision roundoff relative to the domain size. This behaviour is inherent to the residual estimator on this four-quadrant benchmark — it is present in the original `estimate.m`/`refine_mesh.m` pipeline, not a porting artifact — and it means a small fixed neighbourhood of the origin never settles into any asymptotic regime at all: whatever error that neighbourhood contributes is dominated by floating-point noise rather than by the true PDE behaviour, and it contaminates any norm computed over the whole domain.

Table 7 recomputes the energy norm at every sampled mesh of Table 5 — not just the tail — using a much higher-order rule (a 6×6-point Gauss–Legendre product rule via the Duffy square-collapse transform, exact for polynomials of degree 11) and excluding from the integral every element

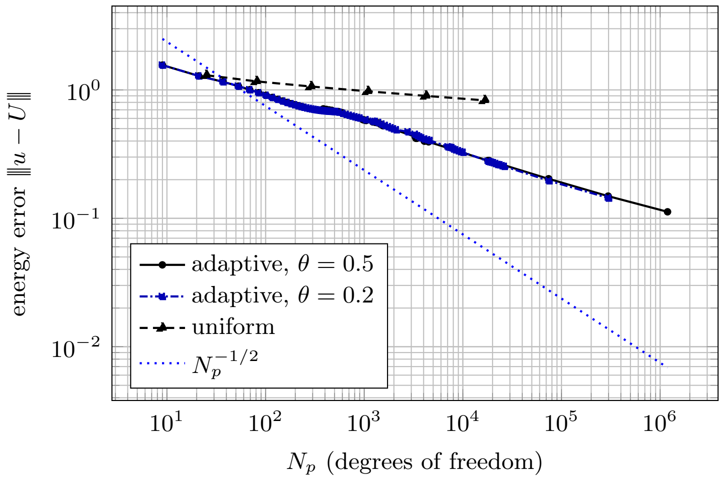


Figure 27: Energy error versus N_p for the two Dörfler parameters: $\theta = 0.5$ (solid, data of Table 5, out to $N_p \approx 1.2 \times 10^6$) and $\theta = 0.2$ (dash-dotted, data of Table 6, out to $N_p \approx 3.0 \times 10^5$). The two curves are nearly indistinguishable throughout their common range and bend away from the optimal $N_p^{-1/2}$ slope (dotted) in essentially the same way: lowering θ towards the theory's "sufficiently small" regime does not, in this benchmark, recover the optimal rate.

with a vertex within a distance $R = 0.05$ of the origin, isolating the part of the domain where both the quadrature and the mesh are behaving as expected. The recomputed rate is noisy while the mesh is still coarse (0.68, 0.49, 0.36, 0.23 across the first four refinement levels, where only a handful of elements are excluded and the sample is small), but from $N_p = 17917$ onward it settles cleanly at 0.48–0.50 for every remaining level, essentially exactly the optimal $\frac{1}{2}$ — in sharp contrast to the 0.20–0.22 plateau of Table 5 at the same N_p. This is not an artifact of the particular exclusion radius chosen: repeating the computation at the three largest meshes with $R \in \{0.02, 0.05, 0.10\}$ gives rates between 0.498 and 0.504 throughout, a spread smaller than the third significant digit.

The Kellogg checkerboard benchmark does not, after all, appear to violate the optimal $N_p^{-1/2}$ rate. The plateau in Tables 5 and 6 is explained by a compounding pair of numerical artifacts localized to a single shrinking neighbourhood of the origin: an increasingly severe quadrature error, riding on top of mesh degeneracy that never reaches an asymptotic regime. Away from that one point, the adaptive method appears to behave exactly as the theory of Sections 8–9 predicts.

6.7 Source code

The complete adaptive Matlab/Octave code, together with the four mesh-data folders, is available in the companion repository

`https://github.com/morinpedro/adaptive-fem`

The whole loop, including mesh generation and bisection, is Matlab/Octave only. It is organized around the driver and one function per module.

- `afem.m` — the driver: reads the data and mesh, runs the Solve–Estimate–Mark–Refine loop.
- `init_data.m` — `[prob_data, adapt] = init_data()`: problem data (a, b, c, f, g_D, g_N, exact solution) for whichever of five built-in `examples` is selected by the `example` variable at the

N_p	$\|u-U\|$, excl. $B_{0.05}(0)$, 6th-order quad.	rate
9	1.239×10^{0}	—
385	9.812×10^{-2}	0.68
967	6.260×10^{-2}	0.49
1549	5.293×10^{-2}	0.36
4491	4.152×10^{-2}	0.23
17917	2.147×10^{-2}	0.48
73769	1.079×10^{-2}	0.49
296873	5.367×10^{-3}	0.50
1189361	2.685×10^{-3}	0.50

Table 7: The same nine meshes as Table 5 ($\theta = 0.5$ run), re-examined with a higher-order quadrature rule and with all elements touching a ball of radius $R = 0.05$ about the origin excluded from the energy integral (at $N_p = 9$ the mesh has no elements that close to the origin, so this row coincides with the whole-domain value). Once the near-origin quadrature error and the degenerate mesh region responsible for it are removed, the rate is noisy at first — as expected before the asymptotic regime sets in, with only a few elements excluded from a coarse mesh — then locks onto the theoretical optimum ${N_p}^{-1/2}$ from $N_p = 17917$ onward, in contrast to the 0.20–0.22 plateau obtained over the whole domain with the standard three-point rule.

top of the file — `'L-shape'` (the reentrant-corner solution used above), `'square-Dirichlet'`, `'big-square-Dirichlet'`, `'square-mixed'`, and `'kellogg'` (the checkerboard-coefficient benchmark of Section 3.10, exercising the piecewise-constant a of Remark 3.1 on the adaptive side) — together with the adaptivity parameters (strategy, tolerance, marking constants, `n_refine`), which apply across all five examples.

- `assemble_and_solve.m` — SOLVE: the fixed-mesh assembly of Section 3 on the current mesh.
- `estimate.m` — ESTIMATE: the residual indicators of Theorem 5.1.
- `mark_elements.m` — MARK: the `GR`, `MS` and `Doerfler` strategies.
- `refine_mesh.m`, `refine_element.m` — REFINE: recursive newest-vertex bisection with completion.
- `get_dirichlet_neumann.m` — reconstructs the Dirichlet vertex list and Neumann segments from `elem_boundaries`.
- `gen_mesh_L_shape.m` — generates a uniform L-shape starting mesh at any level in the element-based format, with the newest-vertex labeling that keeps recursive bisection conforming.
- `kellogg_u_exact.m`, `kellogg_grad_u_exact.m` — the Kellogg singular solution and its gradient (Section 3.10), the same two files used by the fixed-mesh repository, called from the `'kellogg'` case of `init_data.m`.
- `L_shape_dirichlet/`, `square_all_dirichlet/`, `square_mixed/`, `big_square_all_dirichlet/` — the four initial-mesh data folders (vertex coordinates, element vertices/neighbours/boundary flags). Five `examples` share these four folders: the `'kellogg'` case reuses `big_square_all_dirichlet/`, the same starting mesh as `'big-square-Dirichlet'`, only with a different coefficient a and boundary data. Apart from Kellogg's, which lives in the two files above, every example's

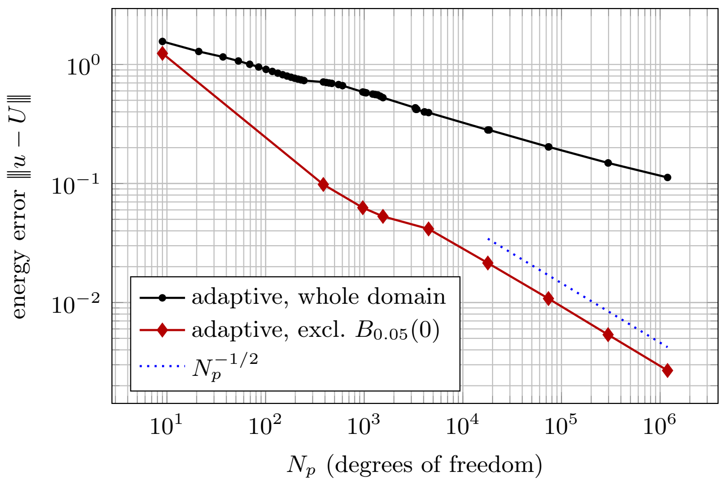


Figure 28: The whole-domain energy error of Figure 26 (solid, standard three-point quadrature, visibly bending away from the optimal slope into the plateau of Table 5) against the corrected values of Table 7 (diamonds: same nine meshes, higher-order quadrature, near-origin elements excluded). The corrected points sit well below the whole-domain curve — excising the contaminated neighbourhood of the origin removes a large, spurious contribution to the error — and, after an initial noisy stretch on the coarsest meshes, settle onto the $N_p^{-1/2}$ reference line (dotted, anchored at $N_p = 17917$) almost exactly, in contrast to the whole-domain curve's visible bend away from it.

exact solution and gradient are defined inline, as anonymous functions, inside its `case` block of `init_data.m`.

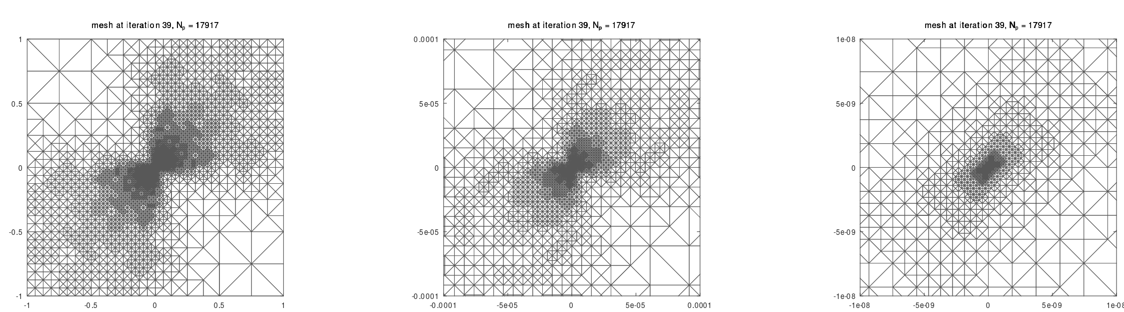


Figure 29: Adaptive mesh produced by `afem.m` on the Kellogg problem, at iteration 39 with 17917 vertices.Full mesh (left), zoom to $[-10^{-4}, 10^{4}]^2$ (middle) and zoom to $[-10^{-8}, 10^{8}]^2$ (right). The concentration of refinement here is much stronger than the one observed in Fig. 25.

7 Basic Properties of AFEM

Before proving convergence of AFEM, we assemble the structural properties of the adaptive loop

$$\textsc{Solve} \to \textsc{Estimate} \to \textsc{Mark} \to \textsc{Refine}$$

on which the contraction argument of Section 8 rests. Two of them — the *localized upper bound* and the *estimator reduction* — quantify how a single step of AFEM changes the discrete solution and the estimator. We discuss them below.

7.1 The adaptive loop

The loop of Section 6 produces a sequence of nested conforming meshes

$$\mathscr{T}_0 \le \mathscr{T}_1 \le \mathscr{T}_2 \le \cdots, \qquad \mathbb{V}_k := \mathbb{V}_{\mathscr{T}_k} \subset \mathbb{V}_{\mathscr{T}_{k+1}} =: \mathbb{V}_{k+1},$$

with Galerkin solutions $U_k \in \mathbb{V}_k$, element indicators $\mathscr{E}_k(U_k, T)$, and marked sets $\mathscr{M}_k \subset \mathscr{T}_k$ selected by Dörfler marking (47). Nestedness is the structural consequence of refinement by bisection: no element is ever coarsened, so $\mathbb{V}_k \subset \mathbb{V}_{k+1}$ and the discrete spaces form an increasing hierarchy. Throughout the analysis $\|v\|_\Omega = \mathscr{B}[v,v]^{1/2}$ denotes the energy norm (equivalent to $|\cdot|_{H^1_0(\Omega)}$), and we abbreviate

$$e_k = \|u - U_k\|_\Omega, \qquad E_k = \|U_{k+1} - U_k\|_\Omega, \qquad \mathscr{E}_k = \mathscr{E}_k(U_k, \mathscr{T}_k), \qquad \mathrm{osc}_k = \mathrm{osc}_k(U_k).$$

The single-step increment $U_{k+1} - U_k \in \mathbb{V}_{k+1}$ plays a central role below.

The discrete neighborhoods of an element T and a side s depend on which triangulation we are considering them. Therefore, from now on, for an element $T \in \mathscr{T}_k$ and a side s of $\mathscr{T}_k$, we will write $\omega_k(T)$, (resp. $\omega_k(s)$) to denote the set of stars of $\mathscr{T}_k$ which contain T (resp. s).

7.2 Building blocks

We collect the structural properties established in Section 5, specialized to the nested meshes $\mathscr{T}_k \le \mathscr{T}_{k+1}$:

1. **Upper bound** (reliability, Thm. 5.1):
$$e_k^2 \le C_1\, \mathscr{E}_k^2. \tag{48}$$
2. **Lower bound** (efficiency, Thm. 5.5):
$$C_2\, \mathscr{E}_k^2 \le e_k^2 + \mathrm{osc}_k^2\,. \tag{49}$$
3. **Localized upper bound** (Lemma 7.1):
$$E_k^2 \le C_1\, \mathscr{E}_k^2(U_k, \mathscr{R}_k^+), \tag{50}$$
where $\mathscr{R}_k^+$ is the discrete $\mathscr{T}_k$-neighborhood of the refined set $\mathscr{R}_k = \mathscr{T}_k \setminus \mathscr{T}_{k+1}$.
4. **Discrete lower bound** (Rem. 5.7):
$$C_2\, \mathscr{E}_k^2(U_k, T) \le \|U_{k+1} - U_k\|^2_{\omega_k(T)} + \mathrm{osc}_k^2(U_k, \omega_k(T)), \tag{51}$$
provided $\textsc{Refine}$ adds an interior node to each marked $T \in \mathscr{M}_k$.

5. **Pythagoras orthogonality:**

$$e_k^2 = \|u - U_{k+1}\|_\Omega^2 + \|U_{k+1} - U_k\|_\Omega^2 = e_{k+1}^2 + E_k^2, \tag{52}$$

because $\mathscr{B}[u - U_{k+1}, U_{k+1} - U_k] = 0$ for $U_{k+1} - U_k \in \mathbb{V}_{k+1}$.

Properties 1, 2, 4 come from Section 5 and property 5 is Galerkin orthogonality in the energy inner product; property 3 is proved next, followed by the estimator reduction that the general contraction proof requires.

7.3 Localized upper bound

Even though we cannot expect local upper bounds between the continuous and discrete solution, the following crucial result shows that this is not the case between discrete solutions on nested meshes $\mathscr{T}_{k+1} \geq \mathscr{T}_k$: what matters is the set of elements of $\mathscr{T}_k$ which are no longer in $\mathscr{T}_{k+1}$, i.e., those that have been refined.

Lemma 7.1 (localized upper bound)**.** *Let $\mathscr{T}_{k+1} \geq \mathscr{T}_k$ with refined set $\mathscr{R}_k = \mathscr{T}_k \setminus \mathscr{T}_{k+1}$, and let $U_k \in \mathbb{V}_k$, $U_{k+1} \in \mathbb{V}_{k+1}$ be the Galerkin solutions. If $\mathscr{R}_k^+$ denotes the discrete $\mathscr{T}_k$-neighborhood of $\mathscr{R}_k$*

$$\mathscr{R}_k^+ = \{T \in \mathscr{T}_k : T \subset \omega_{\mathscr{T}_k}(T') \text{ for some } T' \in \mathscr{R}_k\},$$

then

$$E_k = \|U_{k+1} - U_k\|_\Omega \;\leq\; \sqrt{C_1}\, \mathscr{E}_k(U_k, \mathscr{R}_k^+), \qquad \mathscr{E}_k(U_k, \mathscr{R}_k^+) = \Big(\sum_{T \in \mathscr{R}_k^+} \mathscr{E}_k^2(U_k, T)\Big)^{1/2},$$

with C_1 the reliability constant of (40).

Proof. Set $V = U_{k+1} - U_k \in \mathbb{V}_{k+1}$ and let $I_k V \in \mathbb{V}_k$ be the quasi-interpolant of V onto the coarse space. Galerkin orthogonality on $\mathscr{T}_{k+1}$ gives $\mathscr{B}[U_{k+1}, V] = \langle f, V\rangle$, and on $\mathscr{T}_k$ gives $\mathscr{B}[U_k, I_k V] = \langle f, I_k V\rangle$, so with the residual $\langle R_k, w\rangle = \langle f, w\rangle - \mathscr{B}[U_k, w]$,

$$\|U_{k+1} - U_k\|_\Omega^2 = \mathscr{B}[U_{k+1} - U_k, V] = \langle f, V\rangle - \mathscr{B}[U_k, V] = \langle R_k, V\rangle = \langle R_k, V - I_k V\rangle\,,$$

the last equality by $\langle R_k, I_k V\rangle = 0$. Now $I_k V$ reproduces V on every element whose discrete neighborhood is not touched by refinement, so $V - I_k V$ is supported in $\cup_{T \in R_k^+} T$. Repeating the residual estimate of Theorem 5.1, but summing only over $\mathscr{R}_k^+$,

$$\langle R_k, V - I_k V\rangle \;\leq\; C_1\, \mathscr{E}_k(U_k, \mathscr{R}_k^+)\, \|V\|_\Omega.$$

Dividing by $\|V\|_\Omega$ yields the claim. □

7.4 Estimator reduction

The second key property tracks the estimator across a refinement step. Recall from Section 5 that the estimator is

$$\mathscr{E}_{\mathscr{T}}(U) = \Big(\sum_{T \in \mathscr{T}} \mathscr{E}_{\mathscr{T}}^2(U, T)\Big)^{1/2},$$

where the local *element indicator* is given by

$$\mathscr{E}_{\mathscr{T}}^2(U, T) = h_T^2\, \|r\|_{L^2(T)}^2 + h_T\, \|j\|_{L^2(\partial T)}^2\,,$$

with

$$r|_T := f + \operatorname{div}(A\nabla U)\big|_T, \qquad \text{and} \qquad j|_s = [\![\, A\nabla U \cdot \boldsymbol{n} \,]\!]_s.$$

The estimator reduction combines two effects: newly created elements are *smaller*, which shrinks the factor containing h in the indicators, while the change of discrete solution from U_k to U_{k+1} *perturbs* the indicators in a controlled, Lipschitz manner.

We focus first on the Lipschitz continuity of the estimator with respect to the discrete function.

Lemma 7.2 (Lipschitz dependence on the discrete argument). *Let $W_1, W_2 \in \mathbb{V}_{\mathscr{T}}$, then, for each $T \in \mathscr{T}$,*

$$\Big|\mathscr{E}_{\mathscr{T}}(W_1, T) - \mathscr{E}_{\mathscr{T}}(W_2, T)\Big| \;\lesssim\; \|\nabla(W_1 - W_2)\|_{L^2(\omega_T)}\,,$$

and consequently, there exists a constant C_0, such that,

$$\big|\mathscr{E}_{\mathscr{T}}(W_1) - \mathscr{E}_{\mathscr{T}}(W_2)\big| \;\le\; \sqrt{C_0}\,\|\!|W_1 - W_2|\!\|_\Omega. \tag{53}$$

In the proof of this lemma we will use the so-called *inverse estimates*, which we now state precisely.

Lemma 7.3 (Inverse estimate). *Let $0 \le j \le m$ and $\mathscr{T}$ be shape-regular. Then*

$$|v|_{W_q^m(K)} \le C\, h_K^{\operatorname{sob}(W_p^j) - \operatorname{sob}(W_q^m)}\, |v|_{W_p^j(K)}$$

for all $v \in \mathscr{P}_k(K)$ and all $K \in \mathscr{T}$.

Proof. 1. By scaling it suffices to prove $|\hat v|_{W_q^m(\widehat T)} \le C\, |\hat v|_{W_p^j(\widehat T)}$ for all $\hat v \in \mathscr{P}_k(\widehat T)$. *2.* Use Kolmogorov's theorem: all norms are equivalent in finite-dimensional spaces. The key is to show that if $|\hat v|_{W_p^j(\widehat T)} = 0$ then $|\hat v|_{W_q^m(\widehat T)} = 0$. The details of the proof are left as an exercise. $\square$

Proof. Let $W_1, W_2 \in \mathbb{V}_{\mathscr{T}}(\mathscr{T})$, and $T \in \mathscr{T}$. For the interior residual we proceed as follows:

$$\begin{aligned}\Big|h_T\,\|f + \operatorname{div}(A\nabla W_1)\|_{L^2(T)} - h_T\,\|f + \operatorname{div}(A\nabla W_2)\|_{L^2(T)}\Big| &\le h_T\,\|\operatorname{div}(A\nabla W_1) - \operatorname{div}(A\nabla W_2)\|_{L^2(T)}\\ &= h_T\,\|\operatorname{div}(A\nabla(W_1 - W_2))\|_{L^2(T)}\,.\end{aligned}$$

Now, using inverse estimates, for $W = W_1 - W_2$, we bound

$$\begin{aligned}h_T\,\|\operatorname{div}(A\nabla(W))\|_{L^2(T)} &\le h_T\,\|\operatorname{div}(A)\|_{L^\infty(T)}\,\|\nabla W\|_{L^2(T)} + h_T\,\|A\|_{L^\infty(T)}\,\big\|D^2 W\big\|_{L^2(T)}\\ &\lesssim h_T\,\|\nabla W\|_{L^2(T)} + h_T h_T^{-1}\,\|\nabla W\|_{L^2(T)} \lesssim \|\nabla W\|_{L^2(T)}\,.\end{aligned}$$

For the jump residual, the bound is analogous. On a side s,

$$\begin{aligned}\Big|h_T^{1/2}\,\|[\![\, A\nabla W_1\cdot\boldsymbol{n}\,]\!]\|_{L^2(s)} - h_T^{1/2}\,\|[\![\, A\nabla W_2\cdot\boldsymbol{n}\,]\!]\|_{L^2(s)}\Big| &\le h_T^{1/2}\,\|[\![\, A\nabla W\cdot\boldsymbol{n}\,]\!]\|_{L^2(s)}\\ &\le h_T^{1/2}\,\|A\|_{L^\infty(\Omega)}\Big(\big\|\nabla W^+\big\|_{L^2(s)} + \big\|\nabla W^-\big\|_{L^2(s)}\Big)\,,\end{aligned}$$

where W^+, W^- represent the restriction of W to the elements T^+, T^- at each side of s. By the (scaled) trace theorem,

$$\big\|\nabla W^\pm\big\|_{L^2(s)} \lesssim h_s^{-1/2}\,\big\|\nabla W^\pm\big\|_{L^2(T)} + h_s^{1/2}\,\big\|D^2 W^\pm\big\|_{L^2(T)} \lesssim h_s^{-1/2}\,\big\|\nabla W^\pm\big\|_{L^2(T)}\,.$$

Combining we get

$$\left| h_T^{1/2} \, \|[\![A\nabla W_1 \cdot \boldsymbol{n}]\!]\|_{L^2(s)} - h_T^{1/2} \, \|[\![A\nabla W_2 \cdot \boldsymbol{n}]\!]\|_{L^2(s)} \right| \; \lesssim \; \|\nabla W\|_{L^2(\omega_T)} \,,$$

which, together with the bounds for the interior residual, leads to

$$\left| \mathscr{E}_{\mathscr{T}}(W_1, T) - \mathscr{E}_{\mathscr{T}}(W_2, T) \right| \; \lesssim \; \|\nabla(W_1 - W_2)\|_{L^2(\omega_T)} \,.$$

After adding over all the elements of the mesh, using the fact that each element T belongs to at most $d+2$ patches $\omega_{T'}$ (finite overlapping), we obtain

$$\left| \mathscr{E}_{k+1}(W_1) - \mathscr{E}_{k+1}(W_2) \right| \; \le \; \sqrt{C_0} \, \| W_1 - W_2 \|_\Omega. \qquad \square$$

Proposition 7.4 (estimator reduction)**.** *Let $\mathscr{T}_{k+1} \ge \mathscr{T}_k$ be obtained by bisecting (at least once) every element of the marked set $\mathscr{M}_k$, and set $\lambda := 1 - 2^{-1/d} \in (0,1)$. There is a constant $C_0 > 0$, depending only on shape regularity and the data, such that for every $\delta > 0$,*

$$\mathscr{E}_{k+1}^2(U_{k+1}) \; \le \; (1+\delta) \Big[\mathscr{E}_k^2(U_k) - \lambda \, \mathscr{E}_k^2(U_k, \mathscr{M}_k) \Big] + (1 + \delta^{-1}) \, C_0 \, E_k^2. \tag{54}$$

Proof. Step 1 (reduction for a fixed function). Fix $V \in \mathbb{V}_k \subset \mathbb{V}_{k+1}$. When an element $T \in \mathscr{T}_k$ is bisected, each child $T' \subset T$ has $|T'| = |T|/2$ and $h_{T'} = 2^{-1/d} h_T$ (bisection halves the measure and $h_T = |T|^{1/d}$), and every descendant $T' \subset T$ obtained by one or more bisections satisfies $h_{T'} \le 2^{-1/d} h_T$, whence

$$\sum_{T' \in \mathscr{T}_{k+1}, \, T' \subset T} h_{T'}^2 \, \| f + \operatorname{div}(A\nabla V) \|_{L^2(T')}^2 \le 2^{-2/d} h_T^2 \, \| f + \operatorname{div}(A\nabla V) \|_{L^2(T)}^2 \,.$$

Since V is polynomial on T, it stays polynomial on each descendant T' of T, and the newly created interior edges carry *no* flux jump, so that

$$\sum_{T' \in \mathscr{T}_{k+1}, \, T' \subset T} h_{T'} \, \|[\![A\nabla V \cdot \boldsymbol{n}]\!]\|_{L^2(\partial T')}^2 \le 2^{-1/d} h_T \, \|[\![A\nabla V \cdot \boldsymbol{n}]\!]\|_{L^2(\partial T)}^2 \,.$$

Therefore

$$\sum_{T' \in \mathscr{T}_{k+1}, \, T' \subset T} \mathscr{E}_{k+1}^2(V, T') \; \le \; 2^{-1/d} \, \mathscr{E}_k^2(V, T).$$

Unrefined elements keep their indicator. Summing over $\mathscr{T}_k$ and isolating the marked elements (all of which are refined),

$$\begin{aligned}
\mathscr{E}_{k+1}^2(V) &= \sum_{T \in \mathscr{T}_{k+1}} \mathscr{E}_{k+1}^2(V, T) = \sum_{T \in \mathscr{T}_{k+1} \cap \mathscr{T}_k} \mathscr{E}_{k+1}^2(V, T) + \sum_{T \in \mathscr{T}_{k+1} \setminus \mathscr{T}_k} \mathscr{E}_{k+1}^2(V, T) \\
&\le \sum_{T \in \mathscr{T}_k \cap \mathscr{T}_{k+1}} \mathscr{E}_k^2(V, T) + \sum_{T \in \mathscr{T}_k \setminus \mathscr{T}_{k+1}} \mathscr{E}_k^2(V, T) \\
&\quad - \Big(\sum_{T \in \mathscr{T}_k \setminus \mathscr{T}_{k+1}} \mathscr{E}_k^2(V, T) - \sum_{T \in \mathscr{T}_{k+1} \setminus \mathscr{T}_k} \mathscr{E}_{k+1}^2(V, T) \Big) \\
&= \sum_{T \in \mathscr{T}_k} \mathscr{E}_k^2(V, T) - \Big(\sum_{T \in \mathscr{T}_k \setminus \mathscr{T}_{k+1}} \mathscr{E}_k^2(V, T) - \sum_{T \in \mathscr{T}_k \setminus \mathscr{T}_{k+1}} \sum_{\substack{T' \in \mathscr{T}_{k+1} \\ T' \subset T}} \mathscr{E}_{k+1}^2(V, T') \Big) \\
&\le \sum_{T \in \mathscr{T}_k} \mathscr{E}_k^2(V, T) - \Big(\sum_{T \in \mathscr{T}_k \setminus \mathscr{T}_{k+1}} \mathscr{E}_k^2(V, T) - 2^{-1/d} \sum_{T \in \mathscr{T}_k \setminus \mathscr{T}_{k+1}} \mathscr{E}_k^2(V, T) \Big) \\
&= \sum_{T \in \mathscr{T}_k} \mathscr{E}_k^2(V, T) - (1 - 2^{-1/d}) \sum_{T \in \mathscr{T}_k \setminus \mathscr{T}_{k+1}} \mathscr{E}_k^2(V, T).
\end{aligned}$$

Finally, since all the marked elements in $\mathscr{M}_k$ were refined, $\mathscr{M}_k \subset \mathscr{T}_k \setminus \mathscr{T}_{k+1}$ and

$$\mathscr{E}_k^2(V, \mathscr{M}_k) = \sum_{T \in \mathscr{M}_k} \mathscr{E}_k^2(V, T) \le \sum_{T \in \mathscr{T}_k \setminus \mathscr{T}_{k+1}} \mathscr{E}_k^2(V, T),$$

which implies that

$$\mathscr{E}_{k+1}^2(V) \le \mathscr{E}_k^2(V) - \lambda \mathscr{E}_k^2(V, \mathscr{M}_k). \tag{55}$$

Step 2 (combine with the Lipschitz continuity of the estimator). We recall Young's inequality

$$2ab \le \delta\, a^2 + \delta^{-1}\, b^2, \qquad \forall a, b \in \mathbb{R},\ \delta > 0,$$

which implies that

$$(a+b)^2 \le (1+\delta)\, a^2 + (1+\delta^{-1})\, b^2, \qquad \forall a, b \in \mathbb{R},\ \delta > 0.$$

Apply now (53) with $W_1 = U_{k+1}$, $W_2 = U_k$, on $\mathscr{T} = \mathscr{T}_{k+1}$, and (55) with $V = U_k$:

$$\begin{aligned}
\mathscr{E}_{k+1}^2(U_{k+1}) &= \big(\mathscr{E}_{k+1}(U_k) + \mathscr{E}_{k+1}(U_{k+1}) - \mathscr{E}_{k+1}(U_k)\big)^2 \\
&\le \big(\mathscr{E}_{k+1}(U_k) + \sqrt{C_0}\, E_k\big)^2 \\
&\le (1+\delta)\, \mathscr{E}_{k+1}^2(U_k) + (1+\delta^{-1})\, C_0\, E_k^2 \\
&\le (1+\delta) \Big[\mathscr{E}_k^2(U_k) - \lambda\, \mathscr{E}_k^2(U_k, \mathscr{M}_k)\Big] + (1+\delta^{-1})\, C_0\, E_k^2. \qquad \square
\end{aligned}$$

Remark 7.5. As we will see in the next section, estimator reduction is the engine of the *general* contraction proof.

8 Contraction Property

We now prove that the adaptive loop *contracts*: each step reduces a suitable error quantity by a fixed factor $\alpha < 1$, so $U_k \to u$ at a linear rate. We give the argument twice — first in the insightful special case of vanishing oscillation and using the interior node property, and later in full generality.

8.1 Piecewise Constant Data: No Oscillation

When the data are piecewise constant on the initial mesh — $f \in \mathscr{P}_0(\mathscr{T}_0)$ and $A \in \mathscr{P}_0(\mathscr{T}_0)$, so that the interior residual $r_T = f|_T$ is elementwise constant and the jump residual $j_S = [\![A\nabla U \cdot \boldsymbol{n}]\!]_S$ is edgewise constant — the oscillation $\mathrm{osc}_k(U_k)$ *vanishes* on every refinement $\mathscr{T}_k$ of $\mathscr{T}_0$. The estimator is then equivalent to the energy error with no correction, and one can contract e_k *directly*, using the interior node property to convert Dörfler marking into an energy decrement.

Theorem 8.1 (contraction for piecewise constant data)**.** *Let us assume that* $\mathrm{osc}_k(U_k) = 0$ *and* $\mathscr{M}_k$ *satisfies the Dörfler criterion*

$$\mathscr{E}_k(U_k, \mathscr{M}_k) \geq \theta\, \mathscr{E}_k(U_k, \mathscr{T}_k), \tag{56}$$

for some $0 < \theta < 1$*. If every element* $T \in \mathscr{M}_k$ *has a node of* $\mathscr{T}_{k+1}$ *in its interior, then*

$$e_{k+1} \leq \alpha\, e_k, \qquad \text{with} \quad \alpha = \Big(1 - \theta^2 \frac{C_2}{C_1}\Big)^{1/2} < 1,$$

where $C_1 \geq C_2$ *are the constants in* (48) *and* (49)*, respectively.*

Proof. According to (52), $e_{k+1}^2 = e_k^2 - E_k^2$, it suffices to bound E_k^2 from below. Thanks to the interior node property, (51) holds, whence Dörfler marking (56), and the upper bound (48), in this order, yield

$$E_k^2 \;\geq\; C_2\, \mathscr{E}_k^2(U_k, \mathscr{M}_k) \;\geq\; \theta^2 C_2\, \mathscr{E}_k^2 \;\geq\; \theta^2 \frac{C_2}{C_1}\, e_k^2.$$

Hence $e_{k+1}^2 \leq \big(1 - \theta^2 C_2/C_1\big) e_k^2$. Since $C_2 \leq C_1$ and $0 < \theta < 1$, the factor $\alpha^2 \in (0,1)$. □

Iterating gives *linear convergence*: $e_k \leq \alpha^k e_0 \to 0$. Two features of this argument are restrictive: it assumes $\mathrm{osc}_k(U_k) = 0$, and it relies on the interior node property that guarantees (51) (hence on REFINE placing a node inside each marked element of $\mathscr{M}_k$). The general case of §8.2 removes both assumptions.

Remark 8.2 (role of interior node property). Before proceeding to the proof of the general case, we show that the interior node property (51) is in fact necessary to guarantee an error reduction in one single step. To this end, consider the counterexample of [16]: $A = I$, $f = 1$, $\Omega = (0,1)^2$ with the criss-cross initial mesh $\mathscr{T}_0$. Even two bisections, which make h_T reduce by a factor two, adds *no* interior node does not change the Galerkin solution:

$$U_1 = U_0 = \tfrac{1}{12}\phi_0,$$

with $\phi_0 \in \mathbb{V}_0 \subset \mathbb{V}_1$ the nodal basis function that equals one at $(\frac{1}{2}, \frac{1}{2})$ and vanishes at the boundary of the domain. Then $E_0 = |\!|\!| U_1 - U_0 |\!|\!|_\Omega = 0$ and $e_0 = e_1$: no contraction occurs! Only the further refinement $\mathscr{T}_2$, which introduces interior nodes, yields $U_2 \neq U_1$. The interior node property is what guarantees that marked elements actually contribute to the energy decrement.

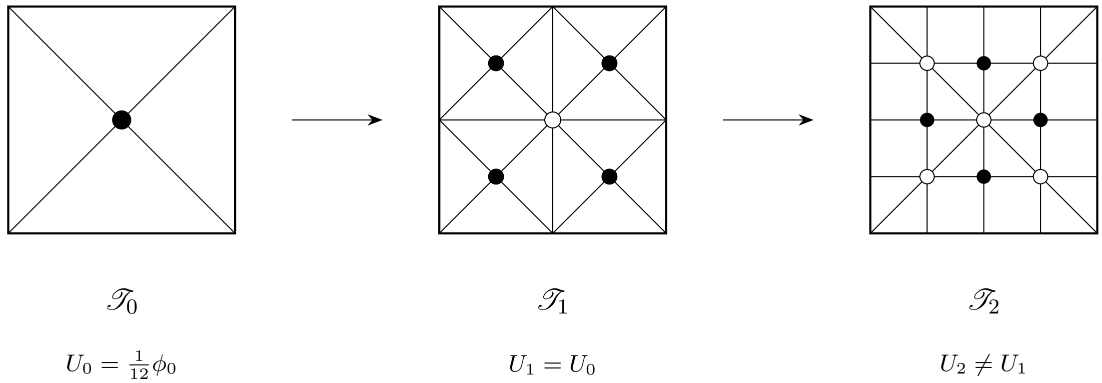


Figure 30: Counterexample of [16] illustrating necessity of the interior node property. The data reads $A = I$, $f = 1$, $\Omega = (0,1)^2$. Going from $\mathscr{T}_0$ to $\mathscr{T}_1$ (with two bisections on each element) adds midpoints on edges and diagonals but *no interior node*; the Galerkin solution is unchanged ($U_1 = U_0$) and the energy does not decrease. Only $\mathscr{T}_2$ introduces interior nodes on the original elements of $\mathscr{T}_0$, giving $U_2 \neq U_1$.

8.2 General Data

For general data the oscillation no longer vanishes and the interior node property seems like an artifact to be able to prove convergence, so the energy error need not contract on its own. The counterexample of Figure 30 explains why: there $U_{k+1} = U_k$, hence $e_{k+1} = e_k$ and the energy error *stalls*. Yet in exactly this situation Lemma 7.4 (estimator reduction) shows that the estimator decreases: with $U_{k+1} = U_k$ (so $E_k = 0$) and $\mathscr{E}_k^2(U_k, \mathscr{M}_k) \geq \theta \mathscr{E}_k^2(U_k)$, (54) reads

$$\mathscr{E}_{k+1}^2(U_{k+1}) \ \leq \ (1+\delta)(1-\lambda\theta^2)\mathscr{E}_k^2(U_k),$$

whence $\mathscr{E}_{k+1}(U_{k+1})$ *strictly* decreases provided $\delta > 0$ is sufficiently small. Neither quantity e_k or $\mathscr{E}_k(U_k)$ contracts on its own — the energy error can stall while the estimator drops, and the estimator alone is not even monotone because it depends on the changing discrete solution — but a weighted combination of the two, the *quasi-error*

$$\zeta_k^2 := e_k^2 + \gamma\, \mathscr{E}_k^2 \qquad (\gamma > 0 \text{ to be chosen}),$$

does contract (see Theorem 8.3). This interplay between a *continuous* quantity (the energy error e_k) and a *discrete* one (the estimator $\mathscr{E}_k(U_k)$), neither of which suffices alone, is a hallmark of the convergence theory of Cascón, Kreuzer, Nochetto and Siebert [9].

Theorem 8.3 (contraction of the quasi-error)**.** *Let $0 < \theta \leq 1$ and suppose each step of AFEM uses Dörfler marking (47) with parameter θ followed by* Refine *bisecting every marked element of $\mathscr{M}_k$ at least once. Then there are constants $\gamma > 0$ and $\alpha \in (0,1)$, depending only on shape regularity, the data, and θ, such that the quasi-error contracts:*

$$e_{k+1}^2 + \gamma\, \mathscr{E}_{k+1}^2 \ = \ \zeta_{k+1}^2 \ \leq \ \alpha^2\, \zeta_k^2 = \alpha^2 \left(e_k^2 + \gamma\, \mathscr{E}_k^2\right).$$

In particular $e_k \leq \zeta_k \leq \alpha^k \zeta_0$, so AFEM converges linearly for any *right-hand side and coefficient — with or without oscillation, and without the interior node property.*

Proof. We set $E_k = \lVert\!| U_{k+1} - U_k |\!\rVert_\Omega, \mathscr{E}_k = \mathscr{E}_k(U_k)$ and proceed in four steps.

Step 1 (assembling). Pythagoras inequality (52) gives

$$e_{k+1}^2 = e_k^2 - E_k^2,$$

and Lemma 7.4 (estimator reduction) gives, for any $\delta > 0$,

$$\mathscr{E}_{k+1}^2 \;\leq\; (1+\delta)\big(\mathscr{E}_k^2 - \lambda\,\mathscr{E}_k^2(U_k, \mathscr{M}_k)\big) + (1+\delta^{-1})C_0\,E_k^2.$$

Multiplying the second inequality by γ, and adding the first, yields

$$\zeta_{k+1}^2 \;\leq\; e_k^2 + \big(\gamma(1+\delta^{-1})C_0 - 1\big)E_k^2 + \gamma(1+\delta)\big(\mathscr{E}_k^2 - \lambda\,\mathscr{E}_k^2(U_k, \mathscr{M}_k)\big).$$

Step 2 (choosing δ and γ). Fix $\delta > 0$ by

$$(1+\delta)(1-\lambda\theta^2) = 1 - \tfrac{1}{2}\lambda\theta^2, \qquad \text{i.e.} \qquad \delta = \frac{\lambda\theta^2}{2(1-\lambda\theta^2)} > 0,$$

and then $\gamma > 0$ by $\gamma(1+\delta^{-1})C_0 = 1$, which cancels the E_k^2 term exactly, whence

$$\zeta_{k+1}^2 s \;\leq\; e_k^2 + \gamma(1+\delta)\big(\mathscr{E}_k^2 - \lambda\,\mathscr{E}_k^2(U_k, \mathscr{M}_k)\big).$$

Step 3 (Dörfler marking). Since $\mathscr{E}_k^2(U_k, \mathscr{M}_k) \geq \theta^2 \mathscr{E}_k^2$, according to (47), invoking the choice of δ,

$$\begin{aligned}
\zeta_{k+1}^2 \;&\leq\; e_k^2 + \gamma(1+\delta)(1-\lambda\theta^2)\,\mathscr{E}_k^2 \\
&= e_k^2 + \gamma\big(1 - \tfrac{1}{2}\lambda\theta^2\big)\mathscr{E}_k^2 = e_k^2 - \tfrac{1}{4}\gamma\lambda\theta^2\,\mathscr{E}_k^2 + \gamma\big(1 - \tfrac{1}{4}\lambda\theta^2\big)\mathscr{E}_k^2.
\end{aligned}$$

Step 4 (reliability). Applying the upper bound (48), namely $e_k^2 \leq C_1 \mathscr{E}_k^2$, results in

$$\zeta_{k+1}^2 \;\leq\; \Big(1 - \frac{\gamma\lambda\theta^2}{4C_1}\Big)e_k^2 + \Big(1 - \frac{\lambda\theta^2}{4}\Big)\gamma\,\mathscr{E}_k^2 \;\leq\; \alpha^2\,\zeta_k^2,$$

with contraction parameter

$$\alpha^2 := \max\Big\{1 - \frac{\gamma\lambda\theta^2}{4C_1},\, 1 - \frac{\lambda\theta^2}{4}\Big\} < 1.$$

This concludes the proof. □

Remark 8.4 (ingredients of the proofs). The simple case of vanishing oscillation contracts e_k itself but needs the interior node property and $\mathrm{osc}_k(U_k) = 0$. In contrast, the general case contracts the quasi-error ζ_k and needs neither. The general proof rests on exactly four ingredients: Dörfler marking (47), the Pythagoras identity (52), the a posteriori *upper* bound (48) (reliability), and the estimator reduction (54) of Lemma 7.4. Notably it does *not* use the lower bound (49) (efficiency): that property re-enters only in the convergence rate analysis of Section 9.

Remark 8.5 (quasi-error vs. total error). Reliability and efficiency together give the *total error*

$$\mathscr{E}_k^2 \simeq e_k^2 + \mathrm{osc}_k^2\,. \tag{57}$$

We stress that the quasi-error $\zeta_k^2 = e_k^2 + \gamma\,\mathscr{E}_k^2$ is equivalent to the total error, and that the latter is the quantity that governs the optimal decay *rates* of Section 9. Hence, efficiency, though unused in the contraction property, is exactly what ties the contracted quantity to the approximation classes. Note also that MARK is driven by the estimator $\mathscr{E}_k$ *alone*: a single Dörfler marking suffices for a contraction property between consecutive steps.

9 Convergence Rates

We now complement the convergence analysis of Section 8 with optimal convergence rates: the error decay matches the rate of the best possible mesh with the same number of degrees of freedom. As with the contraction property, we proceed in two stages: first, we examine piecewise-constant data, where oscillation vanishes and the energy error alone measures the rate; second, we study general data, where the natural quantity is the *total error*. We make the following assumptions:

Assumption 9.1 (marking parameter)**.** There exists a parameter $0 < \theta_* \leq 1$ such that the Dörfler marking parameter satisfies $\theta \in (0, \theta_*)$.

Assumption 9.2 (minimal cardinality)**.** MARK selects a set $\mathscr{M}$ with *minimal* cardinality.

Assumption 9.3 (initial labeling)**.** The labeling of the initial mesh $\mathscr{T}_0$ satisfies (37) for $d = 2$ or its multidimensional counterpart for $d > 2$ [20, 17].

A few comments about these assumptions are now in order.

Remark 9.4 (threshold $\theta_* < 1$)*.* It is reasonable to be cautious in making marking decisions if the constants C_1 and C_2 in (48) and (49) are very disparate, and thus the ratio C_2/C_1 is far from 1. This justifies the upper bound $\theta_* < 1$ in Assumption 9.1.

Remark 9.5 (minimal $\mathscr{M}$)*.* According to the error equidistribution principle (34) and the local lower bound $C_2 \mathscr{E}_{\mathscr{T}}(U, T) \lesssim \|u - U\|_{\omega_T}$ without oscillation (42), it is natural to mark elements with largest error indicators $\mathscr{E}_{\mathscr{T}}(U, T)$. This leads to a minimal set $\mathscr{M}$, as stated in Assumption 9.2, and turns out to be crucial to link AFEM with optimal meshes.

Remark 9.6 (initial mesh $\mathscr{T}_0$)*.* Assumption 9.3 guarantees the validity of Theorem 4.6 (complexity of bisection). This means that REFINE produces conforming refinements $\mathscr{T}_k \geq \mathscr{T}_0$ such that

$$\#\mathscr{T}_k - \#\mathscr{T}_0 \leq \Lambda_0 \sum_{j=0}^{k-1} \#\mathscr{M}_j.$$

9.1 Quasi-Optimal Cardinality: Vanishing Oscillation

Throughout this subsection f and A are piecewise constant on $\mathscr{T}_0$, so $\operatorname{osc}_k(U_k) \equiv 0$ on every refinement $\mathscr{T}_k$ of $\mathscr{T}_0$ and the estimator is two-sided equivalent to the energy error

$$C_2 \, \mathscr{E}_k^2 \leq e_k^2 \leq C_1 \, \mathscr{E}_k^2,$$

according to (48) and (49). The achievable rate is then measured purely by the energy error.

Approximation classes. Let $\mathbb{T}_N$ be the set of *all* conforming refinements of $\mathscr{T}_0$ (by bisection) with at most N elements more than $\mathscr{T}_0$, and define the best possible element error with N elements

$$\sigma_N(u) = \inf_{\mathscr{T} \in \mathbb{T}_N} \inf_{V \in \mathbb{V}_{\mathscr{T}}} \|u - V\|_\Omega. \tag{58}$$

We say $u \in \mathscr{A}_s$ if

$$|u|_s := \sup_{N \in \mathbb{N}} N^s \, \sigma_N(u) < \infty, \qquad \text{i.e.} \qquad \sigma_N(u) \leq |u|_s \, N^{-s}. \tag{59}$$

Thus $\mathscr{A}_s$ collects all functions that can be approximated with rate N^{-s}. Finding a mesh $\mathscr{T} \in \mathbb{T}_N$ that delivers such a rate, or close to it, has in general *exponential* complexity in terms of N. The purpose of this section is to show that AFEM is able to learn the underlying regularity of u, namely s, from the Galerkin solution U via a posteriori error estimates and construct a quasi-optimal mesh with *linear* complexity. Membership in $\mathscr{A}_s$ is all that is needed for the subsequent development, but it is worth realizing that this concept is related to Besov regularity and that a characterization of $\mathscr{A}_s$ is available in [2, 3].

Lemma 9.7 (membership in $\mathscr{A}_{1/2}$ for piecewise constant (A, f))**.** *Let $d = 2$, and $u \in H_0^1(\Omega) \cap W_p^2(\Omega)$ for $p > 1$ satisfy $-\operatorname{div}(A\nabla u) = f$ on a polygon Ω (including reentrant corners). If Assumption 9.3 (initial labeling) is enforced, then one has $u \in \mathscr{A}_{1/2}$ with*

$$|u|_{1/2} \lesssim |u|_{W_p^2(\Omega)}.$$

Proof. Apply Theorem 4.9 (optimal greedy approximation) and definitions (58) and (59) of $\mathscr{A}_s$. $\square$

The key point of this section is that AFEM *realizes* the convergence rate $N^{-1/2}$ *without* explicit knowledge of the singularity, or equivalently that $s = 1/2$.

Exercise 9.8. Show that $u \in \mathscr{A}_s$ if and only if for every $\varepsilon > 0$ there is a conforming refinement $\mathscr{T}_\varepsilon \geq \mathscr{T}_0$ and $V_\varepsilon \in \mathscr{V}_{\mathscr{T}_\varepsilon}$ with $\|u - V_\varepsilon\|_\Omega \leq \varepsilon$ and $\#\mathscr{T}_\varepsilon - \#\mathscr{T}_0 \lesssim |u|_s^{1/s}\, \varepsilon^{-1/s}$.

Optimal marking. A crucial insight for the simplest scenario, the Laplacian and piecewise constant forcing f, is due to Stevenson [19]:

any marking strategy that reduces the energy error relative to the current value must contain a substantial portion of $\mathscr{E}_{\mathscr{T}}(U)$, and so it can be related to Dörfler Marking. (60)

This allows one to compare meshes produced by AFEM with optimal ones and to conclude a quasi-optimal error decay. We discuss this now, but it is not enough to handle the model problem with variable data A and f.

Lemma 9.9 (Dörfler marking: vanishing oscillation)**.** *Let AFEM satisfy Assumption 9.1 (marking parameter) with $\theta_* = \sqrt{C_2/C_1}$ and Assumption 9.3 (initial labeling), and let $\mu := 1 - \theta^2/\theta_*^2 > 0$. If a conforming refinement $\mathscr{T}_* \geq \mathscr{T}_k$ with Galerkin solution U_* satisfies*

$$\|u - U_*\|_\Omega^2 \leq \mu\, \|u - U_k\|_\Omega^2,$$

then the refined set $\mathscr{R} = \mathscr{T}_k \setminus \mathscr{T}_*$ satisfies the Dörfler property $\mathscr{E}_k(U_k, \mathscr{R}) \geq \theta\, \mathscr{E}_k(U_k, \mathscr{T}_k)$.

Proof. Using lower bound (49), the energy reduction hypothesis, and Pythagoras equality (52), yields

$$(1-\mu) C_2\, \mathscr{E}_k^2 \leq (1-\mu)\, e_k^2 = e_k^2 - \mu\, e_k^2 \leq \|u - U_k\|_\Omega^2 - \|u - U_*\|_\Omega^2 = \|U_k - U_*\|_\Omega^2.$$

Applying now the localized upper bound (50) implies

$$(1-\mu) C_2\, \mathscr{E}_k^2 \leq \|U_k - U_*\|_\Omega^2 \leq C_1\, \mathscr{E}_k^2(U_k, \mathscr{R}),$$

whence

$$(1-\mu)\tfrac{C_2}{C_1} \mathscr{E}_k^2 \leq \mathscr{E}_k^2(U_k, \mathscr{R}) \quad \Rightarrow \quad \mathscr{E}_k(U_k, \mathscr{R}) \geq \theta\, \mathscr{E}_k,$$

with $\theta^2 = (1-\mu) C_2/C_1 < \theta_*^2$. This concludes the proof. $\square$

Cardinality of the marked set. To examine the cardinality of $\mathscr{M}_k$ in terms of $\|u - U_k\|_\Omega$ we must relate AFEM with the approximation class $\mathscr{A}_s$. Even though this might appear like an undoable task, the key to unravel this connection is given by Lemma 9.9. Before we embark into this topic, we need the concept of *overlay*: given two conforming refinements $\mathscr{T}_1, \mathscr{T}_2$ of $\mathscr{T}_0$, the overlay $\mathscr{T} = \mathscr{T}_1 \oplus \mathscr{T}_2$ is the smallest conforming refinement of $\mathscr{T}_1$ and $\mathscr{T}_2$. The key property of $\mathscr{T}$ is

$$\#\mathscr{T} \le \#\mathscr{T}_1 + \#\mathscr{T}_2 - \#\mathscr{T}_0. \tag{61}$$

Lemma 9.10 (cardinality of $\mathscr{M}_k$: vanishing oscillation)**.** *Let AFEM satisfy Assumption 9.1 (marking parameter) with $\theta_* = \sqrt{C_2/C_1}$ and Assumptions 9.2 (minimal cardinality) and 9.3 (initial labeling). If $u \in \mathscr{A}_s$ with $0 < s \le 1/2$, then for all $k \ge 1$ one has*

$$\#\mathscr{M}_k \ \lesssim \ |u|_s^{1/s} \, \|u - U_k\|_\Omega^{-1/s}.$$

Proof. Apply the Exercise 9.8 with $\varepsilon^2 = \mu\, e_k^2$ to get a mesh $\mathscr{T}_\varepsilon$ achieving $\|u - U_\varepsilon\|_\Omega \le \varepsilon$ with $\#\mathscr{T}_\varepsilon - \#\mathscr{T}_0 \lesssim |u|_s^{1/s}\, \varepsilon^{-1/s}$. Let $\mathscr{T}_* = \mathscr{T}_k \oplus \mathscr{T}_\varepsilon$ be the overlay of $\mathscr{T}_k$ and $\mathscr{T}_\varepsilon$, which satisfies

$$\#\mathscr{T}_* \le \#\mathscr{T}_k + \#\mathscr{T}_\varepsilon - \#\mathscr{T}_0,$$

according to (61). If $U_* \in \mathbb{V}_{\mathscr{T}_*}$ is the Galerkin solution over $\mathscr{T}_*$, we have

$$\|u - U_*\|_\Omega^2 \le \|u - U_\varepsilon\|_\Omega^2 \le \varepsilon^2 = \mu e_k^2.$$

Therefore, $\mathscr{T}_* \ge \mathscr{T}_k$ meets the hypothesis of Lemma 9.9, whence its refined set $\mathscr{R}_*$ satisfies the Dörfler property $\mathscr{E}_k(U_k, \mathscr{R}) \ge \theta\, \mathscr{E}_k(U_k, \mathscr{T}_k)$. Exploiting that the cardinality of $\mathscr{M}_k$ is minimal gives

$$\#\mathscr{M}_k \le \#\mathscr{R} = \#\mathscr{T}_* - \#\mathscr{T}_k \le \#\mathscr{T}_\varepsilon - \#\mathscr{T}_0 \lesssim |u|_s^{1/s}\, \varepsilon^{-1/s},$$

which is the desired bound. $\square$

Quasi-optimality. We are now in a position to prove that the energy error $\|u - U_k\|_\Omega$ decays with the optimal rate N^{-s} for u in the approximation class $\mathscr{A}_s$, without a priori knowledge of s. This is an early, yet fundamental and practical, instance of learning from computation.

Theorem 9.11 (quasi-optimal convergence rate for piecewise constant data)**.** *Let $\mathrm{osc}_k(U_k) = 0$, and AFEM satisfy Assumption 9.1 (marking parameter) with threshold*

$$\theta_* = \sqrt{C_2/C_1} \tag{62}$$

and Assumptions 9.2 (minimal cardinality) and 9.3 (initial labeling). If the interior node property (51) is enforced and $u \in \mathscr{A}_s$ with $0 < s \le 1/2$, then AFEM produces the error decay

$$\|u - U_k\|_\Omega \ \lesssim \ |u|_s \left(\#\mathscr{T}_k\right)^{-s} \qquad \forall\, k \ge 1.$$

provided $\#\mathscr{T}_k \ge 2\#\mathscr{T}_0$. That is, AFEM converges with the best possible rate N^{-s} for the class $\mathscr{A}_s$.

Proof. Combining Theorem 4.6 (complexity of bisection) and Lemma 9.10 (cardinality of $\mathscr{M}_k$) we infer that

$$\frac{1}{2}\#\mathscr{T}_k \le \#\mathscr{T}_k - \#\mathscr{T}_0 \ \lesssim \ \sum_{j=0}^{k-1} \#\mathscr{M}_j \ \lesssim \ |u|_s^{1/s} \sum_{j=0}^{k-1} e_j^{-1/s}.$$

On the other hand, Theorem 8.1 (contraction property), valid because $\mathrm{osc}_k(U_k) = 0$, gives geometric decay of the error,

$$e_k \le \alpha^{k-j} e_j \quad \forall\, j \le k \quad \Rightarrow \quad e_j \ge \alpha^{-(k-j)} e_k;$$

hence $e_j^{-1/s} \le \alpha^{(k-j)/s} e_k^{-1/s}$. Therefore, the geometric series $\sum_j (\alpha^{1/s})^{k-j}$ converges because $\alpha < 1$, and $\#\mathscr{T}_k \lesssim |u|_s^{1/s}\, e_k^{-1/s}$, as asserted. $\square$

9.2 Quasi-Optimal Cardinality: General Data

For general f, A the oscillation no longer vanishes, and the energy error alone no longer determines the achievable rate. What AFEM actually reduces is the quasi-error of Section 8, which by reliability (48) and efficiency (49) is equivalent to the *total error* $E_{\mathscr{T}}(U)$

$$E_{\mathscr{T}}(U)^2 := \ |\!|\!| u - U |\!|\!|_{\Omega}^2 + \operatorname{osc}_{\mathscr{T}}^2(U) \ \simeq \ \mathscr{E}_{\mathscr{T}}^2(U) \ \simeq \ |\!|\!| u - U |\!|\!|_{\Omega}^2 + \gamma\, \mathscr{E}_{\mathscr{T}}^2(U); \tag{63}$$

see Remark 8.5 for details. It is *here*, not in the contraction, that efficiency (49) is indispensable.

Quasi-best approximation. It turns out that AFEM satisfies the following quasi-best approximation property, or Céa-type estimate, for the total error. This means that AFEM delivers, up a multiplicative constant, the best possible total error for a given mesh $\mathscr{T}$.

Lemma 9.12 (quasi-best approximation)**.** *There exists a constant $\Lambda_1 > 0$, depending only on data (A, f) and shape regularity, such that the Galerkin solution $U \in \mathbb{V}_{\mathscr{T}}$ satisfies*

$$E_{\mathscr{T}}(U) \leq \Lambda_1 \inf_{V \in \mathbb{V}_{\mathscr{T}}} E_{\mathscr{T}}(V)$$

Proof. Propose as exercise. □

Hence the total error is the correct notion of approximability for AFEM and its decay justifies the following definition of approximation class $\mathbb{A}_s$.

Approximation classes on the triple (u, f, A). Because oscillation depends on data (f, A), the approximation class must be defined for the triple (u, f, A), not for u alone. With Π_N as before, let

$$\sigma_N(u, f, A) = \inf_{\mathscr{T} \in \Pi_N} \inf_{V \in \mathbb{V}_{\mathscr{T}}} \Big(|\!|\!| u - V |\!|\!|_{\Omega}^2 + \operatorname{osc}_{\mathscr{T}}^2(V) \Big)^{1/2}, \tag{64}$$

and define the approximation class $\mathbb{A}_s$ as

$$(u, f, A) \in \mathbb{A}_s : \quad |(u, f, A)|_s := \sup_N \Big(N^s \sigma_N(u, f, A) \Big) < \infty. \tag{65}$$

We now discuss a few examples of membership in $\mathbb{A}_s$ and highlight some important open questions. We first investigate the class $\mathbb{A}_s$ for A piecewise constant over $\mathscr{T}_0$. In this simplified scenario, the oscillation $\operatorname{osc}_{\mathscr{T}}(U)$ of (44) reduces to *data oscillation*

$$\operatorname{osc}_{\mathscr{T}}(f) := \| h(f - \bar{f}) \|_{L^2(\Omega)},$$

where $\bar{f}$ stands for the piecewise mean-value of f over $\mathscr{T}$. We now study the decay rate of $\operatorname{osc}_{\mathscr{T}}(f)$ in terms of $\#\mathscr{T}$. To this end, we define the approximation class $\mathscr{B}_t$ for f.

$$\mathscr{B}_t := \Big\{ g \in L^2(\Omega) : \ |g|_{\mathscr{B}_t} := \sup_{N > 0} \Big(N^t \inf_{\mathscr{T} \in \mathbb{T}_N} \operatorname{osc}_{\mathscr{T}}(g) \Big) < \infty \Big\}.$$

We have the following result about $\mathscr{B}_t$ for $d = 2$, namely $t = \frac{1+s}{2}$ provided $f \in W_p^s(\Omega), 0 \leq s \leq 1$.

Lemma 9.13 (decay rate of $\operatorname{osc}_{\mathscr{T}}(f)$)**.** *Let $0 \leq s \leq 1$ satisfy $1 + s - 2/p \geq 0, 1 \leq p \leq \infty$, and let $f \in W_p^s(\Omega)$. If Assumption 9.3 (initial labeling) is enforced and $N \geq 2\,\#\mathscr{T}_0$, then*

$$\inf_{\mathscr{T} \in \mathbb{T}_N} \operatorname{osc}_{\mathscr{T}}(f) \lesssim |f|_{W_p^s(\Omega)}\, N^{-\frac{1+s}{2}}.$$

Proof. Since $\text{sob}(W_p^s) - \text{sob}(L^2) = (s - \frac{2}{p}) - (0 - \frac{2}{2}) \geq 0$, we have $W_p^s(\Omega) \subset L^2(\Omega)$. Invoking Lemma 1.18 (Poincaré-Friedrichs inequality, zero mean) yields

$$\|f - \bar{f}_T\|_{L^2(T)} \lesssim h_T^{s-2/p+1} |f|_{W_p^s(\Omega)} \quad \Rightarrow \quad \text{osc}_{\mathscr{T}}(f, T) \lesssim h_T^r |f|_{W_p^s(T)}$$

where $r := 2 + s - 2/p$. We use the algorithm GREEDY of Section 4.4 with tolerance δ for the quantity $\text{osc}_{\mathscr{T}}(f, T)$, and proceed as in the proof of Theorem 4.9 (optimal greedy approximation) to deduce

$$\delta \approx (\#\mathscr{T} - \#\mathscr{T}_0)^{-\frac{1+s}{2}-\frac{1}{2}} |f|_{W_p^s(\Omega)} |\Omega|^{-\frac{1}{p}} \approx (\#\mathscr{T})^{-\frac{1+s}{2}-\frac{1}{2}} |f|_{W_p^s(\Omega)} |\Omega|^{-\frac{1}{p}}$$

because $\#\mathscr{T} \geq 2\,\#\mathscr{T}_0$. Therefore

$$\text{osc}_{\mathscr{T}}^2(f) = \sum_{T \in \mathscr{T}} \text{osc}_{\mathscr{T}}^2(f, T) \leq \delta^2 \#\mathscr{T} \approx (\#\mathscr{T})^{-(1+s)} |f|_{W_p^s(\Omega)}^2 |\Omega|^{-\frac{2}{p}},$$

as asserted. □

We then have the following characterization of $\mathbb{A}_s$ in terms of the approximation classes $\mathscr{A}_s$ for u and $\mathscr{B}_t$ for f, the former being defined as in (58) and (59) [1, 2, 19]. The result below is general, but in the present context we have $0 < s \leq \frac{1}{2}$ and $t \geq \frac{1}{2}$, according to Lemma 9.13. Therefore, the decay rate s is solely dictated by u.

Lemma 9.14 (equivalence of classes). *Let A be piecewise constant over $\mathscr{T}_0$ and Assumption 9.3 (initial labeling) be enforced. Then $(u, f, A) \in \mathbb{A}_s$ if and only if $(u, f) \in \mathscr{A}_s \times \mathscr{B}_s$ and*

$$|(u, f, A)|_s \approx |u|_{\mathscr{A}_s} + |f|_{\mathscr{B}_s}. \tag{66}$$

Proof. It is obvious from the definition of $\mathbb{A}_s$ that $(u, f, A) \in \mathbb{A}_s$ implies that $(u, f) \in \mathscr{A}_s \times \mathscr{B}_s$. In order to prove the reverse implication, let $(u, f) \in \mathscr{A}_s \times \mathscr{B}_s$ and $\mathscr{T}_1, \mathscr{T}_2 \in \mathbb{T}_N$ be meshes so that $\|\!|u - U|\!\|_\Omega \lesssim |u|_{\mathscr{A}_s} N^{-s}$ and $\text{osc}_{\mathscr{T}_2}(f) \lesssim |f|_{\mathscr{B}_s} N^{-s}$, respectively. According to (61), the overlay $\mathscr{T} = \mathscr{T}_1 \oplus \mathscr{T}_2$ is a mesh with $\#\mathscr{T} - \#\mathscr{T}_0 \leq 2N$, whence $\mathscr{T} \in \mathbb{T}_{2N}$ and

$$\|\!|u - U|\!\| + \text{osc}_{\mathscr{T}}(f) \lesssim \big(|u|_{\mathscr{A}_s} + |f|_{\mathscr{B}_s}\big) N^{-s},$$

because both $\|\!|u - U|\!\|$ and $\text{osc}_{\mathscr{T}}(f)$ are monotone with refinement. Hence, $(u, f, A) \in \mathbb{A}_s$ and (66) is valid, as asserted. □

Corollary 9.15 (membership in $\mathbb{A}_{1/2}$ for piecewise constant A). *Let $u \in W_p^2(\Omega) \cap H_0^1(\Omega)$, $f \in L^2(\Omega)$, A be piecewise constant, and Assumption 9.3 (initial labeling) be enforced. Then*

$$|(u, f, A)|_{1/2} \lesssim |u|_{W_p^2(\Omega)} + \|f\|_{L^2(\Omega)}. \tag{67}$$

Proof. We recall Theorem 4.9 (optimal greedy approximation), which states for $d = 2$ and $p > 1$

$$u \in W_p^2(\Omega) \cap H_0^1(\Omega) \qquad \Rightarrow \qquad u \in \mathscr{A}_{1/2}, \quad |u|_{\mathscr{A}_{1/2}} \lesssim |u|_{W_p^2(\Omega)}.$$

This, combined with Lemma 9.13 (decay rate for $\text{osc}_{\mathscr{T}}(f)$) and Lemma 9.14 (equivalence of classes), implies for piecewise constant A the bound (67), as stated. □

We finish this discussion with a result for variable A, which assumes that A is piecewise Lipschitz over $\mathscr{T}_0$: notice the nonlinear interaction encoded in $\text{osc}_{\mathscr{T}}(U)$ between A and ∇U! We extend the analysis to discontinuous A in Section 9.3.

Lemma 9.16 (decay rate of $\mathrm{osc}_{\mathscr{T}}(U)$)**.** *Let $A \in W^1_\infty(\Omega, \mathscr{T}_0) \cap W^{1+s_A}_q(\Omega; \mathscr{T}_0)$ with $s_A - \frac{2}{q} \geq 0, 1 \leq q \leq \infty$ and let $f \in L^2(\Omega) \cap W^{s_f}_p(\Omega; \mathscr{T}_0)$ with $s_f - \frac{2}{p} + 1 \geq 0, 1 \leq p \leq \infty$. If Assumption 9.3 (initial labeling) and $N \geq 2\#\mathscr{T}_0$, then the oscillation $\mathrm{osc}_{\mathscr{T}}(U)$ has a decay rate of order $-(1+s)/2$ with $s = min\{s_A, s_f\}$, i.e.*

$$\inf_{\mathscr{T}\in\mathbb{T}_N} \mathrm{osc}_{\mathscr{T}}(U) \lesssim \Big(|A|_{W^{1+s_A}_q(\Omega;\mathscr{T}_0)} \|f\|_{L^2(\Omega)} + |f|_{W^{s_f}_p(\Omega;\mathscr{T}_0)}\Big) N^{-(1+s)/2}. \qquad (68)$$

Proof. We readily see that for all edges $S = T_1 \cap T_2$

$$\begin{aligned} h_S^{1/2}\|[\![A\nabla U]\!]\|_{L^2(\partial S)} &\leq h_S^{1/2}\|A_1 - \overline{A}_1\|_{L^\infty(S)}\|\nabla U_1\|_{L^2(S)} \\ &\quad + h_S^{1/2}\|A_2 - \overline{A}_2\|_{L^\infty(S)}\|\nabla U_2\|_{L^2(S)} \lesssim \|A - \overline{A}\|_{L^\infty(\omega_S)}\|\nabla U\|_{L^2(\omega_S)}, \end{aligned}$$

where we have used that ∇U is piecewise constant to convert the integral over S into an integral over the discrete neighborhood $\omega_S = T_1 \cup T_2$ of S. If we now set

$$\mathrm{osc}_{\mathscr{T}}(A, T) = h_T\|\operatorname{div} A - \overline{\operatorname{div} A}\|_{L^\infty(T)} + \|A - \overline{A}\|_{L^\infty(\omega_T)}$$

definition (43) of oscillation, yields

$$\mathrm{osc}_{\mathscr{T}}(U, T) = h_T\|r - \bar{r}_T\|_{L^2(T)} + h_T^{1/2}\|j - \bar{j}\|_{L^2(\partial T)} \lesssim \mathrm{osc}_{\mathscr{T}}(f, T) + \mathrm{osc}_{\mathscr{T}}(A, T)\|\nabla U\|_{L^2(\omega_T)}.$$

Applying Lemma 1.18 (Poincaré-Friedrichs inequality, zero mean) with scaling exercise implies

$$\mathrm{osc}_{\mathscr{T}}(A, T) \lesssim h_T^{1+s_A-2/q}|A|_{W^{s_A}_q(T)}$$

where we allow $s_A = 0, q = \infty$ in which case $W^1_\infty(T)$ stands for $W^{s_A}_q(T)$. We now apply the argument of Theorem 4.9 (optimal greedy approximation) to the local quantities $\mathrm{osc}_{\mathscr{T}}(A, T)$ with tolerance δ to deduce

$$\delta \approx (\#\mathscr{T})^{-(1+s_A)/2}|A|_{W^{1+s_A}_q(\Omega;\mathscr{T}_0)}$$

provided $\#\mathscr{T} \geq 2\#\mathscr{T}_0$, whence for $\mathrm{osc}_{\mathscr{T}}(A) = \max_{T\in\mathscr{T}} \mathrm{osc}_{\mathscr{T}}(A, T)$ and $N \geq 2\#\mathscr{T}_0$

$$\sup_{\mathscr{T}\in\mathbb{T}_N} \mathrm{osc}_{\mathscr{T}}(A) \leq \delta \approx |A|_{W^{1+s_A}_q(\Omega;\mathscr{T}_0)} N^{-(1+s_A)/2}.$$

On the other hand, we already know from Lemma 9.13 (decay rate of $\mathrm{osc}_{\mathscr{T}}(f)$) that

$$\sup_{\mathscr{T}\in\mathbb{T}_N} \mathrm{osc}_{\mathscr{T}}(f) \lesssim |f|_{W^{s_f}_p(\Omega;\mathscr{T}_0)} N^{-(1+s_f)/2},$$

where we now assume that f is piecewise $W^{s_f}_p$ over $\mathscr{T}_0$. Given $N \geq 2\#\mathscr{T}_0$ there exist meshes $\mathscr{T}_A, \mathscr{T}_f \in \mathbb{T}_N$ satisfying the last two inequalities. If we finally let $\mathscr{T} = \mathscr{T}_A \oplus \mathscr{T}_f$ be the overlay of $\mathscr{T}_A, \mathscr{T}_f$, and apply (61), we obtain $\#\mathscr{T} - \#\mathscr{T}_0 \leq 2N$ and

$$\mathrm{osc}_{\mathscr{T}}(U) \leq \mathrm{osc}_{\mathscr{T}}(f) + \mathrm{osc}_{\mathscr{T}}(A)\|\nabla U\|_{L^2(\Omega)} \lesssim \Big(|f|_{W^{s_f}_p(\Omega;\mathscr{T}_0)} + |A|_{W^{1+s_A}_q(\Omega;\mathscr{T}_0)}\|f\|_{L^2(\Omega)}\Big) N^{-(1+s)/2}$$

with $s = \min\{s_A, s_f\}$ because $\|\nabla U\|_{L^2(\Omega)} \lesssim \|f\|_{L^2(\Omega)}$. This gives (68). $\square$

Lemma 9.16 reveals that $\mathrm{osc}_{\mathscr{T}}(U)$ decays faster than order $1/2$ provided $s > 0$. Since $1/2$ is the best decay we can expect for u with piecewise linear elements (i.e. polynomial degree $k = 1$), we realize that what dominates the performance of AFEM is the underlying regularity of u. We quantify this statement below.

Theorem 9.17 (membership in $\mathbb{A}_{1/2}$ with variable A)**.** *Let $d = 2$, $k = 1$, and $p > 1$. If $f \in L^2(\Omega)$, $A \in W^1_\infty(\Omega, \mathscr{T}_0)$, $u \in H^1_0(\Omega) \cap W^2_p(\Omega; \mathscr{T}_0)$, and Assumption 9.3 (initial labeling) is enforced, then $(u, f, A) \in \mathbb{A}_{1/2}$ and*

$$|(u, f, A)|_{1/2} \lesssim \|D^2 u\|_{L^p(\Omega;\mathscr{T}_0)} + \|f\|_{L^2(\Omega)} + \|A\|_{W^1_\infty(\Omega;\mathscr{T}_0)} \|f\|_{L^2(\Omega)}.$$

Proof. Given $N \geq 2\#\mathscr{T}_0$, let $\mathscr{T}_1, \mathscr{T}_2$ be the meshes given by Theorem 4.9 (optimal greedy approximation) for $u \in W^2_p(\Omega; \mathscr{T}_0)$ and by Lemma 9.16 (decay rate of $\operatorname{osc}_{\mathscr{T}}(U)$) for $\operatorname{osc}_{\mathscr{T}}(U)$. Let $\mathscr{T} = \mathscr{T}_1 \oplus \mathscr{T}_2$ be the overlay of $\mathscr{T}_1, \mathscr{T}_2$, which satisfies $\#\mathscr{T} - \#\mathscr{T}_0 \leq 2N$ according to (61) and

$$|\!|\!| u - U |\!|\!|_\Omega + \operatorname{osc}_{\mathscr{T}}(U) \lesssim \Big(\|D^2 u\|_{L^p(\Omega;\mathscr{T}_0)} + \|f\|_{L^2(\Omega)} + \|A\|_{W^1_\infty(\Omega;\mathscr{T}_0)} \|f\|_{L^2(\Omega)} \Big) N^{-1/2}$$

This implies the asserted estimate according to definitions (64) and (65). $\square$

Exercise 9.18. Show that $(u, f, A) \in \mathbb{A}_s$ if and only if for every $\varepsilon > 0$ there is a conforming refinement $\mathscr{T}_\varepsilon \geq \mathscr{T}_0$ and $V_\varepsilon \in \mathbb{V}(\mathscr{T}_\varepsilon)$ with $E_{\mathscr{T}}(V_\varepsilon) \leq \varepsilon$ and $\#\mathscr{T}_\varepsilon - \#\mathscr{T}_0 \lesssim |(u, f, A)|_s^{1/s} \varepsilon^{-1/s}$.

Quasi-optimal rates. We now remove the restriction $\operatorname{osc}_{\mathscr{T}}(U) = 0$ of Section 9.1, and thereby make use of the preceding discussion of the total error $E_{\mathscr{T}}(U)$ defined in (63). In order to accommodate the nonlinear class $\mathbb{A}_s$ for the triple (u, f, A), that accounts for the presence of general data (f, A), we need to make an even more stringent assumption on the threshold θ_*.

Lemma 9.19 (Dörfler marking: general data)**.** *Let Assumption 9.1 (marking parameter) hold with*

$$\theta_* = \sqrt{\frac{C_2}{1 + C_1(1 + 2C_0)}}, \tag{69}$$

where C_0 is the constant in Lemma 7.2 (Lipschitz dependence on the discrete argument) and $C_1 \geq C_2$ are the constants in (48) and (49) respectively. If $\mu := \frac{1}{2}(1 - \frac{\theta^2}{\theta_*^2}) > 0$, $\mathscr{T}_* \geq \mathscr{T}_k$ is a conforming refinement of $\mathscr{T}_k$, and the Galerkin solution $U_* \in \mathbb{V}_{\mathscr{T}_*}$ satisfy

$$|\!|\!| u - U_* |\!|\!|^2 + \operatorname{osc}^2_{\mathscr{T}_*}(U_*) \leq \mu\big(|\!|\!| u - U_k |\!|\!|^2 + \operatorname{osc}^2_{\mathscr{T}_k}(U_k) \big), \tag{70}$$

then the discrete neighborhood $\mathscr{R}^+$ of the refined set $\mathscr{R} = \mathscr{T}_k \backslash \mathscr{T}_*$ satisfies the Dörfler property

$$\mathscr{E}_{\mathscr{T}_k}(U_k, \mathscr{R}^+) \geq \theta\, \mathscr{E}_{\mathscr{T}_k}(U_k). \tag{71}$$

Proof. We split the proof into four steps.
Step 1: In view of the global lower bound (49) and (70), we can write

$$\begin{aligned}
(1 - 2\mu)\, C_2\, \mathscr{E}^2_{\mathscr{T}_k}(U_k) &\leq (1 - 2\mu)\big(|\!|\!| u - U_k |\!|\!|^2_\Omega + \operatorname{osc}^2_{\mathscr{T}_k}(U_k) \big) \\
&\leq \big(|\!|\!| u - U_k |\!|\!|^2_\Omega - 2 |\!|\!| u - U_* |\!|\!|^2_\Omega \big) + \big(\operatorname{osc}^2_{\mathscr{T}_k}(U_k) - 2 \operatorname{osc}^2_{\mathscr{T}_*}(U_*) \big).
\end{aligned}$$

Step 2: Combining the Pythagoras orthogonality relation (52)

$$|\!|\!| u - U |\!|\!|^2_\Omega - |\!|\!| u - U_* |\!|\!|^2_\Omega = |\!|\!| U - U_* |\!|\!|^2_\Omega.$$

with the localized upper bound (50) yields

$$|\!|\!| u - U |\!|\!|^2_\Omega - 2 |\!|\!| u - U_* |\!|\!|^2_\Omega \leq |\!|\!| U - U_* |\!|\!|^2 \leq C_1\, \mathscr{E}^2_{\mathscr{T}_k}(U_k, \mathscr{R}^+).$$

Step 3: To deal with oscillation we decompose the elements of $\mathscr{T}_k$ into two disjoint sets: $\mathscr{R}^+$ and $\mathscr{T}_k \setminus \mathscr{R}^+$. In the former case, we have

$$\operatorname{osc}^2_{\mathscr{T}_k}(U_k, \mathscr{R}^+) - 2\operatorname{osc}^2_{\mathscr{T}_*}(U_*, \mathscr{R}^+) \le \operatorname{osc}^2_{\mathscr{T}}(U_k, \mathscr{R}^+) \le \mathscr{E}^2_{\mathscr{T}_k}(U_k, \mathscr{R}^+),$$

because $\operatorname{osc}_{\mathscr{T}}(U,T) \le \mathscr{E}_{\mathscr{T}}(U,T)$ for all $\mathscr{T} \in \mathbb{T}$. On the other hand, we use that $\mathscr{T}_k \setminus \mathscr{R}^+ \subset \mathscr{T}_k \cap \mathscr{T}_*$ and apply a variant of Lemma 7.2 (Lipschitz dependence on the discrete argument) for $\operatorname{osc}_{\mathscr{T}_k}(U_k, T)$ with $T \in \mathscr{T}_k \cap \mathscr{T}_*$ together with Lemma 7.1 (localized upper bound) to get

$$\operatorname{osc}^2_{\mathscr{T}_k}(U_k, \mathscr{T}_k \setminus \mathscr{R}^+) - 2\operatorname{osc}^2_{\mathscr{T}_*}(U_*, \mathscr{T}_k \setminus \mathscr{R}^+) \le 2C_0 |\!|\!| U_k - U_* |\!|\!|^2_\Omega \le 2C_0C_1 \mathscr{E}^2_{\mathscr{T}_k}(U_k, \mathscr{R}^+),$$

where the factor 2 accounts for the fact that (53) does not carry squares. Adding these two estimates gives

$$\operatorname{osc}^2_{\mathscr{T}_k}(U_k) - 2\operatorname{osc}^2_{\mathscr{T}_*}(U_*) \le (1 + 2C_0C_1)\mathscr{E}^2_{\mathscr{T}_k}(U_k, \mathscr{R}^+).$$

Step 4: Returning to Step 1, we realize that

$$(1 - 2\mu)\, C_2\, \mathscr{E}^2_{\mathscr{T}_k}(U_k) \le \big(1 + C_1(1 + 2C_0)\big)\, \mathscr{E}^2_{\mathscr{T}_k}(U_k, \mathscr{R}^+),$$

which is the asserted estimate (71) in disguise. □

Lemma 9.20 (cardinality of $\mathscr{M}_k$ for general data)**.** *Let Assumption 9.1 (marking parameter) be enforced with θ_* given by (69), and let Assumptions 9.2 (minimal cardinality) and 9.3 (initial-labeling) hold. If the triple $(u, f, A) \in \mathbb{A}_s$ for $0 < s \le 1/2$, then*

$$\#\mathscr{M}_k \lesssim |(u, f, A)|_s^{1/s} \big(|\!|\!| u - U_k |\!|\!|_\Omega + \operatorname{osc}_k(U_k) \big)^{-1/s} \quad \forall\, k \ge 0. \tag{72}$$

Proof. We split the proof into three steps.
Step 1: Let Λ_1 be the constant in Lemma 9.12 (quasi-best approximation), and set

$$\varepsilon^2 := \mu\, \Lambda_1^{-2} \big(|\!|\!| u - U_k |\!|\!|^2_\Omega + \operatorname{osc}^2_k(U_k) \big), \qquad \mu = \tfrac{1}{2}\big(1 - \theta^2/\theta_*^2\big) > 0 \text{ as in Lemma 9.19.}$$

Since $(u, f, A) \in \mathbb{A}_s$, Exercise 9.18 provides a conforming refinement $\mathscr{T}_\varepsilon \ge \mathscr{T}_0$ and $V_\varepsilon \in \mathbb{V}_{\mathscr{T}_\varepsilon}$ with

$$|\!|\!| u - V_\varepsilon |\!|\!|^2_\Omega + \operatorname{osc}^2_{\mathscr{T}_\varepsilon}(V_\varepsilon) \le \varepsilon^2 \qquad \text{and} \qquad \#\mathscr{T}_\varepsilon - \#\mathscr{T}_0 \lesssim |(u, f, A)|_s^{1/s}\, \varepsilon^{-1/s}.$$

Since $\mathscr{T}_\varepsilon$ may be unrelated to $\mathscr{T}_k$, we introduce the overlay $\mathscr{T}_* = \mathscr{T}_k \oplus \mathscr{T}_\varepsilon$ so that $\mathscr{T}_* \ge \mathscr{T}_k, \mathscr{T}_\varepsilon$.
Step 2: We claim the total error over $\mathscr{T}_*$ reduces by the factor μ relative to that over $\mathscr{T}_k$. Indeed, since $\mathscr{T}_* \ge \mathscr{T}_\varepsilon$, whence $\mathbb{V}_{\mathscr{T}_*} \supset \mathbb{V}_{\mathscr{T}_\varepsilon}$, Lemma 9.12 (quasi-best approximation) gives

$$|\!|\!| u - U_* |\!|\!|^2_\Omega + \operatorname{osc}^2_{\mathscr{T}_*}(U_*) \le \Lambda_1^2 \big(|\!|\!| u - V_\varepsilon |\!|\!|^2_\Omega + \operatorname{osc}^2_{\mathscr{T}_\varepsilon}(V_\varepsilon) \big) \le \Lambda_1^2\, \varepsilon^2 = \mu\big(|\!|\!| u - U_k |\!|\!|^2_\Omega + \operatorname{osc}^2_k(U_k) \big).$$

Thus $\mathscr{T}_* \ge \mathscr{T}_k$ satisfies hypothesis (70) of Lemma 9.19 (Dörlfer marking: general data), whence the discrete neighborhood $\mathscr{R}^+$ of the refined set $\mathscr{R}_k = \mathscr{T}_k \setminus \mathscr{T}_*$ satisfies the Dörfler property (71) with parameter $\theta < \theta_*$, namely

$$\mathscr{E}_{\mathscr{T}_k}(U_k, \mathscr{R}^+) \ge \theta \mathscr{E}_{\mathscr{T}_k}(U_k).$$

Step 3: By Assumption 9.2 (minimal cardinality), MARK selects a set $\mathscr{M}_k$ of minimal cardinality with the Dörfler property. Therefore, using (61) to account for the overlay,

$$\#\mathscr{M}_k \le \#\mathscr{R}^+ \lesssim \#\mathscr{R} \le \#\mathscr{T}_* - \#\mathscr{T}_k \le \#\mathscr{T}_\varepsilon - \#\mathscr{T}_0 \lesssim |(u, f, A)|_s^{1/s}\, \varepsilon^{-1/s}.$$

Recalling the definition of ε yields the asserted estimate (72). □

We are now ready to prove the main result of this section, which combines Theorem 4.6 (complexity of bisection) with Lemma 9.20 (cardinality of $\mathscr{M}_k$ for general data).

Theorem 9.21 (quasi-optimality for general data). *Let Assumptions 9.1 (marking parameter) with θ_* given by (69), 9.2 (minimal cardinality) and 9.3 (initial labeling) hold. If $(u, f, A) \in \mathbb{A}_s$ with $0 < s \le 1/2$, and $\#\mathscr{T}_k \ge 2\#\mathscr{T}_0$, then AFEM gives rise to a sequence $\{\mathscr{T}_k, \mathbb{V}_{\mathscr{T}_k}, U_k\}_{k\ge 0}$ such that*

$$\|\!|u - U_k|\!\|_\Omega + \operatorname{osc}_k(U_k) \ \lesssim \ |(u, f, A)|_s \left(\#\mathscr{T}_k\right)^{-s} \qquad \forall\, k \ge 1.$$

Proof. Since no confusion arises, we abbreviate $\operatorname{osc}_j = \operatorname{osc}_j(U_j)$ and $\mathscr{E}_j = \mathscr{E}_j(U_j)$.
Step 1: By Assumption 9.3, Theorem 4.6 applies, and together with the cardinality bound (72) of Lemma 9.20 we obtain

$$\frac{1}{2}\,\#\mathscr{T}_k \le \#\mathscr{T}_k - \#\mathscr{T}_0 \ \lesssim \ \sum_{j=0}^{k-1} \#\mathscr{M}_j \ \lesssim \ |(u, f, A)|_s^{1/s} \sum_{j=0}^{k-1} \left(\|\!|u - U_j|\!\|_\Omega^2 + \operatorname{osc}_j^2\right)^{-1/(2s)}.$$

Step 2: Let $\gamma > 0$ be the scaling factor of Theorem 8.3 (contraction of the quasi-error). The global lower bound (49) together with $\operatorname{osc}_j \le \mathscr{E}_j$ yields

$$\|\!|u - U_j|\!\|_\Omega^2 + \gamma \operatorname{osc}_j^2 \le \|\!|u - U_j|\!\|_\Omega^2 + \gamma\, \mathscr{E}_j^2 \le \left(1 + \frac{\gamma}{C_2}\right)\left(\|\!|u - U_j|\!\|_\Omega^2 + \operatorname{osc}_j^2\right),$$

so that $\left(\|\!|u - U_j|\!\|_\Omega^2 + \operatorname{osc}_j^2\right)^{-1/(2s)} \lesssim \left(\|\!|u - U_j|\!\|_\Omega^2 + \gamma\, \mathscr{E}_j^2\right)^{-1/(2s)}$.
Step 3: Theorem 8.3 gives, for $0 \le j < k$,

$$\|\!|u - U_k|\!\|_\Omega^2 + \gamma\, \mathscr{E}_k^2 \le \alpha^{2(k-j)}\left(\|\!|u - U_j|\!\|_\Omega^2 + \gamma\, \mathscr{E}_j^2\right),$$

whence, combining Steps 1–3,

$$\#\mathscr{T}_k \ \lesssim \ |(u, f, A)|_s^{1/s} \left(\|\!|u - U_k|\!\|_\Omega^2 + \gamma\, \mathscr{E}_k^2\right)^{-1/(2s)} \sum_{j=0}^{k-1} \alpha^{(k-j)/s}.$$

Since $\sum_{j=0}^{k-1} \alpha^{(k-j)/s} \le \sum_{i=1}^{\infty} \alpha^{i/s} < \infty$ because $\alpha < 1$, and $\|\!|u - U_k|\!\|_\Omega^2 + \gamma\, \mathscr{E}_k^2 \gtrsim \|\!|u - U_k|\!\|_\Omega^2 + \operatorname{osc}_k^2$ once more by $\operatorname{osc}_k \le \mathscr{E}_k$, we conclude that

$$\#\mathscr{T}_k \lesssim |(u, f, A)|_s^{1/s} \left(\|\!|u - U_k|\!\|_\Omega^2 + \operatorname{osc}_k^2\right)^{-1/(2s)}.$$

Rearranging this inequality gives the asserted rate. $\square$

We conclude this section with two applications of Theorem 9.21.

Corollary 9.22 (W_p^2-regularity with piecewise constant A). *Let $d = 2$, the polynomial degree be 1, $f \in L^2(\Omega)$, and let A be piecewise constant over $\mathscr{T}_0$. Let Assumption 9.1 (marking parameter) be enforced with threshold θ_* given by (69) and Assumptions 9.2 (minimal cardinality) and 9.3 (initial-labeling) hold. If $u \in W_p^2(\Omega; \mathscr{T}_0)$ for $p > 1$, then AFEM gives rise to a sequence $\{\mathscr{T}_k, \mathbb{V}_{\mathscr{T}_k}, U_k\}_{k\ge 0}$ with $\operatorname{osc}_k(U_k) = \operatorname{osc}_k(f) = \|h_k(f - \overline{f})\|_{L^2(\Omega)}$ and, for all $k \ge 1$,*

$$\|\!|u - U_k|\!\|_\Omega + \operatorname{osc}_k(f) \ \lesssim \ \left(\|D^2 u\|_{L^p(\Omega;\mathscr{T}_0)} + \|f\|_{L^2(\Omega)}\right)\left(\#\mathscr{T}_k\right)^{-1/2},$$

provided $\#\mathscr{T}_k \ge 2\,\#\mathscr{T}_0$.

Proof. Combine Corollary 9.15 (membership in $\mathbb{A}_{1/2}$ for piecewise constant A) with Theorem 9.21 (quasi-optimality for general data). $\square$

Corollary 9.23 (W_p^2-regularity with variable A)**.** *In addition to the assumptions of Corollary 9.22, let A be piecewise Lipschitz over $\mathscr{T}_0$, i.e. $A \in W_\infty^1(\Omega;\mathscr{T}_0)$. Then AFEM gives rise to a sequence $\{\mathscr{T}_k, \mathbb{V}_{\mathscr{T}_k}, U_k\}_{k\geq 0}$ such that, for all $k \geq 1$,*

$$|\!|\!| u - U_k |\!|\!|_\Omega + \operatorname{osc}_k(U_k) \;\lesssim\; \left(\|D^2 u\|_{L^p(\Omega;\mathscr{T}_0)} + \|f\|_{L^2(\Omega)} + \|A\|_{W^1_\infty(\Omega;\mathscr{T}_0)}\|f\|_{L^2(\Omega)}\right)\left(\#\mathscr{T}_k\right)^{-1/2},$$

provided $\#\mathscr{T}_k \geq 2\,\#\mathscr{T}_0$.

Proof. Combine Theorem 9.17 (membership in $\mathbb{A}_{1/2}$ with variable A) with Theorem 9.21 (quasi-optimality for general data). $\square$

Remark 9.24 (highlights of the general theory). It removes two restrictive hypotheses of §9.1, namely *vanishing oscillation* and the *interior node property*. This comes at the price of

- a more stringent threshold θ_* in (69) than in (62);
- measuring approximability of the triple (u, f, A) by the total error $E_{\mathscr{T}}(U) = |\!|\!| u - U |\!|\!|_\Omega + \operatorname{osc}_{\mathscr{T}}(U)$;
- driving the whole analysis by the quasi-error $|\!|\!| u - U |\!|\!|_\Omega + \mathscr{E}_{\mathscr{T}}(U)$.

Remarkably, a *single* Dörfler marking on the estimator still suffices; no separate marking of oscillation is required.

Remark 9.25 (the complete picture). The three pillars of adaptivity combine cleanly:

- *reliability + efficiency* (Section 5) make the estimator a faithful proxy for the error;
- the *contraction property* of the total error (Section 8) gives convergence;
- *optimal marking + complexity of bisection* (Section 9 and Section 4) yield the best-possible convergence rate.

Crucially, AFEM achieves rate optimality without any a priori knowledge of the solution singular structure — it discovers the quasi-optimal meshes automatically from the computed indicators.

9.3 New Paradigm of AFEM

The preceding analysis is carried out under the premise that

$$A \in W^1_\infty(\Omega;\mathscr{T}_0), \quad f \in L^2(\Omega).$$

This does not allow for discontinuities of A within elements of any conforming refinement $\mathscr{T} \geq \mathscr{T}_0$ of the initial mesh $\mathscr{T}_0$, which entails resolution of A over $\mathscr{T}_0$. It does not allow either for more general forcing terms f, which appear natural in applications. We now discuss how to circumvent these restrictions; consult [3] for details.

We start upon stressing the different role played by f and A. The relation between f and u is linear whereas that between A and u is nonlinear, multiplicative to be precise.

Forcing $f \in H^{-1}(\Omega)$ Dealing with forcing in $H^{-1}(\Omega)$ is important for two reasons. First, such loads appear naturally in applications. For instance, we could have $f \in L^p(\Omega)$ with $1 < p < 2$ or f could be a line Dirac mass along a Lipschitz curve for $d = 2$. Second, this is the natural norm to evaluate the residual of the Galerkin solution $U \in \mathbb{V}_{\mathscr{T}}$, namely

$$R = f - \operatorname{div}(A\nabla U) \in H^{-1}(\Omega),$$

because

$$M^{-1}\|R\|_{H^{-1}(\Omega)} \leq |u - U|_{H_0^1(\Omega)} \leq \alpha^{-1}\|R\|_{H^{-1}(\Omega)}.$$

Since $\langle R, \phi_z \rangle$ for all interior nodes $z \in \mathscr{N}$, it is possible to localize the nonlocal H^{-1}-norm for R, indeed we have

$$\frac{1}{1+d} \sum_{z \in \mathscr{N}} \|R\|^2_{H^{-1}(\omega_z)} \leq \|R\|^2_{H^{-1}(\Omega)} \leq (d+1)C_{loc} \sum_{z \in \mathscr{N}} \|R\|^2_{H^{-1}(\omega_z)},$$

where C_{loc} depends on d and the shape regularity constant of $\mathscr{T}$. The goal is thus to approximate $\|R\|_{H^{-1}(\omega_z)}$ avoiding over and underestimation. The standard evaluation of R using weighted L^2-norms may overestimate the error $|u - U|_{H_0^1(\Omega)}$, as discussed in Remark 5.8 (overestimation): the oscillation $\operatorname{osc}_{\mathscr{T}}(U)$ dominates the error but not conversely. To remedy this drawback, we need to evaluate $\|R\|_{H^{-1}(\omega_z)}$ without replacing the norm $H^{-1}(\omega_z)$ by $L^2(\omega_z)$. We elucidate this issue next.

In order to understand the structure of R, we point out that for A piecewise constant over $\mathscr{T}$, and polynomial degree $n = 1$, $\operatorname{div}(A\nabla U)$ is a distribution that consists of Dirac masses supported on the skeleton of $\mathscr{T}$ and have constant densities on edges. It is thus natural to introduce a projection $\mathscr{P}_{\mathscr{T}} : H^{-1}(\Omega) \to \mathbb{F}_{\mathscr{T}}$ where $\mathbb{F}_{\mathscr{T}}$ is the discrete space

$$\mathbb{F}_{\mathscr{T}} := \Big\{ \ell \in H^{-1}(\Omega) : \quad \langle \ell, w \rangle = \sum_{S \in \mathscr{S}} \int_S q_S w + \sum_{T \in \mathscr{T}} \int_T q_T w \quad \forall w \in H_0^1(\Omega) \Big\},$$

where q_S, q_T are constants [13, 14] for polynomial degree $n = 1$; see also [3] for any polynomial degree. Since $\mathscr{P}\ell = \ell$ for all $\ell \in \mathbb{F}_{\mathscr{T}}$ we deduce

$$\mathscr{P}_{\mathscr{T}}(\operatorname{div}(A\nabla U)) = \operatorname{div}(A\nabla U).$$

This implies

$$R - \mathscr{P}_{\mathscr{T}} R = f - \mathscr{P}_{\mathscr{T}} f.$$

We now define the Galerkin solution $U \in \mathbb{V}_{\mathscr{T}}$ to satisfy (10) with f replaced by $\mathscr{P}_{\mathscr{T}} f$. This function U is computable because data $(A, \mathscr{P}_{\mathscr{T}} f)$ is discrete. The residual $R_{\mathscr{T}}$ associated to U reads

$$R_{\mathscr{T}} = \mathscr{P}_{\mathscr{T}} f + \operatorname{div}(A\nabla U) = \mathscr{P}_{\mathscr{T}}(f + \operatorname{div}(A\nabla U)) = -\mathscr{P}_{\mathscr{T}}(\operatorname{div}(A(\nabla(u - U))$$

is also computable, and satisfies

$$\|R_{\mathscr{T}}\|_{H^{-1}(\Omega)} \lesssim |u - U|_{H_0^1(\Omega)}.$$

Moreover

$$\begin{aligned}
\operatorname{osc}_{\mathscr{T}}(f) &= \|f - \mathscr{P}_{\mathscr{T}} f\|_{H^{-1}(\Omega)} \\
&= \|\operatorname{div}(A\nabla(u - U))) - \mathscr{P}_{\mathscr{T}}(\operatorname{div}(A\nabla(u - U)))\|_{H^{-1}(\Omega)} \lesssim |u - U|_{H_0^1(\Omega)}
\end{aligned}$$

On the other hand, we also have the converse inequality, namely

$$|u - U|_{H_0^1(\Omega)} \lesssim \|\operatorname{div}(A\nabla(u - U))\|_{H^{-1}(\Omega)} = \|f + \operatorname{div}(A\nabla(U)\|_{H^{-1}(\Omega)} \leq \|R_{\mathscr{T}}\|_{H^{-1}(\Omega)} + \operatorname{osc}_{\mathscr{T}}(f).$$

Therefore, thus construction circumvents overestimation but at the expense to being able to compute $\mathscr{P}_{\mathscr{T}} f$ for $f \in H^{-1}(\Omega)$; we this have an error dominated estimator. The oscillation $\operatorname{osc}_{\mathscr{T}}(f)$ is only computable for special functions f, such as those mentioned above, and is discussed in [13, 14] for polynomial degree $n = 1$ and [3] for any polynomial degree.

Discontinuous diffusion coefficient A Consider a perturbation $\widehat{\mathscr{D}} = (\widehat{A}, \widehat{f})$ of data $\mathscr{D} = (A, f)$, and corresponding continuous solutions $\widehat{u}, u \in H_0^1(\Omega)$. The most natural perturbation estimate measures the H^1-error between $\widehat{u}$ and u as follows:

$$|u - \widehat{u}|_{H_0^1(\Omega)} \leq \widehat{\alpha}\Big(\|f - \widehat{f}\|_{H^{-1}(\Omega)} + \alpha^{-1}\|A - \widehat{A}\|_{L^\infty(\Omega)}\|f\|_{H^{-1}(\Omega)}\Big),$$

where $\alpha, \widehat{\alpha} > 0$ are the smallest eigenvalues of $A, \widehat{A}$, or equivalently the respective coercivity constants. This is problematic if A is discontinuous unless the discontinuities of A are perfectly matched by $\widehat{A}$, something we try to avoid because it is not practical. This leads to the *perturbation estimate*

$$|u - \widehat{u}|_{H_0^1(\Omega)} \leq \widehat{\alpha}\Big(\|f - \widehat{f}\|_{H^{-1}(\Omega)} + \|\nabla u\|_{L^p(\Omega)}\|A - \widehat{A}\|_{L^q(\Omega)}\Big), \quad q = \frac{2p}{p-2}$$

for provided $u \in W_p^1(\Omega)$ for $p \geq 2$. Note that when $p = 2$ this estimate reduces to the previous one because $\|\nabla u\|_{L^2(\Omega)} \leq \alpha^{-1}\|f\|_{H^{-1}(\Omega)}$. It is important to realize that the approximation $\widehat{A}$ must have a uniform lowest eigenvalue $\widehat{\alpha} > 0$ and that the solution u must satisfy $u \in W_p^1(\Omega)$. The latter can be guaranteed provided $f \in W_p^{-1}(\Omega)$ even for $A \in L^\infty(\Omega)$; we refer to Meyers [15] and [7, 3] for further details. The former leads to the question of approximation of A in $L^q(\Omega)$ with the restriction that $\widehat{A}$ be uniformly positive definite. This delicate question is discussed in [7, 3] for any polynomial degree n, but it is trivial if $n = 0$ and $\widehat{A}$ is given by elementwise mean values of A.

Two-step AFEM We now present the algorithm advocated in [3], first proposed by Stevenson [20] and further explored in [7, 10, 4, 5, 6].

Algorithm 9.26 (AFEM-TS)**.** Given an initial mesh $\mathscr{T}_0$ satisfying Assumption 9.3 (initial-labeling), an initial tolerance ε_0, and a parameter ω sufficiently small, iterate

AFEM-TS $(\mathscr{T}_0, \varepsilon_0, \omega)$
  $k = 0$
  $[\widehat{\mathscr{T}_k}, \widehat{\mathscr{D}}_k] = \text{DATA}\,(\mathscr{T}_k, \mathscr{D}_k, \omega\,\varepsilon_k)$
  $[\mathscr{T}_{k+1}, U_{k+1}] = \text{GALERKIN}\,(\widehat{\mathscr{T}_k}, \widehat{\mathscr{D}}_k, \varepsilon_k)$
  $\varepsilon_{k+1} = \frac{1}{2}\varepsilon_k;\ k \leftarrow k+1$

The two components of data $\mathscr{D} = (A, f)$ are first approximated by discrete data $\widehat{\mathscr{D}} = (\widehat{A}, \widehat{f})$ within the module

$$[\widehat{\mathscr{T}}, \widehat{\mathscr{D}}] = \text{DATA}\,(\mathscr{T}, \mathscr{D}, \tau)$$

to accuracy $\tau = \omega\varepsilon$ significantly smaller than ε. This is achieved by an algorithm similar to Algorithm 4.7 (greedy algorithm). The resulting admissible refinement $\widehat{\mathscr{T}}$ of $\mathscr{T}$ and discrete data $\widehat{\mathscr{D}}$ over $\widehat{\mathscr{T}}$ are next taken by GALERKIN to reduce the PDE error to the desired tolerance ε, namely the module

$$[\mathscr{T}, U_{\mathscr{T}}] = \text{GALERKIN}\,(\widehat{\mathscr{T}}, \widehat{\mathscr{D}}, \varepsilon)$$

constructs a conforming refinement $\mathscr{T}$ of $\widehat{\mathscr{T}}$ with discrete data $\widehat{\mathscr{D}}$ over $\widehat{\mathscr{T}}$ such that $\mathscr{E}_{\mathscr{T}}(U_{\mathscr{T}}) \leq \varepsilon$. We point out that if data is discrete, i.e. $\mathscr{D} = \widehat{\mathscr{D}}$, then DATA is skipped and AFEM-TS reduces to the one-step procedure GALERKIN.

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